

1 Agroforestry SmartAgroforst: Poplar biomass modelling

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3 Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

4 *Keywords:* Agroforestry, Short-rotation coppice, Biomass supply chain, Product
5 cascading

```
6 ## [1] "age"          "scenario"      "Merch. stem"  "Merch. branch"  
7 ## [5] "Residue"  
  
8 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...  
9 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)  
10 ##   C1: Path establishment
```

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11  ## C2: Path connectivity
12  ## C3: Biomass yield
13  ## C4: Inventory balance
14  ## C5: Storage capacity
15  ## C6: Processing capacity
16  ## C7: Demand satisfaction
17  ## Total constraints: 27183, nnz: 977322

18  ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
19  ## Set parameter Username
20  ## Set parameter LicenseID to value 2805714
21  ## Set parameter TimeLimit to value 300
22  ## Set parameter MIPGap to value 0.05
23  ## Set parameter MIPFocus to value 1
24  ## Set parameter Cuts to value 3
25  ## Academic license - for non-commercial use only - expires 2027-04-10
26  ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
27  ##
28  ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
29  ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
30  ##
31  ## Non-default parameters:
32  ## TimeLimit 300
33  ## MIPGap 0.05
34  ## MIPFocus 1
35  ## Cuts 3
36  ## LogToConsole 0
37  ##
38  ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
39  ## Model fingerprint: 0x2903b5f1
40  ## Variable types: 246642 continuous, 66265 integer (66265 binary)
41  ## Coefficient statistics:

```

```

42 ## Matrix range      [1e+00, 6e+05]
43 ## Objective range   [1e-01, 5e+07]
44 ## Bounds range     [1e+00, 5e+07]
45 ## RHS range        [1e+00, 7e+07]
46 ## Found heuristic solution: objective -0.0000000
47 ## Presolve removed 3513 rows and 47344 columns
48 ## Presolve time: 2.66s
49 ## Presolved: 23670 rows, 265563 columns, 844537 nonzeros
50 ## Variable types: 204919 continuous, 60644 integer (60644 binary)
51 ## Found heuristic solution: objective 1789027.8702
52 ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
53 ## Showing barrier log only...
54 ##
55 ## Root barrier log...
56 ##
57 ## Ordering time: 0.41s
58 ##
59 ## Barrier statistics:
60 ## AA' NZ      : 5.969e+05
61 ## Factor NZ   : 7.942e+06 (roughly 200 MB of memory)
62 ## Factor Ops  : 8.060e+09 (less than 1 second per iteration)
63 ## Threads    : 10
64 ##
65 ##              Objective              Residual
66 ## Iter      Primal      Dual      Primal      Dual      Compl      Time
67 ##   0  -1.27238033e+17  2.30657289e+17  2.46e+09  2.33e+06  7.99e+13  4s
68 ##   1  -5.04199417e+16  2.04695645e+17  9.71e+08  5.34e+06  3.23e+13  4s
69 ##   2  -6.81438217e+15  1.58107025e+17  1.26e+08  4.55e+05  4.44e+12  4s
70 ##   3  -1.88672858e+15  8.72316847e+16  3.22e+07  3.35e+04  1.18e+12  4s
71 ##   4  -3.97794604e+14  4.46059668e+16  5.93e+06  1.12e+04  2.59e+11  4s
72 ##   5  -8.49501815e+13  1.88357394e+16  1.15e+06  1.59e+03  6.94e+10  4s

```

73	##	6	-2.68767003e+13	9.79902059e+15	1.88e+05	6.98e+02	2.79e+10	4s
74	##	7	-4.01671381e+12	6.59281020e+15	1.67e+04	4.40e+02	1.33e+10	4s
75	##	8	-1.80913625e+12	3.16542978e+15	7.17e+03	1.52e+02	6.17e+09	5s
76	##	9	-7.25487760e+11	1.00688933e+15	2.48e+03	2.11e+01	1.91e+09	5s
77	##	10	-3.85787441e+11	1.05717787e+14	1.02e+03	6.88e-08	2.08e+08	5s
78	##	11	-3.41807499e+11	7.76593023e+13	8.42e+02	9.73e-08	1.54e+08	5s
79	##	12	-3.12174733e+11	5.32639810e+13	7.21e+02	5.58e-08	1.07e+08	5s
80	##	13	-2.32575974e+11	2.36218750e+13	4.34e+02	6.89e-08	5.03e+07	5s
81	##	14	-1.37689645e+11	4.50041646e+12	1.49e+02	7.45e-08	1.19e+07	5s
82	##	15	-1.10219518e+11	2.41466677e+12	6.99e+01	1.01e-07	6.38e+06	5s
83	##	16	-1.04872786e+11	2.20177064e+12	5.66e+01	1.49e-07	5.68e+06	5s
84	##	17	-8.20518347e+10	1.99548634e+12	4.23e+01	2.14e-07	4.90e+06	5s
85	##	18	-5.69333690e+10	1.21912826e+12	2.75e+01	1.19e-07	3.03e+06	6s
86	##	19	-3.80307534e+10	1.15364221e+12	1.77e+01	1.68e-07	2.62e+06	6s
87	##	20	-2.66490448e+10	4.52218888e+11	1.23e+01	8.01e-08	1.13e+06	6s
88	##	21	-1.32031252e+10	3.17549827e+11	6.06e+00	6.89e-08	7.09e+05	6s
89	##	22	-1.09058418e+10	2.36203225e+11	5.05e+00	8.01e-08	5.23e+05	6s
90	##	23	-7.59270974e+09	1.64975714e+11	3.64e+00	6.52e-08	3.55e+05	6s
91	##	24	-5.57530507e+09	1.19898544e+11	2.80e+00	5.59e-08	2.52e+05	6s
92	##	25	-2.92433908e+09	7.60099060e+10	1.73e+00	4.10e-08	1.53e+05	6s
93	##	26	-1.42927532e+09	3.18124346e+10	1.12e+00	2.61e-08	6.32e+04	6s
94	##	27	-8.26573224e+08	2.27715950e+10	8.98e-01	2.05e-08	4.43e+04	6s
95	##	28	-3.51216792e+08	1.77726999e+10	7.28e-01	1.49e-08	3.38e+04	7s
96	##	29	4.19161754e+08	1.34506875e+10	4.61e-01	9.31e-09	2.40e+04	7s
97	##	30	8.47139246e+08	9.96479607e+09	3.24e-01	1.12e-08	1.67e+04	7s
98	##	31	1.04456073e+09	7.97935798e+09	2.50e-01	9.31e-09	1.27e+04	7s
99	##	32	1.57500376e+09	5.96973266e+09	9.39e-02	5.59e-09	8.01e+03	7s
100	##	33	1.69966300e+09	5.11733794e+09	7.45e-02	7.45e-09	6.23e+03	7s
101	##	34	1.80541556e+09	4.08300426e+09	6.11e-02	5.59e-09	4.15e+03	7s
102	##	35	1.94792749e+09	3.74581284e+09	4.37e-02	4.66e-09	3.27e+03	7s
103	##	36	2.02573440e+09	3.44169122e+09	3.53e-02	4.66e-09	2.58e+03	7s

104	##	37	2.13289912e+09	3.23990877e+09	2.60e-02	4.66e-09	2.01e+03	7s
105	##	38	2.29852752e+09	3.05507689e+09	1.13e-02	5.59e-09	1.38e+03	8s
106	##	39	2.34435278e+09	2.87751063e+09	8.16e-03	4.66e-09	9.71e+02	8s
107	##	40	2.39685087e+09	2.78967023e+09	4.91e-03	4.66e-09	7.15e+02	8s
108	##	41	2.43002273e+09	2.72706746e+09	3.18e-03	3.73e-09	5.41e+02	8s
109	##	42	2.44143411e+09	2.68874987e+09	2.71e-03	3.73e-09	4.50e+02	8s
110	##	43	2.46127511e+09	2.62199767e+09	1.95e-03	3.73e-09	2.93e+02	8s
111	##	44	2.47996227e+09	2.59242469e+09	1.31e-03	3.73e-09	2.05e+02	8s
112	##	45	2.49144551e+09	2.57776214e+09	9.40e-04	3.73e-09	1.57e+02	8s
113	##	46	2.49915231e+09	2.56895493e+09	7.11e-04	4.66e-09	1.27e+02	9s
114	##	47	2.50783351e+09	2.55419595e+09	4.62e-04	3.73e-09	8.44e+01	9s
115	##	48	2.51492977e+09	2.54333803e+09	2.69e-04	2.79e-09	5.17e+01	9s
116	##	49	2.51954656e+09	2.53898070e+09	1.50e-04	3.73e-09	3.54e+01	9s
117	##	50	2.52106929e+09	2.53445249e+09	1.11e-04	2.79e-09	2.44e+01	9s
118	##	51	2.52241955e+09	2.53258097e+09	7.68e-05	3.73e-09	1.85e+01	9s
119	##	52	2.52297697e+09	2.53008212e+09	6.33e-05	6.52e-09	1.29e+01	9s
120	##	53	2.52360802e+09	2.52938349e+09	4.89e-05	5.59e-09	1.05e+01	9s
121	##	54	2.52447787e+09	2.52813409e+09	2.82e-05	9.31e-09	6.66e+00	9s
122	##	55	2.52505777e+09	2.52646722e+09	1.47e-05	3.73e-09	2.57e+00	9s
123	##	56	2.52545835e+09	2.52614036e+09	6.40e-06	2.79e-09	1.24e+00	10s
124	##	57	2.52558967e+09	2.52598969e+09	3.79e-06	5.59e-09	7.29e-01	10s
125	##	58	2.52567903e+09	2.52590647e+09	2.05e-06	8.38e-09	4.14e-01	10s
126	##	59	2.52574052e+09	2.52585516e+09	9.55e-07	3.49e-09	2.09e-01	10s
127	##	60	2.52576636e+09	2.52581369e+09	6.02e-07	3.73e-09	8.62e-02	10s
128	##	61	2.52577980e+09	2.52580177e+09	3.77e-07	9.31e-09	4.00e-02	10s
129	##	62	2.52578500e+09	2.52579399e+09	2.44e-07	3.73e-09	1.64e-02	10s
130	##	63	2.52578709e+09	2.52579207e+09	3.17e-07	4.66e-09	9.07e-03	10s
131	##	64	2.52578906e+09	2.52579109e+09	9.92e-07	4.66e-09	3.70e-03	10s
132	##	65	2.52579012e+09	2.52579068e+09	1.78e-06	4.39e-09	1.02e-03	10s
133	##	66	2.52579052e+09	2.52579057e+09	1.74e-07	3.83e-09	9.37e-05	11s
134	##	67	2.52579056e+09	2.52579056e+09	4.15e-10	4.19e-09	2.88e-07	11s

```

135  ##      68      2.52579056e+09  2.52579056e+09  5.69e-09 4.66e-09  6.03e-12    11s
136  ##
137  ## Barrier solved model in 68 iterations and 10.84 seconds (14.04 work units)
138  ## Optimal objective 2.52579056e+09
139  ##
140  ##
141  ## Root crossover log...
142  ##
143  ##      19877 DPushes remaining with DInf 0.0000000e+00          11s
144  ##           0 DPushes remaining with DInf 0.0000000e+00          12s
145  ##
146  ##      351 PPushes remaining with PInf 0.0000000e+00          12s
147  ##           0 PPushes remaining with PInf 0.0000000e+00          12s
148  ##
149  ## Push phase complete: Pinf 0.0000000e+00, Dinf 2.9696090e+05    12s
150  ##
151  ##
152  ## Root simplex log...
153  ##
154  ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
155  ##      18588      2.5257906e+09  0.0000000e+00  2.969609e+05    12s
156  ##      18857      2.5257906e+09  0.0000000e+00  0.0000000e+00    12s
157  ## Concurrent spin time: 4.50s (can be avoided by choosing Method=3)
158  ##
159  ## Solved with barrier
160  ##
161  ## Root relaxation: objective 2.525791e+09, 18857 iterations, 14.02 seconds (10.30 work units)
162  ## Total elapsed time = 31.78s (DegenMoves)
163  ## Total elapsed time = 37.17s (DegenMoves)
164  ##
165  ##      Nodes      |      Current Node      |      Objective Bounds      |      Work

```

```

166  ## Expl Unexpl | Obj Depth IntInf | Incumbent BestBd Gap | It/Node Time
167  ##
168  ##      0      0 2.5258e+09      0 922 1789027.87 2.5258e+09      -      -    37s
169  ## H      0      0                1.739327e+09 2.5258e+09 45.2%      -    40s
170  ## H      0      0                1.751460e+09 2.5258e+09 44.2%      -    40s
171  ## H      0      0                1.780348e+09 2.5258e+09 41.9%      -    40s
172  ## H      0      0                1.833217e+09 2.5258e+09 37.8%      -    41s
173  ## H      0      0                1.889184e+09 2.5258e+09 33.7%      -    42s
174  ## H      0      0                1.896653e+09 2.5258e+09 33.2%      -    42s
175  ## H      0      0                1.907201e+09 2.5258e+09 32.4%      -    42s
176  ## H      0      0                1.923624e+09 2.5258e+09 31.3%      -    42s
177  ## H      0      0                2.089427e+09 2.5258e+09 20.9%      -    43s
178  ## H      0      0                2.093311e+09 2.5258e+09 20.7%      -    43s
179  ## H      0      0                2.093782e+09 2.5258e+09 20.6%      -    44s
180  ## H      0      0                2.096091e+09 2.5258e+09 20.5%      -    44s
181  ## H      0      0                2.096141e+09 2.5258e+09 20.5%      -    44s
182  ## H      0      0                2.096510e+09 2.5258e+09 20.5%      -    44s
183  ##      0      0 2.5252e+09      0 1066 2.0965e+09 2.5252e+09 20.4%      -   139s
184  ## H      0      0                2.363785e+09 2.5252e+09 6.83%      -   188s
185  ##      0      0 2.5250e+09      0 1052 2.3638e+09 2.5250e+09 6.82%      -   207s
186  ##      0      0 2.5250e+09      0 1051 2.3638e+09 2.5250e+09 6.82%      -   211s
187  ##      0      0 2.5250e+09      0 1052 2.3638e+09 2.5250e+09 6.82%      -   212s
188  ##      0      0 2.5244e+09      0 1153 2.3638e+09 2.5244e+09 6.79%      -   270s
189  ##
190  ## Cutting planes:
191  ## Gomory: 3
192  ## MIR: 148
193  ## Flow cover: 574
194  ##
195  ## Explored 1 nodes (45282 simplex iterations) in 300.03 seconds (367.78 work units)
196  ## Thread count was 24 (of 24 available processors)

```

```

197  ##
198  ## Solution count 10: 2.36379e+09 2.36379e+09 2.09651e+09 ... 1.89665e+09
199  ##
200  ## Time limit reached
201  ## Warning: max constraint violation (1.7987e-06) exceeds tolerance
202  ## Warning: max bound violation (2.6978e-06) exceeds tolerance
203  ## Best objective 2.363785191489e+09, best bound 2.524373955466e+09, gap 6.7937%
204  ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 2363785191.4891 | Runtime: 300.5 s

205  ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...
206  ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)
207  ##   C1: Path establishment
208  ##   C2: Path connectivity
209  ##   C3: Biomass yield
210  ##   C4: Inventory balance
211  ##   C5: Storage capacity
212  ##   C6: Processing capacity
213  ##   C7: Demand satisfaction
214  ##   Total constraints: 27183, nnz: 977322
215  ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
216  ## Set parameter Username
217  ## Set parameter LicenseID to value 2805714
218  ## Set parameter TimeLimit to value 300
219  ## Set parameter MIPGap to value 0.05
220  ## Set parameter MIPFocus to value 1
221  ## Set parameter Cuts to value 3
222  ## Academic license - for non-commercial use only - expires 2027-04-10
223  ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
224  ##
225  ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
226  ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
227  ##

```



```

228  ## Non-default parameters:
229  ## TimeLimit 300
230  ## MIPGap 0.05
231  ## MIPFocus 1
232  ## Cuts 3
233  ## LogToConsole 0
234  ##
235  ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
236  ## Model fingerprint: 0xc6b5c589
237  ## Variable types: 246642 continuous, 66265 integer (66265 binary)
238  ## Coefficient statistics:
239  ##   Matrix range      [1e+00, 6e+05]
240  ##   Objective range   [6e-02, 2e+07]
241  ##   Bounds range      [1e+00, 5e+07]
242  ##   RHS range         [1e+00, 7e+07]
243  ## Found heuristic solution: objective -0.0000000
244  ## Presolve removed 3524 rows and 47233 columns
245  ## Presolve time: 2.73s
246  ## Presolved: 23659 rows, 265674 columns, 844553 nonzeros
247  ## Variable types: 204919 continuous, 60755 integer (60755 binary)
248  ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
249  ## Showing barrier log only...
250  ##
251  ## Root barrier log...
252  ##
253  ## Ordering time: 0.38s
254  ##
255  ## Barrier statistics:
256  ##   AA' NZ      : 5.958e+05
257  ##   Factor NZ   : 7.925e+06 (roughly 200 MB of memory)
258  ##   Factor Ops  : 8.035e+09 (less than 1 second per iteration)

```

259 ## Threads : 10

260 ##

261	##		Objective		Residual			
262	##	Iter	Primal	Dual	Primal	Dual	Compl	Time
263	##	0	-1.12060832e+17	9.70823041e+15	2.46e+09	4.79e+02	3.55e+12	4s
264	##	1	-4.79486965e+16	8.62883139e+15	1.05e+09	2.36e+05	1.55e+12	4s
265	##	2	-6.52517216e+15	6.66883085e+15	1.38e+08	1.37e+04	2.14e+11	4s
266	##	3	-1.32225662e+15	3.82115033e+15	2.65e+07	1.07e+03	4.46e+10	4s
267	##	4	-2.27571769e+14	1.94563791e+15	4.30e+06	3.73e+02	9.38e+09	4s
268	##	5	-3.67110966e+13	6.94735367e+14	5.47e+05	4.66e-09	2.11e+09	4s
269	##	6	-6.23214799e+12	4.70884337e+14	3.28e+04	2.79e-09	9.88e+08	4s
270	##	7	-1.89211088e+12	3.05105462e+14	9.52e+03	2.79e-09	5.88e+08	4s
271	##	8	-7.99107595e+11	1.00868890e+14	3.70e+03	3.96e-09	1.92e+08	5s
272	##	9	-3.72752571e+11	2.89111827e+13	1.44e+03	3.73e-09	5.50e+07	5s
273	##	10	-2.35923535e+11	3.61928641e+12	7.56e+02	5.59e-09	7.48e+06	5s
274	##	11	-1.95116201e+11	2.58842082e+12	5.86e+02	6.05e-09	5.44e+06	5s
275	##	12	-1.39377091e+11	1.25077334e+12	3.64e+02	5.24e-09	2.78e+06	5s
276	##	13	-9.52983524e+10	7.49570707e+11	1.99e+02	7.68e-09	1.69e+06	5s
277	##	14	-6.85921889e+10	5.02942754e+11	1.03e+02	7.92e-09	1.14e+06	5s
278	##	15	-5.36456961e+10	4.52098134e+11	6.94e+01	9.78e-09	9.89e+05	5s
279	##	16	-4.44396890e+10	3.75985196e+11	5.72e+01	1.27e-08	8.23e+05	5s
280	##	17	-3.63891849e+10	3.03473713e+11	4.56e+01	9.20e-09	6.65e+05	5s
281	##	18	-3.20530022e+10	2.85796009e+11	3.95e+01	1.26e-08	6.19e+05	5s
282	##	19	-2.06052633e+10	2.37074876e+11	2.62e+01	1.40e-08	4.96e+05	6s
283	##	20	-1.14532689e+10	1.58616497e+11	1.49e+01	9.31e-09	3.23e+05	6s
284	##	21	-9.15016614e+09	1.33223660e+11	1.22e+01	8.85e-09	2.70e+05	6s
285	##	22	-3.72939111e+09	8.41562398e+10	5.25e+00	8.15e-09	1.64e+05	6s
286	##	23	-2.28161911e+09	7.04755670e+10	3.39e+00	8.38e-09	1.35e+05	6s
287	##	24	-1.14180093e+09	4.94628338e+10	1.77e+00	7.33e-09	9.29e+04	6s
288	##	25	-9.11735379e+08	3.58562679e+10	1.45e+00	6.05e-09	6.73e+04	6s
289	##	26	-7.15657037e+08	2.48411032e+10	1.22e+00	3.73e-09	4.67e+04	6s

290	##	27	-3.64527653e+08	1.88365345e+10	8.40e-01	3.73e-09	3.50e+04	6s
291	##	28	1.80300098e+08	1.17474976e+10	2.81e-01	2.79e-09	2.11e+04	6s
292	##	29	3.98167351e+08	8.97557930e+09	1.52e-01	2.33e-09	1.56e+04	7s
293	##	30	5.16076030e+08	5.50585811e+09	9.21e-02	2.10e-09	9.08e+03	7s
294	##	31	5.59248404e+08	5.14038383e+09	7.66e-02	2.33e-09	8.33e+03	7s
295	##	32	5.97451744e+08	4.26894815e+09	6.28e-02	2.33e-09	6.68e+03	7s
296	##	33	6.11605207e+08	4.00861409e+09	5.82e-02	2.79e-09	6.18e+03	7s
297	##	34	6.96307198e+08	2.70762368e+09	3.15e-02	2.33e-09	3.66e+03	7s
298	##	35	7.46838478e+08	2.24189869e+09	1.91e-02	2.79e-09	2.72e+03	7s
299	##	36	7.79777781e+08	1.87870293e+09	1.20e-02	2.56e-09	2.00e+03	7s
300	##	37	7.92862575e+08	1.50441771e+09	9.76e-03	2.10e-09	1.29e+03	7s
301	##	38	8.20924844e+08	1.32115302e+09	5.59e-03	2.79e-09	9.10e+02	7s
302	##	39	8.24063876e+08	1.22742583e+09	5.24e-03	2.74e-09	7.34e+02	8s
303	##	40	8.27148095e+08	1.20415872e+09	4.92e-03	2.44e-09	6.86e+02	8s
304	##	41	8.51142607e+08	1.12204674e+09	2.55e-03	2.88e-09	4.93e+02	8s
305	##	42	8.58458451e+08	1.02646626e+09	2.08e-03	1.86e-09	3.06e+02	8s
306	##	43	8.67524337e+08	1.00941629e+09	1.55e-03	2.97e-09	2.58e+02	8s
307	##	44	8.74531579e+08	9.75370686e+08	1.17e-03	2.95e-09	1.83e+02	8s
308	##	45	8.82031223e+08	9.53159193e+08	8.19e-04	1.95e-09	1.29e+02	8s
309	##	46	8.87321046e+08	9.36856862e+08	5.82e-04	1.98e-09	9.01e+01	8s
310	##	47	8.91888864e+08	9.21328768e+08	3.93e-04	3.81e-09	5.36e+01	8s
311	##	48	8.95996559e+08	9.14172942e+08	2.31e-04	3.00e-09	3.31e+01	9s
312	##	49	8.96927654e+08	9.10358183e+08	1.96e-04	3.38e-09	2.44e+01	9s
313	##	50	8.97742586e+08	9.08999694e+08	1.66e-04	3.30e-09	2.05e+01	9s
314	##	51	8.98902955e+08	9.06966777e+08	1.23e-04	4.68e-09	1.47e+01	9s
315	##	52	9.00051222e+08	9.05585815e+08	8.15e-05	2.33e-09	1.01e+01	9s
316	##	53	9.01120175e+08	9.03890370e+08	4.33e-05	2.02e-09	5.04e+00	9s
317	##	54	9.01706009e+08	9.03100627e+08	2.40e-05	4.16e-09	2.54e+00	9s
318	##	55	9.02132871e+08	9.02777871e+08	1.07e-05	2.13e-09	1.17e+00	9s
319	##	56	9.02274096e+08	9.02657944e+08	6.16e-06	2.23e-09	6.98e-01	9s
320	##	57	9.02374375e+08	9.02580642e+08	3.07e-06	2.76e-09	3.75e-01	9s

```

321  ##      58      9.02417786e+08  9.02535563e+08  1.98e-06  2.14e-09  2.14e-01  10s
322  ##      59      9.02455004e+08  9.02505106e+08  1.00e-06  1.86e-09  9.12e-02  10s
323  ##      60      9.02465015e+08  9.02488371e+08  1.03e-06  2.33e-09  4.25e-02  10s
324  ##      61      9.02473097e+08  9.02479703e+08  1.00e-06  1.89e-09  1.20e-02  10s
325  ##      62      9.02475305e+08  9.02478166e+08  2.32e-06  2.19e-09  5.21e-03  10s
326  ##      63      9.02475979e+08  9.02476941e+08  1.18e-06  2.10e-09  1.75e-03  10s
327  ##      64      9.02476585e+08  9.02476780e+08  2.68e-07  2.57e-09  3.55e-04  10s
328  ##      65      9.02476769e+08  9.02476772e+08  1.49e-10  2.25e-09  4.95e-06  10s
329  ##      66      9.02476771e+08  9.02476771e+08  2.45e-10  2.79e-09  9.91e-10  10s
330  ##
331  ## Barrier solved model in 66 iterations and 10.30 seconds (13.83 work units)
332  ## Optimal objective 9.02476771e+08
333  ##
334  ##
335  ## Root crossover log...
336  ##
337  ##      19782 DPushes remaining with DInf 0.0000000e+00 10s
338  ##           0 DPushes remaining with DInf 0.0000000e+00 11s
339  ##
340  ##      373 PPushes remaining with PInf 4.3962109e-04 11s
341  ##           0 PPushes remaining with PInf 0.0000000e+00 11s
342  ##
343  ## Push phase complete: Pinf 0.0000000e+00, Dinf 2.8878007e+04 11s
344  ##
345  ##
346  ## Root simplex log...
347  ##
348  ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
349  ##      19248      9.0247677e+08  0.000000e+00  2.887801e+04  11s
350  ##      19687      9.0247677e+08  0.000000e+00  0.000000e+00  11s
351  ## Concurrent spin time: 2.22s (can be avoided by choosing Method=3)

```

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352  ##
353  ## Solved with barrier
354  ##
355  ## Root relaxation: objective 9.024768e+08, 19687 iterations, 10.62 seconds (10.18 work units)
356  ## Total elapsed time = 22.26s (DegenMoves)
357  ## Total elapsed time = 25.25s (DegenMoves)
358  ##
359  ##      Nodes      |      Current Node      |      Objective Bounds      |      Work
360  ##  Expl Unexpl |  Obj  Depth IntInf |  Incumbent    BestBd   Gap | It/Node Time
361  ##
362  ##      0      0 9.0248e+08      0 1048   -0.00000 9.0248e+08      -      -   25s
363  ## H      0      0                        4.680643e+08 9.0248e+08 92.8%      -   25s
364  ## H      0      0                        5.179789e+08 9.0248e+08 74.2%      -   25s
365  ## H      0      0                        5.294540e+08 9.0248e+08 70.5%      -   25s
366  ## H      0      0                        5.428690e+08 9.0248e+08 66.2%      -   26s
367  ## H      0      0                        5.567192e+08 9.0248e+08 62.1%      -   28s
368  ## H      0      0                        5.745607e+08 9.0248e+08 57.1%      -   28s
369  ## H      0      0                        5.914132e+08 9.0248e+08 52.6%      -   29s
370  ## H      0      0                        6.365745e+08 9.0248e+08 41.8%      -   29s
371  ## H      0      0                        6.500415e+08 9.0248e+08 38.8%      -   30s
372  ## H      0      0                        6.559930e+08 9.0248e+08 37.6%      -   30s
373  ## H      0      0                        6.585352e+08 9.0248e+08 37.0%      -   31s
374  ## H      0      0                        6.585582e+08 9.0248e+08 37.0%      -   31s
375  ## H      0      0                        6.585930e+08 9.0248e+08 37.0%      -   31s
376  ## H      0      0                        6.594971e+08 9.0248e+08 36.8%      -   31s
377  ## H      0      0                        6.618684e+08 9.0248e+08 36.4%      -   31s
378  ## H      0      0                        6.619308e+08 9.0248e+08 36.3%      -   31s
379  ## H      0      0                        6.680343e+08 9.0248e+08 35.1%      -   31s
380  ## H      0      0                        6.683745e+08 9.0248e+08 35.0%      -   31s
381  ## H      0      0                        6.694517e+08 9.0248e+08 34.8%      -   31s
382  ## H      0      0                        6.709365e+08 9.0248e+08 34.5%      -   31s

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383 ## H      0      0                6.716410e+08 9.0248e+08 34.4%      -   32s
384 ## H      0      0                6.724927e+08 9.0248e+08 34.2%      -   32s
385 ## H      0      0                6.779968e+08 9.0248e+08 33.1%      -   32s
386 ## H      0      0                6.794248e+08 9.0248e+08 32.8%      -   32s
387 ##      0      0 9.0223e+08      0 1151 6.7942e+08 9.0223e+08 32.8%      -   77s
388 ## H      0      0                6.834142e+08 9.0223e+08 32.0%      -   85s
389 ## H      0      0                8.615114e+08 9.0223e+08 4.73%      -  115s
390 ##
391 ## Cutting planes:
392 ##   Gomory: 3
393 ##   MIR: 68
394 ##   Flow cover: 278
395 ##
396 ## Explored 1 nodes (27401 simplex iterations) in 115.52 seconds (209.84 work units)
397 ## Thread count was 24 (of 24 available processors)
398 ##
399 ## Solution count 10: 8.61511e+08 6.83414e+08 6.79425e+08 ... 6.61868e+08
400 ##
401 ## Optimal solution found (tolerance 5.00e-02)
402 ## Best objective 8.615113892192e+08, best bound 9.022317287105e+08, gap 4.7266%
403 ## [solve_lp_with_gurobi] Status: OPTIMAL | ObjVal: 861511389.2192 | Runtime: 115.9 s
404 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...
405 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)
406 ##   C1: Path establishment
407 ##   C2: Path connectivity
408 ##   C3: Biomass yield
409 ##   C4: Inventory balance
410 ##   C5: Storage capacity
411 ##   C6: Processing capacity
412 ##   C7: Demand satisfaction
413 ##   Total constraints: 27183, nnz: 977322

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414 ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
415 ## Set parameter Username
416 ## Set parameter LicenseID to value 2805714
417 ## Set parameter TimeLimit to value 300
418 ## Set parameter MIPGap to value 0.05
419 ## Set parameter MIPFocus to value 1
420 ## Set parameter Cuts to value 3
421 ## Academic license - for non-commercial use only - expires 2027-04-10
422 ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
423 ##
424 ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
425 ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
426 ##
427 ## Non-default parameters:
428 ## TimeLimit 300
429 ## MIPGap 0.05
430 ## MIPFocus 1
431 ## Cuts 3
432 ## LogToConsole 0
433 ##
434 ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
435 ## Model fingerprint: 0xf00d3f02
436 ## Variable types: 246642 continuous, 66265 integer (66265 binary)
437 ## Coefficient statistics:
438 ## Matrix range [1e+00, 6e+05]
439 ## Objective range [6e-02, 4e+07]
440 ## Bounds range [1e+00, 5e+07]
441 ## RHS range [1e+00, 7e+07]
442 ## Found heuristic solution: objective -0.0000000
443 ## Presolve removed 3524 rows and 47233 columns
444 ## Presolve time: 2.71s

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445 ## Presolved: 23659 rows, 265674 columns, 844553 nonzeros
446 ## Variable types: 204919 continuous, 60755 integer (60755 binary)
447 ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
448 ## Showing barrier log only...
449 ##
450 ## Root barrier log...
451 ##
452 ## Ordering time: 0.40s
453 ##
454 ## Barrier statistics:
455 ## AA' NZ      : 5.958e+05
456 ## Factor NZ   : 7.925e+06 (roughly 200 MB of memory)
457 ## Factor Ops  : 8.035e+09 (less than 1 second per iteration)
458 ## Threads    : 10
459 ##
460 ##              Objective              Residual
461 ## Iter      Primal      Dual      Primal      Dual      Compl      Time
462 ##   0  -2.23057568e+17  1.82489736e+16  2.46e+09  4.79e+02  6.70e+12  4s
463 ##   1  -9.54537205e+16  1.62224370e+16  1.05e+09  4.46e+05  2.93e+12  4s
464 ##   2  -1.29842867e+16  1.25385631e+16  1.38e+08  2.59e+04  4.04e+11  4s
465 ##   3  -2.65761621e+15  7.18691221e+15  2.69e+07  2.01e+03  8.52e+10  4s
466 ##   4  -4.57688772e+14  3.67310852e+15  4.38e+06  7.01e+02  1.79e+10  4s
467 ##   5  -7.31493154e+13  1.31438728e+15  5.55e+05  7.45e-09  4.01e+09  4s
468 ##   6  -1.22740047e+13  8.84638213e+14  3.28e+04  7.45e-09  1.86e+09  4s
469 ##   7  -5.10370023e+12  7.51863884e+14  1.33e+04  5.59e-09  1.46e+09  4s
470 ##   8  -1.92646094e+12  2.42271453e+14  4.73e+03  9.31e-09  4.64e+08  5s
471 ##   9  -7.57128983e+11  6.93189232e+13  1.61e+03  7.45e-09  1.32e+08  5s
472 ##  10  -4.40087356e+11  6.31088677e+12  7.92e+02  1.12e-08  1.32e+07  5s
473 ##  11  -3.52369672e+11  4.67179036e+12  6.05e+02  1.58e-08  9.89e+06  5s
474 ##  12  -2.29695696e+11  2.26897320e+12  3.54e+02  1.30e-08  5.02e+06  5s
475 ##  13  -1.42458226e+11  1.47934850e+12  1.84e+02  1.58e-08  3.21e+06  5s

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476	##	14	-9.14886217e+10	5.65735026e+11	8.80e+01	1.84e-08	1.38e+06	5s
477	##	15	-7.62432482e+10	5.19069809e+11	6.78e+01	2.61e-08	1.23e+06	5s
478	##	16	-5.55401473e+10	4.37384477e+11	4.87e+01	3.28e-08	1.01e+06	5s
479	##	17	-4.51709760e+10	2.84864672e+11	3.83e+01	2.42e-08	6.88e+05	5s
480	##	18	-3.02282752e+10	2.68945620e+11	2.42e+01	3.54e-08	6.00e+05	5s
481	##	19	-2.60665063e+10	2.37025319e+11	2.07e+01	3.54e-08	5.24e+05	6s
482	##	20	-1.68150628e+10	1.83606426e+11	1.31e+01	3.35e-08	3.91e+05	6s
483	##	21	-1.18554348e+10	1.21585747e+11	9.05e+00	2.07e-08	2.58e+05	6s
484	##	22	-8.42543774e+09	1.03968787e+11	6.33e+00	2.05e-08	2.14e+05	6s
485	##	23	-6.59807549e+09	6.83038026e+10	4.87e+00	1.79e-08	1.41e+05	6s
486	##	24	-4.19008413e+09	4.30437589e+10	2.88e+00	1.47e-08	8.76e+04	6s
487	##	25	-3.18213249e+09	3.53444071e+10	2.07e+00	1.21e-08	7.09e+04	6s
488	##	26	-2.19963270e+09	2.20571789e+10	1.30e+00	9.31e-09	4.42e+04	6s
489	##	27	-1.88684745e+09	1.46687328e+10	1.10e+00	8.38e-09	3.00e+04	6s
490	##	28	-1.21891141e+09	1.16449133e+10	7.01e-01	6.52e-09	2.33e+04	7s
491	##	29	-9.30773531e+08	1.00775362e+10	5.44e-01	5.59e-09	1.99e+04	7s
492	##	30	-4.91445770e+08	9.75126224e+09	3.16e-01	5.59e-09	1.86e+04	7s
493	##	31	-1.81212239e+08	7.15043723e+09	1.79e-01	5.59e-09	1.33e+04	7s
494	##	32	7.03755913e+06	5.21832806e+09	1.07e-01	5.59e-09	9.46e+03	7s
495	##	33	1.31053601e+08	2.67934045e+09	6.56e-02	4.66e-09	4.62e+03	7s
496	##	34	2.07036668e+08	1.93337611e+09	4.59e-02	6.05e-09	3.13e+03	7s
497	##	35	2.94900281e+08	1.61601116e+09	2.55e-02	5.59e-09	2.40e+03	7s
498	##	36	3.08524349e+08	1.41884943e+09	2.27e-02	5.47e-09	2.01e+03	7s
499	##	37	3.32688444e+08	1.20138135e+09	1.82e-02	6.17e-09	1.58e+03	7s
500	##	38	3.45491093e+08	1.03453163e+09	1.60e-02	6.98e-09	1.25e+03	8s
501	##	39	3.58388921e+08	9.14740422e+08	1.40e-02	6.29e-09	1.01e+03	8s
502	##	40	4.10110403e+08	8.04857946e+08	6.79e-03	6.75e-09	7.16e+02	8s
503	##	41	4.37561514e+08	7.32270307e+08	4.03e-03	6.26e-09	5.35e+02	8s
504	##	42	4.55362734e+08	6.58231179e+08	2.54e-03	5.52e-09	3.68e+02	8s
505	##	43	4.61660582e+08	6.25620698e+08	2.13e-03	5.70e-09	2.98e+02	8s
506	##	44	4.67362406e+08	5.94396436e+08	1.78e-03	5.88e-09	2.31e+02	8s

```

507 ## 45 4.76599599e+08 5.59094180e+08 1.24e-03 5.59e-09 1.50e+02 8s
508 ## 46 4.84035809e+08 5.42764872e+08 8.58e-04 5.08e-09 1.07e+02 8s
509 ## 47 4.90574314e+08 5.26451208e+08 5.37e-04 4.86e-09 6.51e+01 9s
510 ## 48 4.94763067e+08 5.17722947e+08 3.45e-04 5.22e-09 4.17e+01 9s
511 ## 49 4.97700706e+08 5.13556087e+08 2.18e-04 5.47e-09 2.88e+01 9s
512 ## 50 4.98892296e+08 5.11169316e+08 1.65e-04 6.75e-09 2.23e+01 9s
513 ## 51 4.99721102e+08 5.08838610e+08 1.29e-04 5.94e-09 1.66e+01 9s
514 ## 52 5.00711765e+08 5.07341817e+08 8.70e-05 5.59e-09 1.20e+01 9s
515 ## 53 5.01576325e+08 5.05340270e+08 5.16e-05 5.33e-09 6.83e+00 9s
516 ## 54 5.02432959e+08 5.04227744e+08 2.00e-05 4.35e-09 3.26e+00 9s
517 ## 55 5.02610752e+08 5.03571007e+08 1.31e-05 5.05e-09 1.74e+00 9s
518 ## 56 5.02774284e+08 5.03348173e+08 7.37e-06 4.80e-09 1.04e+00 9s
519 ## 57 5.02893261e+08 5.03204301e+08 3.16e-06 4.48e-09 5.65e-01 10s
520 ## 58 5.02946610e+08 5.03097919e+08 1.52e-06 4.40e-09 2.75e-01 10s
521 ## 59 5.02975150e+08 5.03054063e+08 7.76e-07 5.09e-09 1.43e-01 10s
522 ## 60 5.02989679e+08 5.03020706e+08 4.25e-07 5.09e-09 5.64e-02 10s
523 ## 61 5.02994373e+08 5.03011718e+08 3.07e-07 3.73e-09 3.15e-02 10s
524 ## 62 5.02998416e+08 5.03005947e+08 1.60e-07 4.19e-09 1.37e-02 10s
525 ## 63 5.02999153e+08 5.03002590e+08 1.36e-07 4.23e-09 6.24e-03 10s
526 ## 64 5.03000453e+08 5.03001630e+08 6.85e-07 5.84e-09 2.14e-03 10s
527 ## 65 5.03000803e+08 5.03000918e+08 7.65e-07 4.69e-09 2.10e-04 10s
528 ## 66 5.03000835e+08 5.03000836e+08 6.05e-09 4.15e-09 5.58e-07 10s
529 ## 67 5.03000835e+08 5.03000835e+08 3.86e-09 5.05e-09 1.96e-11 10s
530 ##
531 ## Barrier solved model in 67 iterations and 10.40 seconds (14.02 work units)
532 ## Optimal objective 5.03000835e+08
533 ##
534 ##
535 ## Root crossover log...
536 ##
537 ## 20136 DPushes remaining with DInf 0.0000000e+00 10s

```

```

538 ##          0 DPushes remaining with DInf 0.0000000e+00          11s
539 ##
540 ##          326 PPushes remaining with PInf 5.6635499e-05          11s
541 ##          0 PPushes remaining with PInf 0.0000000e+00          11s
542 ##
543 ## Push phase complete: Pinf 0.0000000e+00, Dinf 7.8869834e+05    11s
544 ##
545 ##
546 ## Root simplex log...
547 ##
548 ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
549 ##      20103      5.0300084e+08      0.000000e+00      7.886983e+05      11s
550 ##      20684      5.0300084e+08      0.000000e+00      0.000000e+00      11s
551 ## Concurrent spin time: 2.33s (can be avoided by choosing Method=3)
552 ##
553 ## Solved with barrier
554 ##
555 ## Root relaxation: objective 5.030008e+08, 20684 iterations, 10.89 seconds (10.34 work units)
556 ## Total elapsed time = 20.83s (DegenMoves)
557 ##
558 ##      Nodes      |      Current Node      |      Objective Bounds      |      Work
559 ##      Expl Unexpl |      Obj  Depth IntInf |      Incumbent      BestBd      Gap |      It/Node Time
560 ##
561 ##          0          0 5.0300e+08          0 944      -0.00000 5.0300e+08          -          -      23s
562 ## H          0          0                          2.622174e+08 5.0300e+08 91.8%          -      23s
563 ## H          0          0                          2.691127e+08 5.0300e+08 86.9%          -      24s
564 ## H          0          0                          2.706323e+08 5.0300e+08 85.9%          -      24s
565 ## H          0          0                          2.811011e+08 5.0300e+08 78.9%          -      24s
566 ## H          0          0                          2.830642e+08 5.0300e+08 77.7%          -      24s
567 ## H          0          0                          2.844875e+08 5.0300e+08 76.8%          -      25s
568 ## H          0          0                          2.874458e+08 5.0300e+08 75.0%          -      25s

```

569	## H	0	0			2.909978e+08	5.0300e+08	72.9%	-	25s
570	## H	0	0			2.911169e+08	5.0300e+08	72.8%	-	25s
571	## H	0	0			2.934585e+08	5.0300e+08	71.4%	-	25s
572	## H	0	0			2.959723e+08	5.0300e+08	69.9%	-	25s
573	## H	0	0			2.960096e+08	5.0300e+08	69.9%	-	25s
574	## H	0	0			2.961586e+08	5.0300e+08	69.8%	-	25s
575	## H	0	0			2.980525e+08	5.0300e+08	68.8%	-	25s
576	## H	0	0			3.001546e+08	5.0300e+08	67.6%	-	25s
577	## H	0	0			4.233591e+08	5.0300e+08	18.8%	-	31s
578	##	0	0	5.0276e+08	0 989	4.2336e+08	5.0276e+08	18.8%	-	54s
579	##	0	0	5.0272e+08	0 929	4.2336e+08	5.0272e+08	18.7%	-	70s
580	##	0	0	5.0272e+08	0 937	4.2336e+08	5.0272e+08	18.7%	-	71s
581	##	0	0	5.0272e+08	0 929	4.2336e+08	5.0272e+08	18.7%	-	71s
582	##	0	0	5.0238e+08	0 1134	4.2336e+08	5.0238e+08	18.7%	-	118s
583	## H	0	0			4.462810e+08	5.0238e+08	12.6%	-	134s
584	##	0	0	5.0228e+08	0 1246	4.4628e+08	5.0228e+08	12.5%	-	151s
585	##	0	0	5.0226e+08	0 1260	4.4628e+08	5.0226e+08	12.5%	-	160s
586	##	0	0	5.0225e+08	0 1291	4.4628e+08	5.0225e+08	12.5%	-	164s
587	##	0	0	5.0225e+08	0 1275	4.4628e+08	5.0225e+08	12.5%	-	166s
588	##	0	0	5.0225e+08	0 1276	4.4628e+08	5.0225e+08	12.5%	-	166s
589	##	0	0	5.0198e+08	0 1370	4.4628e+08	5.0198e+08	12.5%	-	233s
590	##	0	0	5.0185e+08	0 1299	4.4628e+08	5.0185e+08	12.5%	-	282s
591	##	0	0	5.0182e+08	0 1380	4.4628e+08	5.0182e+08	12.4%	-	294s
592	##	0	0	5.0182e+08	0 1367	4.4628e+08	5.0182e+08	12.4%	-	298s
593	##	0	0	-	0	4.4628e+08	5.0182e+08	12.4%	-	300s
594	##									
595	##	Cutting planes:								
596	##	Gomory: 1								
597	##	MIR: 201								
598	##	Flow cover: 865								
599	##									

```

600 ## Explored 1 nodes (75103 simplex iterations) in 300.04 seconds (521.02 work units)
601 ## Thread count was 24 (of 24 available processors)
602 ##
603 ## Solution count 10: 4.46281e+08 4.46281e+08 4.23359e+08 ... 2.91117e+08
604 ##
605 ## Time limit reached
606 ## Warning: max constraint violation (1.2186e-06) exceeds tolerance
607 ## Warning: max bound violation (1.7222e-06) exceeds tolerance
608 ## Best objective 4.462810080392e+08, best bound 5.018176758359e+08, gap 12.4443%
609 ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 446281008.0392 | Runtime: 302.1 s

610 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...
611 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)
612 ##   C1: Path establishment
613 ##   C2: Path connectivity
614 ##   C3: Biomass yield
615 ##   C4: Inventory balance
616 ##   C5: Storage capacity
617 ##   C6: Processing capacity
618 ##   C7: Demand satisfaction
619 ##   Total constraints: 27183, nnz: 977322
620 ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
621 ## Set parameter Username
622 ## Set parameter LicenseID to value 2805714
623 ## Set parameter TimeLimit to value 300
624 ## Set parameter MIPGap to value 0.05
625 ## Set parameter MIPFocus to value 1
626 ## Set parameter Cuts to value 3
627 ## Academic license - for non-commercial use only - expires 2027-04-10
628 ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
629 ##
630 ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]

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631 ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
632 ##
633 ## Non-default parameters:
634 ## TimeLimit 300
635 ## MIPGap 0.05
636 ## MIPFocus 1
637 ## Cuts 3
638 ## LogToConsole 0
639 ##
640 ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
641 ## Model fingerprint: 0x7d700b6a
642 ## Variable types: 246642 continuous, 66265 integer (66265 binary)
643 ## Coefficient statistics:
644 ## Matrix range [1e+00, 6e+05]
645 ## Objective range [3e-02, 2e+07]
646 ## Bounds range [1e+00, 5e+07]
647 ## RHS range [1e+00, 7e+07]
648 ## Found heuristic solution: objective -0.0000000
649 ## Presolve removed 3524 rows and 47233 columns
650 ## Presolve time: 2.77s
651 ## Presolved: 23659 rows, 265674 columns, 844553 nonzeros
652 ## Variable types: 204919 continuous, 60755 integer (60755 binary)
653 ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
654 ## Showing barrier log only...
655 ##
656 ## Root barrier log...
657 ##
658 ## Ordering time: 0.42s
659 ##
660 ## Barrier statistics:
661 ## AA' NZ : 5.958e+05

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662 ## Factor NZ : 7.925e+06 (roughly 200 MB of memory)
663 ## Factor Ops : 8.035e+09 (less than 1 second per iteration)
664 ## Threads : 10
665 ##
666 ##

```

		Objective		Residual			
##	Iter	Primal	Dual	Primal	Dual	Compl	Time
##	0	-1.12900047e+17	1.25311858e+16	2.46e+09	4.78e+02	4.52e+12	4s
##	1	-4.82833078e+16	1.11314958e+16	1.05e+09	3.01e+05	1.97e+12	4s
##	2	-6.60383891e+15	8.60161718e+15	1.38e+08	1.76e+04	2.71e+11	4s
##	3	-1.34136382e+15	4.92294307e+15	2.59e+07	1.39e+03	5.57e+10	4s
##	4	-2.47126027e+14	2.47286883e+15	4.46e+06	4.87e+02	1.22e+10	4s
##	5	-3.93925218e+13	8.79028526e+14	5.55e+05	6.98e-09	2.69e+09	4s
##	6	-6.91439018e+12	5.92582636e+14	3.39e+04	3.73e-09	1.25e+09	4s
##	7	-2.09955927e+12	4.07950289e+14	9.36e+03	4.19e-09	7.85e+08	4s
##	8	-9.48083743e+11	1.09471225e+14	3.52e+03	4.66e-09	2.09e+08	5s
##	9	-5.09176877e+11	1.76349045e+13	1.33e+03	5.59e-09	3.47e+07	5s
##	10	-3.79930554e+11	3.21631595e+12	7.36e+02	6.52e-09	7.16e+06	5s
##	11	-3.23204634e+11	1.70494194e+12	5.37e+02	7.68e-09	4.15e+06	5s
##	12	-2.51438874e+11	1.04879916e+12	3.22e+02	8.38e-09	2.66e+06	5s
##	13	-1.87475976e+11	9.75300235e+11	1.90e+02	1.21e-08	2.29e+06	5s
##	14	-1.09362782e+11	4.52991201e+11	1.05e+02	8.85e-09	1.13e+06	5s
##	15	-8.20442332e+10	4.39665708e+11	7.36e+01	1.26e-08	1.02e+06	5s
##	16	-6.25897278e+10	4.01120861e+11	5.98e+01	1.46e-03	9.02e+05	5s
##	17	-3.21696076e+10	2.66769823e+11	3.18e+01	9.31e-09	5.76e+05	5s
##	18	-1.86445752e+10	1.73530721e+11	1.92e+01	9.08e-09	3.68e+05	6s
##	19	-9.74192581e+09	1.03589451e+11	1.06e+01	8.15e-09	2.15e+05	6s
##	20	-7.88791868e+09	9.89791250e+10	8.86e+00	9.08e-09	2.02e+05	6s
##	21	-7.42776204e+09	6.79239418e+10	8.41e+00	8.85e-09	1.43e+05	6s
##	22	-6.03643919e+09	5.40970351e+10	6.99e+00	7.22e-09	1.13e+05	6s
##	23	-4.81043161e+09	4.35337518e+10	5.64e+00	6.98e-09	9.08e+04	6s
##	24	-4.24569506e+09	3.96372445e+10	5.02e+00	6.98e-09	8.22e+04	6s

693	##	25	-2.90469036e+09	2.76845368e+10	3.39e+00	5.59e-09	5.67e+04	6s
694	##	26	-2.23904339e+09	1.99815314e+10	2.53e+00	5.12e-09	4.09e+04	6s
695	##	27	-1.93386960e+09	1.55405055e+10	2.16e+00	3.73e-09	3.20e+04	7s
696	##	28	-1.18341156e+09	1.08250051e+10	1.10e+00	3.26e-09	2.18e+04	7s
697	##	29	-7.52301398e+08	7.56322279e+09	7.14e-01	2.79e-09	1.51e+04	7s
698	##	30	-5.30732157e+08	5.69694981e+09	5.24e-01	2.79e-09	1.13e+04	7s
699	##	31	-4.48527313e+08	4.94094943e+09	4.64e-01	2.79e-09	9.76e+03	7s
700	##	32	-3.88377790e+08	4.28996286e+09	4.21e-01	3.26e-09	8.47e+03	7s
701	##	33	-2.23191026e+08	3.69677529e+09	3.04e-01	2.79e-09	7.10e+03	7s
702	##	34	-1.10154322e+08	2.72079983e+09	2.33e-01	2.33e-09	5.12e+03	7s
703	##	35	-2.32582289e+07	2.53265609e+09	1.82e-01	3.49e-09	4.63e+03	7s
704	##	36	8.89093737e+07	1.85191198e+09	1.23e-01	2.79e-09	3.19e+03	8s
705	##	37	1.75229506e+08	1.35148388e+09	8.34e-02	2.56e-09	2.13e+03	8s
706	##	38	2.38822105e+08	1.15874221e+09	6.04e-02	1.86e-09	1.67e+03	8s
707	##	39	2.80153338e+08	9.56923756e+08	4.71e-02	2.56e-09	1.23e+03	8s
708	##	40	3.07468501e+08	8.20580212e+08	3.93e-02	2.50e-09	9.28e+02	8s
709	##	41	3.46294338e+08	7.24928330e+08	2.85e-02	2.02e-09	6.85e+02	8s
710	##	42	3.80210114e+08	6.70203391e+08	1.98e-02	2.33e-09	5.25e+02	8s
711	##	43	4.02603755e+08	6.07429142e+08	1.46e-02	2.15e-09	3.71e+02	8s
712	##	44	4.25947959e+08	5.69208453e+08	9.55e-03	2.35e-09	2.59e+02	9s
713	##	45	4.38017176e+08	5.33482583e+08	7.13e-03	4.23e-09	1.73e+02	9s
714	##	46	4.48549975e+08	5.13936632e+08	5.13e-03	4.22e-09	1.18e+02	9s
715	##	47	4.57402233e+08	4.98751341e+08	3.51e-03	2.30e-09	7.47e+01	9s
716	##	48	4.65776692e+08	4.91438889e+08	2.01e-03	2.93e-09	4.64e+01	9s
717	##	49	4.67148247e+08	4.88886630e+08	1.77e-03	2.68e-09	3.93e+01	9s
718	##	50	4.69235934e+08	4.85277416e+08	1.42e-03	3.09e-09	2.90e+01	9s
719	##	51	4.70900131e+08	4.83914044e+08	1.15e-03	3.14e-09	2.35e+01	9s
720	##	52	4.73875709e+08	4.82887627e+08	6.50e-04	2.42e-09	1.63e+01	9s
721	##	53	4.75068041e+08	4.81137378e+08	4.50e-04	2.85e-09	1.10e+01	9s
722	##	54	4.76516861e+08	4.79281555e+08	2.11e-04	2.22e-09	5.00e+00	10s
723	##	55	4.77291663e+08	4.78540426e+08	8.77e-05	2.06e-09	2.26e+00	10s


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724 ## 56 4.77630873e+08 4.78133980e+08 3.51e-05 2.07e-09 9.11e-01 10s
725 ## 57 4.77744779e+08 4.78007533e+08 1.85e-05 2.70e-09 4.76e-01 10s
726 ## 58 4.77786637e+08 4.77949078e+08 1.26e-05 2.20e-09 2.94e-01 10s
727 ## 59 4.77829488e+08 4.77914452e+08 6.65e-06 2.89e-09 1.54e-01 10s
728 ## 60 4.77860978e+08 4.77888740e+08 2.33e-06 2.33e-09 5.02e-02 10s
729 ## 61 4.77872224e+08 4.77884605e+08 1.34e-06 3.23e-09 2.24e-02 10s
730 ## 62 4.77877178e+08 4.77882020e+08 6.26e-07 2.50e-09 8.78e-03 10s
731 ## 63 4.77878236e+08 4.77880111e+08 6.14e-07 2.33e-09 3.39e-03 10s
732 ## 64 4.77879375e+08 4.77879807e+08 5.80e-08 2.15e-09 7.84e-04 10s
733 ## 65 4.77879514e+08 4.77879561e+08 1.11e-08 2.68e-09 8.44e-05 11s
734 ## 66 4.77879544e+08 4.77879549e+08 1.58e-09 2.33e-09 9.73e-06 11s
735 ## 67 4.77879547e+08 4.77879549e+08 1.24e-08 3.58e-09 3.00e-06 11s
736 ## 68 4.77879549e+08 4.77879549e+08 8.60e-09 2.79e-09 3.23e-09 11s
737 ##
738 ## Barrier solved model in 68 iterations and 11.05 seconds (14.70 work units)
739 ## Optimal objective 4.77879549e+08
740 ##
741 ##
742 ## Root crossover log...
743 ##
744 ## 19735 DPushes remaining with DInf 0.0000000e+00 11s
745 ## 0 DPushes remaining with DInf 0.0000000e+00 12s
746 ##
747 ## 371 PPushes remaining with PInf 4.7517212e-04 12s
748 ## 0 PPushes remaining with PInf 0.0000000e+00 12s
749 ##
750 ## Push phase complete: Pinf 0.0000000e+00, Dinf 9.6642686e+04 12s
751 ##
752 ##
753 ## Root simplex log...
754 ##

```

```

755 ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
756 ##      19206      4.7787926e+08      0.000000e+00      9.664269e+04      12s
757 ##      19588      4.7787955e+08      0.000000e+00      0.000000e+00      12s
758 ## Concurrent spin time: 2.49s (can be avoided by choosing Method=3)
759 ##
760 ## Solved with barrier
761 ##
762 ## Root relaxation: objective 4.778795e+08, 19588 iterations, 11.46 seconds (10.89 work units)
763 ## Total elapsed time = 23.30s (DegenMoves)
764 ## Total elapsed time = 26.43s (DegenMoves)
765 ##
766 ##      Nodes      |      Current Node      |      Objective Bounds      |      Work
767 ## Expl Unexpl | Obj Depth IntInf | Incumbent      BestBd      Gap | It/Node Time
768 ##
769 ##      0      0 4.7788e+08      0 1101      -0.00000 4.7788e+08      -      -      26s
770 ## H      0      0      1.940584e+08 4.7788e+08      146%      -      27s
771 ## H      0      0      2.072988e+08 4.7788e+08      131%      -      27s
772 ## H      0      0      2.086620e+08 4.7788e+08      129%      -      27s
773 ## H      0      0      2.169433e+08 4.7788e+08      120%      -      27s
774 ## H      0      0      2.260177e+08 4.7788e+08      111%      -      28s
775 ## H      0      0      2.351394e+08 4.7788e+08      103%      -      28s
776 ## H      0      0      2.620500e+08 4.7788e+08      82.4%      -      28s
777 ## H      0      0      2.691231e+08 4.7788e+08      77.6%      -      28s
778 ## H      0      0      2.714720e+08 4.7788e+08      76.0%      -      29s
779 ## H      0      0      2.726302e+08 4.7788e+08      75.3%      -      29s
780 ## H      0      0      2.748115e+08 4.7788e+08      73.9%      -      29s
781 ## H      0      0      2.809229e+08 4.7788e+08      70.1%      -      29s
782 ## H      0      0      2.822366e+08 4.7788e+08      69.3%      -      29s
783 ## H      0      0      2.854823e+08 4.7788e+08      67.4%      -      29s
784 ## H      0      0      2.870237e+08 4.7788e+08      66.5%      -      29s
785 ## H      0      0      2.873568e+08 4.7788e+08      66.3%      -      29s

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786	##	H	0	0		2.884404e+08	4.7788e+08	65.7%	-	29s		
787	##	H	0	0		2.895105e+08	4.7788e+08	65.1%	-	30s		
788	##		0	0	4.7675e+08	0	1178	2.8951e+08	4.7675e+08	64.7%	-	78s
789	##	H	0	0		2.983581e+08	4.7675e+08	59.8%	-	83s		
790	##	H	0	0		3.016960e+08	4.7675e+08	58.0%	-	83s		
791	##	H	0	0		3.022132e+08	4.7675e+08	57.8%	-	83s		
792	##	H	0	0		3.094243e+08	4.7675e+08	54.1%	-	84s		
793	##	H	0	0		3.098882e+08	4.7675e+08	53.8%	-	84s		
794	##	H	0	0		3.121055e+08	4.7675e+08	52.8%	-	84s		
795	##	H	0	0		3.141556e+08	4.7675e+08	51.8%	-	84s		
796	##	H	0	0		3.146035e+08	4.7675e+08	51.5%	-	84s		
797	##	H	0	0		3.148318e+08	4.7675e+08	51.4%	-	84s		
798	##	H	0	0		3.158197e+08	4.7675e+08	51.0%	-	84s		
799	##	H	0	0		3.171487e+08	4.7675e+08	50.3%	-	84s		
800	##	H	0	0		3.174035e+08	4.7675e+08	50.2%	-	84s		
801	##	H	0	0		3.179388e+08	4.7675e+08	49.9%	-	84s		
802	##	H	0	0		3.187178e+08	4.7675e+08	49.6%	-	84s		
803	##	H	0	0		3.208858e+08	4.7675e+08	48.6%	-	85s		
804	##	H	0	0		3.209804e+08	4.7675e+08	48.5%	-	85s		
805	##	H	0	0		3.210072e+08	4.7675e+08	48.5%	-	85s		
806	##	H	0	0		3.222787e+08	4.7675e+08	47.9%	-	85s		
807	##	H	0	0		3.446183e+08	4.7675e+08	38.3%	-	93s		
808	##	H	0	0		3.492607e+08	4.7675e+08	36.5%	-	96s		
809	##		0	0	4.7655e+08	0	1197	3.4926e+08	4.7655e+08	36.4%	-	110s
810	##		0	0	4.7653e+08	0	1155	3.4926e+08	4.7653e+08	36.4%	-	113s
811	##		0	0	4.7653e+08	0	1172	3.4926e+08	4.7653e+08	36.4%	-	114s
812	##		0	0	4.7653e+08	0	1161	3.4926e+08	4.7653e+08	36.4%	-	114s
813	##		0	0	4.7527e+08	0	1296	3.4926e+08	4.7527e+08	36.1%	-	234s
814	##	H	0	0		3.531835e+08	4.7527e+08	34.6%	-	267s		
815	##	H	0	0		3.532405e+08	4.7527e+08	34.5%	-	292s		
816	##		0	0	-	0	3.5324e+08	4.7527e+08	34.5%	-	300s	

```

817 ##
818 ## Cutting planes:
819 ##   Gomory: 4
820 ##   MIR: 177
821 ##   Flow cover: 794
822 ##
823 ## Explored 1 nodes (47141 simplex iterations) in 300.04 seconds (345.23 work units)
824 ## Thread count was 24 (of 24 available processors)
825 ##
826 ## Solution count 10: 3.53241e+08 3.53241e+08 3.53183e+08 ... 3.18718e+08
827 ##
828 ## Time limit reached
829 ## Warning: max constraint violation (1.5195e-06) exceeds tolerance
830 ## Warning: max bound violation (4.2085e-06) exceeds tolerance
831 ## Best objective 3.532405223800e+08, best bound 4.752655177399e+08, gap 34.5444%
832 ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 353240522.3800 | Runtime: 300.6 s

833 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...
834 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)
835 ##   C1: Path establishment
836 ##   C2: Path connectivity
837 ##   C3: Biomass yield
838 ##   C4: Inventory balance
839 ##   C5: Storage capacity
840 ##   C6: Processing capacity
841 ##   C7: Demand satisfaction
842 ##   Total constraints: 27183, nnz: 977322
843 ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
844 ## Set parameter Username
845 ## Set parameter LicenseID to value 2805714
846 ## Set parameter TimeLimit to value 300
847 ## Set parameter MIPGap to value 0.05

```

```

848 ## Set parameter MIPFocus to value 1
849 ## Set parameter Cuts to value 3
850 ## Academic license - for non-commercial use only - expires 2027-04-10
851 ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
852 ##
853 ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
854 ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
855 ##
856 ## Non-default parameters:
857 ## TimeLimit 300
858 ## MIPGap 0.05
859 ## MIPFocus 1
860 ## Cuts 3
861 ## LogToConsole 0
862 ##
863 ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
864 ## Model fingerprint: 0x693b45b3
865 ## Variable types: 246642 continuous, 66265 integer (66265 binary)
866 ## Coefficient statistics:
867 ##   Matrix range      [1e+00, 6e+05]
868 ##   Objective range   [3e-02, 4e+07]
869 ##   Bounds range      [1e+00, 5e+07]
870 ##   RHS range         [1e+00, 7e+07]
871 ## Found heuristic solution: objective -0.0000000
872 ## Presolve removed 3524 rows and 47233 columns
873 ## Presolve time: 4.57s
874 ## Presolved: 23659 rows, 265674 columns, 844553 nonzeros
875 ## Variable types: 204919 continuous, 60755 integer (60755 binary)
876 ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
877 ## Showing barrier log only...
878 ##

```

```

879 ## Root barrier log...
880 ##
881 ## Ordering time: 0.61s
882 ##
883 ## Barrier statistics:
884 ## AA' NZ      : 5.958e+05
885 ## Factor NZ   : 7.925e+06 (roughly 200 MB of memory)
886 ## Factor Ops  : 8.035e+09 (less than 1 second per iteration)
887 ## Threads    : 10
888 ##
889 ##              Objective              Residual
890 ## Iter      Primal      Dual      Primal      Dual      Compl      Time
891 ##   0 -2.23905350e+17  2.14045634e+16  2.46e+09  4.78e+02  7.78e+12   6s
892 ##   1 -9.57846332e+16  1.90200229e+16  1.05e+09  5.18e+05  3.40e+12   6s
893 ##   2 -1.30569295e+16  1.46980285e+16  1.38e+08  3.02e+04  4.68e+11   7s
894 ##   3 -2.63416476e+15  8.41845816e+15  2.62e+07  2.35e+03  9.66e+10   7s
895 ##   4 -4.65423563e+14  4.26362142e+15  4.34e+06  8.24e+02  2.07e+10   7s
896 ##   5 -7.47015614e+13  1.51600101e+15  5.47e+05  6.52e-09  4.61e+09   7s
897 ##   6 -1.28786537e+13  1.02760389e+15  3.32e+04  5.59e-09  2.16e+09   7s
898 ##   7 -3.91827934e+12  6.47764038e+14  9.48e+03  5.59e-09  1.25e+09   7s
899 ##   8 -1.65389432e+12  1.83133899e+14  3.53e+03  8.38e-09  3.50e+08   7s
900 ##   9 -8.22785702e+11  6.28076207e+13  1.37e+03  8.38e-09  1.19e+08   8s
901 ##  10 -5.78025482e+11  6.93487381e+12  7.65e+02  1.49e-08  1.46e+07   8s
902 ##  11 -4.94883215e+11  5.01248539e+12  6.00e+02  1.63e-08  1.08e+07   8s
903 ##  12 -3.73267060e+11  2.42846251e+12  3.72e+02  1.30e-08  5.61e+06   8s
904 ##  13 -2.53327568e+11  1.55586923e+12  1.74e+02  1.58e-08  3.55e+06   8s
905 ##  14 -2.08368764e+11  1.27195590e+12  1.18e+02  1.72e-08  2.88e+06   8s
906 ##  15 -1.77421251e+11  7.58789867e+11  1.02e+02  1.96e-08  1.88e+06   8s
907 ##  16 -1.40070922e+11  7.18673983e+11  7.70e+01  2.14e-08  1.69e+06   9s
908 ##  17 -1.04608562e+11  6.05750401e+11  5.80e+01  2.51e-08  1.39e+06   9s
909 ##  18 -7.07167402e+10  4.13085667e+11  3.89e+01  1.96e-08  9.47e+05   9s

```

910	##	19	-3.81865743e+10	2.80011249e+11	2.11e+01	2.00e-08	6.14e+05	9s
911	##	20	-2.74995791e+10	2.36999430e+11	1.56e+01	2.51e-08	5.06e+05	9s
912	##	21	-1.44253149e+10	1.61505451e+11	8.20e+00	2.56e-08	3.31e+05	9s
913	##	22	-1.25775854e+10	1.26275205e+11	7.22e+00	2.14e-08	2.61e+05	10s
914	##	23	-1.13631418e+10	1.14895837e+11	6.53e+00	2.24e-08	2.37e+05	10s
915	##	24	-7.25725546e+09	7.48194874e+10	3.89e+00	2.00e-08	1.52e+05	10s
916	##	25	-5.60975127e+09	4.66828789e+10	2.90e+00	1.12e-08	9.64e+04	10s
917	##	26	-4.86833667e+09	3.72904890e+10	2.46e+00	1.12e-08	7.74e+04	10s
918	##	27	-3.60404139e+09	2.34433160e+10	1.69e+00	6.98e-09	4.92e+04	10s
919	##	28	-2.82713119e+09	1.95274486e+10	1.27e+00	5.59e-09	4.06e+04	10s
920	##	29	-2.13258091e+09	1.51683783e+10	8.72e-01	5.59e-09	3.14e+04	11s
921	##	30	-1.54804278e+09	1.21545960e+10	5.35e-01	5.59e-09	2.49e+04	11s
922	##	31	-1.27653459e+09	9.16116305e+09	4.13e-01	4.66e-09	1.89e+04	11s
923	##	32	-1.03073425e+09	6.51736671e+09	3.00e-01	4.66e-09	1.37e+04	11s
924	##	33	-7.55029187e+08	5.84437047e+09	1.89e-01	5.59e-09	1.20e+04	11s
925	##	34	-5.38728980e+08	4.39842115e+09	1.11e-01	5.59e-09	8.96e+03	12s
926	##	35	-4.46305373e+08	3.46582202e+09	8.29e-02	5.59e-09	7.10e+03	12s
927	##	36	-3.55271319e+08	2.66010817e+09	6.08e-02	5.12e-09	5.47e+03	12s
928	##	37	-2.49009683e+08	2.38772783e+09	3.88e-02	5.36e-09	4.79e+03	12s
929	##	38	-2.07496739e+08	1.98940934e+09	3.21e-02	6.05e-09	3.99e+03	12s
930	##	39	-1.41335100e+08	1.66192001e+09	2.18e-02	5.12e-09	3.28e+03	12s
931	##	40	-9.51199552e+07	1.29470818e+09	1.62e-02	5.12e-09	2.53e+03	13s
932	##	41	-6.14852243e+07	1.00418601e+09	1.27e-02	4.89e-09	1.94e+03	13s
933	##	42	-3.39357775e+07	8.91609873e+08	1.02e-02	5.59e-09	1.68e+03	13s
934	##	43	4.76901832e+06	7.55915679e+08	6.88e-03	6.05e-09	1.37e+03	13s
935	##	44	2.81711002e+07	6.05564466e+08	5.34e-03	4.66e-09	1.05e+03	13s
936	##	45	5.52334547e+07	4.92468106e+08	3.74e-03	5.12e-09	7.95e+02	13s
937	##	46	7.00231078e+07	4.01863933e+08	2.99e-03	4.66e-09	6.03e+02	14s
938	##	47	9.87734908e+07	3.29567594e+08	1.65e-03	4.25e-09	4.20e+02	14s
939	##	48	1.14746120e+08	2.75610738e+08	1.01e-03	5.70e-09	2.93e+02	14s
940	##	49	1.26938940e+08	2.32795771e+08	5.92e-04	5.33e-09	1.93e+02	14s

941	##	50	1.28376554e+08	2.06852327e+08	5.47e-04	4.31e-09	1.43e+02	14s
942	##	51	1.33802728e+08	1.98875976e+08	3.93e-04	5.59e-09	1.18e+02	14s
943	##	52	1.35059021e+08	1.83119408e+08	3.52e-04	5.36e-09	8.74e+01	15s
944	##	53	1.38455696e+08	1.73777372e+08	2.61e-04	5.01e-09	6.42e+01	15s
945	##	54	1.42863585e+08	1.67439347e+08	1.45e-04	6.52e-09	4.47e+01	15s
946	##	55	1.44808024e+08	1.57977142e+08	9.29e-05	3.66e-09	2.39e+01	15s
947	##	56	1.46024837e+08	1.54913548e+08	6.18e-05	4.66e-09	1.62e+01	15s
948	##	57	1.46317643e+08	1.53642389e+08	5.48e-05	5.03e-09	1.33e+01	15s
949	##	58	1.47159649e+08	1.52497558e+08	3.37e-05	7.96e-09	9.71e+00	16s
950	##	59	1.47748359e+08	1.50673115e+08	1.95e-05	4.31e-09	5.32e+00	16s
951	##	60	1.48089338e+08	1.50029347e+08	1.17e-05	4.42e-09	3.53e+00	16s
952	##	61	1.48151795e+08	1.49945729e+08	1.04e-05	6.05e-09	3.26e+00	16s
953	##	62	1.48323160e+08	1.49678105e+08	6.53e-06	5.31e-09	2.47e+00	16s
954	##	63	1.48421384e+08	1.49396877e+08	4.39e-06	6.01e-09	1.77e+00	16s
955	##	64	1.48520621e+08	1.49063620e+08	2.45e-06	6.13e-09	9.88e-01	16s
956	##	65	1.48613049e+08	1.48802503e+08	6.29e-07	5.12e-09	3.45e-01	17s
957	##	66	1.48636463e+08	1.48673322e+08	3.03e-07	4.77e-09	6.71e-02	17s
958	##	67	1.48639924e+08	1.48655325e+08	2.24e-07	3.99e-09	2.80e-02	17s
959	##	68	1.48646200e+08	1.48649899e+08	2.96e-08	4.19e-09	6.73e-03	17s
960	##	69	1.48647059e+08	1.48647575e+08	1.48e-08	4.66e-09	9.40e-04	17s
961	##	70	1.48647244e+08	1.48647264e+08	5.09e-09	4.66e-09	3.63e-05	17s
962	##	71	1.48647252e+08	1.48647252e+08	7.50e-07	5.01e-09	1.43e-07	18s
963	##	72	1.48647252e+08	1.48647252e+08	2.41e-08	4.26e-09	2.80e-11	18s
964	##							
965	##	Barrier solved model in 72 iterations and 17.77 seconds (15.38 work units)						
966	##	Optimal objective 1.48647252e+08						
967	##							
968	##							
969	##	Root crossover log...						
970	##							
971	##	20524 DPushes remaining with DInf 0.0000000e+00						18s


```

972  ##          0 DPushes remaining with DInf 0.0000000e+00          18s
973  ##
974  ##          191 PPushes remaining with PInf 2.2568460e-05          18s
975  ##          0 PPushes remaining with PInf 0.0000000e+00          18s
976  ##
977  ## Push phase complete: Pinf 0.0000000e+00, Dinf 7.7942660e+05    18s
978  ##
979  ##
980  ## Root simplex log...
981  ##
982  ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
983  ##      20482      1.4864725e+08      0.000000e+00      7.794266e+05      18s
984  ##      26047      1.4864725e+08      0.000000e+00      0.000000e+00      20s
985  ## Concurrent spin time: 6.98s (can be avoided by choosing Method=3)
986  ##
987  ## Solved with barrier
988  ##
989  ## Root relaxation: objective 1.486473e+08, 26047 iterations, 22.25 seconds (12.10 work units)
990  ## Total elapsed time = 31.06s (DegenMoves)
991  ##
992  ##      Nodes      |      Current Node      |      Objective Bounds      |      Work
993  ## Expl Unexpl | Obj Depth IntInf | Incumbent      BestBd      Gap | It/Node Time
994  ##
995  ##      0      0 1.4865e+08      0 565      -0.00000 1.4865e+08      -      -      31s
996  ## H      0      0      282698.89717 1.4865e+08      -      -      31s
997  ## H      0      0      5.241634e+07 1.4865e+08      184%      -      31s
998  ## H      0      0      5.448585e+07 1.4865e+08      173%      -      31s
999  ## H      0      0      5.660082e+07 1.4865e+08      163%      -      31s
1000  ## H      0      0      7.400868e+07 1.4865e+08      101%      -      33s
1001  ## H      0      0      9.423467e+07 1.4865e+08      57.7%      -      34s
1002  ##      0      0 1.4756e+08      0 568 9.4235e+07 1.4756e+08      56.6%      -      54s

```

1003	##	0	0	1.4678e+08	0	525	9.4235e+07	1.4678e+08	55.8%	-	71s
1004	##	0	0	1.4675e+08	0	544	9.4235e+07	1.4675e+08	55.7%	-	75s
1005	##	0	0	1.4674e+08	0	550	9.4235e+07	1.4674e+08	55.7%	-	76s
1006	##	0	0	1.4520e+08	0	661	9.4235e+07	1.4520e+08	54.1%	-	125s
1007	## H	0	0				1.166059e+08	1.4520e+08	24.5%	-	130s
1008	##	0	0	1.4490e+08	0	670	1.1661e+08	1.4490e+08	24.3%	-	141s
1009	##	0	0	1.4483e+08	0	670	1.1661e+08	1.4483e+08	24.2%	-	148s
1010	##	0	0	1.4481e+08	0	667	1.1661e+08	1.4481e+08	24.2%	-	153s
1011	##	0	0	1.4481e+08	0	663	1.1661e+08	1.4481e+08	24.2%	-	154s
1012	##	0	0	1.4481e+08	0	662	1.1661e+08	1.4481e+08	24.2%	-	154s
1013	##	0	0	1.4400e+08	0	810	1.1661e+08	1.4400e+08	23.5%	-	214s
1014	##	0	0	1.4371e+08	0	828	1.1661e+08	1.4371e+08	23.2%	-	238s
1015	##	0	0	1.4355e+08	0	825	1.1661e+08	1.4355e+08	23.1%	-	250s
1016	##	0	0	1.4354e+08	0	803	1.1661e+08	1.4354e+08	23.1%	-	255s
1017	##	0	0	1.4354e+08	0	802	1.1661e+08	1.4354e+08	23.1%	-	258s
1018	##	0	0	1.4354e+08	0	824	1.1661e+08	1.4354e+08	23.1%	-	259s
1019	##	0	0	1.4295e+08	0	872	1.1661e+08	1.4295e+08	22.6%	-	300s
1020	##										
1021	##	Cutting planes:									
1022	##	Gomory: 7									
1023	##	MIR: 299									
1024	##	Flow cover: 1022									
1025	##										
1026	##	Explored 1 nodes (68260 simplex iterations) in 300.12 seconds (264.11 work units)									
1027	##	Thread count was 24 (of 24 available processors)									
1028	##										
1029	##	Solution count 8: 1.16606e+08 1.16606e+08 9.42347e+07 ... -0									
1030	##										
1031	##	Time limit reached									
1032	##	Warning: max constraint violation (1.2333e-06) exceeds tolerance									
1033	##	Warning: max bound violation (1.8741e-06) exceeds tolerance									

```

1034 ## Best objective 1.166059184146e+08, best bound 1.429512984060e+08, gap 22.5935%
1035 ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 116605918.4146 | Runtime: 300.5 s

1036 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...
1037 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)
1038 ##   C1: Path establishment
1039 ##   C2: Path connectivity
1040 ##   C3: Biomass yield
1041 ##   C4: Inventory balance
1042 ##   C5: Storage capacity
1043 ##   C6: Processing capacity
1044 ##   C7: Demand satisfaction
1045 ##   Total constraints: 27183, nnz: 977322
1046 ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
1047 ## Set parameter Username
1048 ## Set parameter LicenseID to value 2805714
1049 ## Set parameter TimeLimit to value 300
1050 ## Set parameter MIPGap to value 0.05
1051 ## Set parameter MIPFocus to value 1
1052 ## Set parameter Cuts to value 3
1053 ## Academic license - for non-commercial use only - expires 2027-04-10
1054 ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
1055 ##
1056 ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
1057 ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
1058 ##
1059 ## Non-default parameters:
1060 ## TimeLimit 300
1061 ## MIPGap 0.05
1062 ## MIPFocus 1
1063 ## Cuts 3
1064 ## LogToConsole 0

```

```

1065 ##
1066 ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
1067 ## Model fingerprint: 0x9d328a45
1068 ## Variable types: 246642 continuous, 66265 integer (66265 binary)
1069 ## Coefficient statistics:
1070 ##   Matrix range      [1e+00, 6e+05]
1071 ##   Objective range   [6e-02, 2e+07]
1072 ##   Bounds range      [1e+00, 5e+07]
1073 ##   RHS range         [1e+00, 7e+07]
1074 ## Found heuristic solution: objective -0.0000000
1075 ## Presolve removed 3524 rows and 47233 columns
1076 ## Presolve time: 4.45s
1077 ## Presolved: 23659 rows, 265674 columns, 844553 nonzeros
1078 ## Variable types: 204919 continuous, 60755 integer (60755 binary)
1079 ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
1080 ## Showing barrier log only...
1081 ##
1082 ## Root barrier log...
1083 ##
1084 ## Ordering time: 0.64s
1085 ##
1086 ## Barrier statistics:
1087 ##   AA' NZ      : 5.958e+05
1088 ##   Factor NZ   : 7.925e+06 (roughly 200 MB of memory)
1089 ##   Factor Ops  : 8.035e+09 (less than 1 second per iteration)
1090 ##   Threads    : 10
1091 ##
1092 ##               Objective                Residual
1093 ## Iter      Primal      Dual      Primal    Dual    Compl    Time
1094 ##    0 -1.11991500e+17  9.70589363e+15  2.46e+09  1.44e+03  3.55e+12  6s
1095 ##    1 -4.79189757e+16  8.62683583e+15  1.05e+09  2.36e+05  1.55e+12  6s

```

1096	##	2	-6.52128097e+15	6.66724844e+15	1.38e+08	1.37e+04	2.14e+11	6s
1097	##	3	-1.32148669e+15	3.82018302e+15	2.65e+07	1.07e+03	4.46e+10	6s
1098	##	4	-2.27495602e+14	1.94508487e+15	4.30e+06	3.73e+02	9.38e+09	7s
1099	##	5	-3.66921862e+13	6.94631097e+14	5.47e+05	5.12e-09	2.11e+09	7s
1100	##	6	-6.22599195e+12	4.70741765e+14	3.28e+04	3.96e-09	9.88e+08	7s
1101	##	7	-1.88790171e+12	3.05763832e+14	9.52e+03	3.49e-09	5.90e+08	7s
1102	##	8	-7.95378852e+11	1.00853868e+14	3.70e+03	3.73e-09	1.92e+08	7s
1103	##	9	-3.69025328e+11	2.88458799e+13	1.44e+03	3.79e-09	5.48e+07	7s
1104	##	10	-2.32348300e+11	3.62151079e+12	7.55e+02	6.52e-09	7.48e+06	7s
1105	##	11	-1.91534518e+11	2.59033534e+12	5.86e+02	6.05e-09	5.44e+06	8s
1106	##	12	-1.35772419e+11	1.25290481e+12	3.64e+02	6.52e-09	2.78e+06	8s
1107	##	13	-8.22479847e+10	7.96824573e+11	1.63e+02	7.92e-09	1.72e+06	8s
1108	##	14	-5.85360234e+10	4.15965825e+11	8.31e+01	1.07e-08	9.55e+05	8s
1109	##	15	-4.43924239e+10	3.87823695e+11	5.95e+01	1.21e-08	8.53e+05	8s
1110	##	16	-3.50665249e+10	3.05711913e+11	4.69e+01	2.72e-02	6.74e+05	8s
1111	##	17	-3.02462824e+10	2.67697436e+11	4.02e+01	2.11e-02	5.89e+05	8s
1112	##	18	-2.02776754e+10	2.48000565e+11	2.73e+01	2.07e-02	5.20e+05	9s
1113	##	19	-9.64104905e+09	1.44738110e+11	1.52e+01	1.12e-02	2.97e+05	9s
1114	##	20	-5.12507910e+09	8.68193354e+10	1.03e+01	6.27e-03	1.76e+05	9s
1115	##	21	-1.38298106e+08	6.23997038e+10	4.44e+00	3.61e-03	1.17e+05	9s
1116	##	22	1.24312844e+09	4.58669121e+10	2.83e+00	2.28e-03	8.27e+04	9s
1117	##	23	1.98871431e+09	3.28609901e+10	1.85e+00	1.42e-03	5.68e+04	9s
1118	##	24	2.31217385e+09	2.20697011e+10	1.45e+00	7.19e-04	3.62e+04	10s
1119	##	25	2.51705405e+09	1.77950518e+10	1.23e+00	4.77e-04	2.79e+04	10s
1120	##	26	2.77458503e+09	1.43821338e+10	9.60e-01	2.45e-04	2.11e+04	10s
1121	##	27	3.30246867e+09	1.15684439e+10	3.36e-01	1.01e-04	1.50e+04	10s
1122	##	28	3.45127504e+09	1.00420651e+10	2.27e-01	8.41e-05	1.20e+04	10s
1123	##	29	3.50639938e+09	9.43165452e+09	1.88e-01	6.62e-05	1.08e+04	10s
1124	##	30	3.57026481e+09	7.35299217e+09	1.50e-01	1.57e-05	6.88e+03	10s
1125	##	31	3.65049204e+09	6.77198256e+09	1.07e-01	5.29e-06	5.68e+03	10s
1126	##	32	3.73203281e+09	6.00109657e+09	6.50e-02	2.33e-09	4.13e+03	11s

1127	##	33	3.78064064e+09	5.51693557e+09	4.15e-02	1.86e-09	3.16e+03	11s
1128	##	34	3.80531595e+09	5.01336151e+09	3.16e-02	1.86e-09	2.20e+03	11s
1129	##	35	3.83857289e+09	4.77991060e+09	1.96e-02	1.86e-09	1.71e+03	11s
1130	##	36	3.86567354e+09	4.51949400e+09	1.15e-02	2.79e-09	1.19e+03	11s
1131	##	37	3.89576774e+09	4.36997925e+09	4.94e-03	2.79e-09	8.63e+02	11s
1132	##	38	3.90225178e+09	4.28215145e+09	4.06e-03	2.79e-09	6.92e+02	12s
1133	##	39	3.91494531e+09	4.24986726e+09	2.55e-03	2.33e-09	6.10e+02	12s
1134	##	40	3.92089613e+09	4.17585069e+09	1.97e-03	2.33e-09	4.64e+02	12s
1135	##	41	3.92507525e+09	4.13066288e+09	1.62e-03	1.86e-09	3.74e+02	12s
1136	##	42	3.93021914e+09	4.08241318e+09	1.24e-03	1.86e-09	2.77e+02	12s
1137	##	43	3.93188309e+09	4.06653107e+09	1.14e-03	2.33e-09	2.45e+02	12s
1138	##	44	3.93591892e+09	4.05364457e+09	8.92e-04	2.79e-09	2.14e+02	13s
1139	##	45	3.94312228e+09	4.02689044e+09	4.87e-04	2.79e-09	1.53e+02	13s
1140	##	46	3.94683184e+09	4.00028912e+09	3.34e-04	2.33e-09	9.73e+01	13s
1141	##	47	3.95098787e+09	3.99025383e+09	1.91e-04	2.33e-09	7.15e+01	13s
1142	##	48	3.95270181e+09	3.97802710e+09	1.39e-04	2.33e-09	4.61e+01	13s
1143	##	49	3.95384604e+09	3.97482739e+09	1.08e-04	2.33e-09	3.82e+01	13s
1144	##	50	3.95455805e+09	3.97199263e+09	8.98e-05	2.10e-09	3.18e+01	13s
1145	##	51	3.95554880e+09	3.96858524e+09	6.40e-05	2.33e-09	2.37e+01	14s
1146	##	52	3.95654238e+09	3.96243031e+09	3.97e-05	2.33e-09	1.07e+01	14s
1147	##	53	3.95706936e+09	3.96082660e+09	2.78e-05	1.86e-09	6.84e+00	14s
1148	##	54	3.95766740e+09	3.96014739e+09	1.52e-05	2.33e-09	4.52e+00	14s
1149	##	55	3.95785143e+09	3.95970683e+09	1.12e-05	2.33e-09	3.38e+00	14s
1150	##	56	3.95809673e+09	3.95914394e+09	6.07e-06	2.10e-09	1.91e+00	14s
1151	##	57	3.95820881e+09	3.95892137e+09	3.83e-06	1.86e-09	1.30e+00	15s
1152	##	58	3.95828152e+09	3.95872741e+09	2.42e-06	3.03e-09	8.12e-01	15s
1153	##	59	3.95834759e+09	3.95858562e+09	1.18e-06	2.79e-09	4.33e-01	15s
1154	##	60	3.95838492e+09	3.95850362e+09	5.51e-07	2.33e-09	2.16e-01	15s
1155	##	61	3.95840048e+09	3.95846202e+09	3.63e-07	2.33e-09	1.12e-01	15s
1156	##	62	3.95840779e+09	3.95843861e+09	2.30e-07	2.79e-09	5.61e-02	15s
1157	##	63	3.95841416e+09	3.95842754e+09	1.16e-07	2.10e-09	2.44e-02	15s

```

1158 ## 64 3.95841689e+09 3.95842188e+09 2.22e-07 2.33e-09 9.08e-03 15s
1159 ## 65 3.95841850e+09 3.95842081e+09 1.02e-06 2.10e-09 4.19e-03 16s
1160 ## 66 3.95841907e+09 3.95842013e+09 1.61e-06 2.33e-09 1.92e-03 16s
1161 ## 67 3.95841944e+09 3.95841963e+09 7.46e-06 2.79e-09 3.59e-04 16s
1162 ## 68 3.95841951e+09 3.95841953e+09 1.03e-06 2.79e-09 3.21e-05 16s
1163 ## 69 3.95841952e+09 3.95841952e+09 3.66e-09 4.42e-09 2.92e-07 16s
1164 ## 70 3.95841952e+09 3.95841952e+09 1.40e-09 2.79e-09 2.72e-09 16s
1165 ##
1166 ## Barrier solved model in 70 iterations and 16.28 seconds (14.15 work units)
1167 ## Optimal objective 3.95841952e+09
1168 ##
1169 ##
1170 ## Root crossover log...
1171 ##
1172 ## 19475 DPushes remaining with DInf 0.0000000e+00 16s
1173 ## 0 DPushes remaining with DInf 0.0000000e+00 17s
1174 ##
1175 ## 490 PPushes remaining with PInf 3.6057680e-04 17s
1176 ## 0 PPushes remaining with PInf 0.0000000e+00 17s
1177 ##
1178 ## Push phase complete: Pinf 0.0000000e+00, Dinf 1.6978770e+05 17s
1179 ##
1180 ##
1181 ## Root simplex log...
1182 ##
1183 ## Iteration Objective Primal Inf. Dual Inf. Time
1184 ## 17790 3.9584195e+09 0.000000e+00 1.697877e+05 17s
1185 ## 18271 3.9584195e+09 0.000000e+00 0.000000e+00 18s
1186 ## Concurrent spin time: 5.80s (can be avoided by choosing Method=3)
1187 ##
1188 ## Solved with barrier

```

```

1189  ##
1190  ## Root relaxation: objective 3.958420e+09, 18271 iterations, 18.90 seconds (10.34 work units)
1191  ## Total elapsed time = 36.12s (DegenMoves)
1192  ## Total elapsed time = 40.47s (DegenMoves)
1193  ##
1194  ##      Nodes      |      Current Node      |      Objective Bounds      |      Work
1195  ##  Expl Unexpl |  Obj  Depth IntInf |  Incumbent    BestBd   Gap | It/Node Time
1196  ##
1197  ##      0      0 3.9584e+09      0 1068   -0.00000 3.9584e+09      -      -   41s
1198  ## H      0      0                        2.313089e+09 3.9584e+09  71.1%      -   41s
1199  ## H      0      0                        2.837498e+09 3.9584e+09  39.5%      -   43s
1200  ## H      0      0                        2.933587e+09 3.9584e+09  34.9%      -   44s
1201  ## H      0      0                        2.981892e+09 3.9584e+09  32.7%      -   45s
1202  ## H      0      0                        3.020954e+09 3.9584e+09  31.0%      -   45s
1203  ## H      0      0                        3.103231e+09 3.9584e+09  27.6%      -   46s
1204  ## H      0      0                        3.103236e+09 3.9584e+09  27.6%      -   46s
1205  ## H      0      0                        3.176292e+09 3.9584e+09  24.6%      -   46s
1206  ## H      0      0                        3.182967e+09 3.9584e+09  24.4%      -   46s
1207  ## H      0      0                        3.199640e+09 3.9584e+09  23.7%      -   47s
1208  ## H      0      0                        3.199704e+09 3.9584e+09  23.7%      -   47s
1209  ## H      0      0                        3.200727e+09 3.9584e+09  23.7%      -   48s
1210  ## H      0      0                        3.203864e+09 3.9584e+09  23.6%      -   48s
1211  ## H      0      0                        3.205910e+09 3.9584e+09  23.5%      -   48s
1212  ## H      0      0                        3.207329e+09 3.9584e+09  23.4%      -   48s
1213  ## H      0      0                        3.212199e+09 3.9584e+09  23.2%      -   48s
1214  ## H      0      0                        3.212769e+09 3.9584e+09  23.2%      -   49s
1215  ## H      0      0                        3.216502e+09 3.9584e+09  23.1%      -   49s
1216  ## H      0      0                        3.219618e+09 3.9584e+09  22.9%      -   49s
1217  ## H      0      0                        3.234813e+09 3.9584e+09  22.4%      -   49s
1218  ## H      0      0                        3.240407e+09 3.9584e+09  22.2%      -   49s
1219  ## H      0      0                        3.253717e+09 3.9584e+09  21.7%      -   49s

```


1220	##	H	0	0			3.261512e+09	3.9584e+09	21.4%	-	49s
1221	##	H	0	0			3.261514e+09	3.9584e+09	21.4%	-	49s
1222	##	H	0	0			3.295894e+09	3.9584e+09	20.1%	-	49s
1223	##	H	0	0			3.309585e+09	3.9584e+09	19.6%	-	50s
1224	##	H	0	0			3.312421e+09	3.9584e+09	19.5%	-	50s
1225	##	H	0	0			3.312860e+09	3.9584e+09	19.5%	-	50s
1226	##	H	0	0			3.317465e+09	3.9584e+09	19.3%	-	50s
1227	##	H	0	0			3.329657e+09	3.9584e+09	18.9%	-	50s
1228	##	H	0	0			3.332185e+09	3.9584e+09	18.8%	-	51s
1229	##		0	0	3.9582e+09	0 1156	3.3322e+09	3.9582e+09	18.8%	-	138s
1230	##	H	0	0			3.366809e+09	3.9582e+09	17.6%	-	152s
1231	##	H	0	0			3.370670e+09	3.9582e+09	17.4%	-	152s
1232	##	H	0	0			3.371206e+09	3.9582e+09	17.4%	-	153s
1233	##	H	0	0			3.371262e+09	3.9582e+09	17.4%	-	153s
1234	##	H	0	0			3.389905e+09	3.9582e+09	16.8%	-	153s
1235	##	H	0	0			3.409308e+09	3.9582e+09	16.1%	-	153s
1236	##	H	0	0			3.437943e+09	3.9582e+09	15.1%	-	153s
1237	##	H	0	0			3.707753e+09	3.9582e+09	6.75%	-	195s
1238	##	H	0	0			3.707757e+09	3.9582e+09	6.75%	-	195s
1239	##		0	0	3.9581e+09	0 1183	3.7078e+09	3.9581e+09	6.75%	-	213s
1240	##		0	0	3.9581e+09	0 1216	3.7078e+09	3.9581e+09	6.75%	-	216s
1241	##		0	0	3.9581e+09	0 1195	3.7078e+09	3.9581e+09	6.75%	-	218s
1242	##		0	0	3.9581e+09	0 1191	3.7078e+09	3.9581e+09	6.75%	-	219s
1243	##		0	0	3.9579e+09	0 1312	3.7078e+09	3.9579e+09	6.75%	-	300s
1244	##										
1245	##	Cutting planes:									
1246	##	Gomory: 5									
1247	##	MIR: 218									
1248	##	Flow cover: 711									
1249	##										
1250	##	Explored 1 nodes (43512 simplex iterations) in 300.10 seconds (282.68 work units)									

```

1251 ## Thread count was 24 (of 24 available processors)
1252 ##
1253 ## Solution count 10: 3.70776e+09 3.70776e+09 3.70775e+09 ... 3.36681e+09
1254 ##
1255 ## Time limit reached
1256 ## Best objective 3.707757413852e+09, best bound 3.957876955235e+09, gap 6.7458%
1257 ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 3707757413.8520 | Runtime: 300.5 s

1258 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...
1259 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)
1260 ##   C1: Path establishment
1261 ##   C2: Path connectivity
1262 ##   C3: Biomass yield
1263 ##   C4: Inventory balance
1264 ##   C5: Storage capacity
1265 ##   C6: Processing capacity
1266 ##   C7: Demand satisfaction
1267 ##   Total constraints: 27183, nnz: 977322
1268 ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
1269 ## Set parameter Username
1270 ## Set parameter LicenseID to value 2805714
1271 ## Set parameter TimeLimit to value 300
1272 ## Set parameter MIPGap to value 0.05
1273 ## Set parameter MIPFocus to value 1
1274 ## Set parameter Cuts to value 3
1275 ## Academic license - for non-commercial use only - expires 2027-04-10
1276 ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
1277 ##
1278 ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
1279 ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
1280 ##
1281 ## Non-default parameters:

```

```

1282 ## TimeLimit 300
1283 ## MIPGap 0.05
1284 ## MIPFocus 1
1285 ## Cuts 3
1286 ## LogToConsole 0
1287 ##
1288 ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
1289 ## Model fingerprint: 0xf76a18b3
1290 ## Variable types: 246642 continuous, 66265 integer (66265 binary)
1291 ## Coefficient statistics:
1292 ##   Matrix range      [1e+00, 6e+05]
1293 ##   Objective range   [6e-02, 4e+07]
1294 ##   Bounds range      [1e+00, 5e+07]
1295 ##   RHS range         [1e+00, 7e+07]
1296 ## Found heuristic solution: objective -0.0000000
1297 ## Presolve removed 3524 rows and 47233 columns
1298 ## Presolve time: 4.18s
1299 ## Presolved: 23659 rows, 265674 columns, 844553 nonzeros
1300 ## Variable types: 204919 continuous, 60755 integer (60755 binary)
1301 ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
1302 ## Showing barrier log only...
1303 ##
1304 ## Root barrier log...
1305 ##
1306 ## Ordering time: 0.63s
1307 ##
1308 ## Barrier statistics:
1309 ##   AA' NZ      : 5.958e+05
1310 ##   Factor NZ   : 7.925e+06 (roughly 200 MB of memory)
1311 ##   Factor Ops  : 8.035e+09 (less than 1 second per iteration)
1312 ##   Threads    : 10

```

1313	##								
1314	##		Objective		Residual				
1315	##	Iter	Primal	Dual	Primal	Dual	Compl	Time	
1316	##	0	-2.22988233e+17	1.82464957e+16	2.46e+09	1.44e+03	6.70e+12	6s	
1317	##	1	-9.54239919e+16	1.62203165e+16	1.05e+09	4.46e+05	2.93e+12	6s	
1318	##	2	-1.29803982e+16	1.25368838e+16	1.38e+08	2.59e+04	4.04e+11	6s	
1319	##	3	-2.65683903e+15	7.18588960e+15	2.69e+07	2.01e+03	8.52e+10	6s	
1320	##	4	-4.57568296e+14	3.67253013e+15	4.38e+06	7.01e+02	1.79e+10	6s	
1321	##	5	-7.31353072e+13	1.31417574e+15	5.55e+05	1.12e-08	4.01e+09	7s	
1322	##	6	-1.22699117e+13	8.84461682e+14	3.29e+04	9.31e-09	1.86e+09	7s	
1323	##	7	-5.09630649e+12	7.51469277e+14	1.33e+04	6.52e-09	1.46e+09	7s	
1324	##	8	-1.92210262e+12	2.42305650e+14	4.73e+03	7.45e-09	4.64e+08	7s	
1325	##	9	-7.53485014e+11	6.92618987e+13	1.61e+03	8.85e-09	1.31e+08	7s	
1326	##	10	-4.36492907e+11	6.32011250e+12	7.92e+02	1.30e-08	1.32e+07	7s	
1327	##	11	-3.48807311e+11	4.67981929e+12	6.05e+02	1.30e-08	9.90e+06	7s	
1328	##	12	-2.26219352e+11	2.27474980e+12	3.54e+02	1.02e-08	5.02e+06	8s	
1329	##	13	-1.39311029e+11	1.39381618e+12	1.85e+02	1.30e-08	3.05e+06	8s	
1330	##	14	-9.17301981e+10	5.49630869e+11	9.41e+01	2.47e-08	1.37e+06	8s	
1331	##	15	-6.90645136e+10	5.13711646e+11	6.14e+01	2.51e-08	1.20e+06	8s	
1332	##	16	-5.71847970e+10	4.50390969e+11	5.10e+01	4.93e-03	1.04e+06	8s	
1333	##	17	-4.78622073e+10	3.70724875e+11	4.22e+01	3.00e-02	8.59e+05	8s	
1334	##	18	-2.95198839e+10	2.93685736e+11	2.51e+01	1.83e-02	6.46e+05	8s	
1335	##	19	-1.70692785e+10	2.30620907e+11	1.56e+01	1.37e-02	4.84e+05	9s	
1336	##	20	-9.86441024e+09	1.45183982e+11	9.69e+00	7.77e-03	3.00e+05	9s	
1337	##	21	-5.84551596e+09	1.27803743e+11	6.52e+00	6.80e-03	2.54e+05	9s	
1338	##	22	-3.77059073e+09	7.86499094e+10	4.90e+00	4.26e-03	1.56e+05	9s	
1339	##	23	-9.54485005e+08	5.44705817e+10	2.67e+00	2.71e-03	1.03e+05	9s	
1340	##	24	3.20284527e+08	4.31561023e+10	1.65e+00	1.98e-03	7.88e+04	9s	
1341	##	25	1.15595530e+09	2.75423482e+10	9.86e-01	9.52e-04	4.82e+04	10s	
1342	##	26	1.55827481e+09	1.90915484e+10	7.51e-01	4.93e-04	3.19e+04	10s	
1343	##	27	1.99996921e+09	1.68410435e+10	5.27e-01	3.45e-04	2.70e+04	10s	

1344	##	28	2.23208883e+09	1.50623538e+10	4.11e-01	2.72e-04	2.33e+04	10s
1345	##	29	2.36945133e+09	1.45588327e+10	3.47e-01	2.52e-04	2.21e+04	10s
1346	##	30	2.60061062e+09	1.14896068e+10	2.61e-01	1.88e-04	1.61e+04	10s
1347	##	31	2.87815982e+09	9.46341773e+09	1.51e-01	1.20e-04	1.20e+04	10s
1348	##	32	2.96791818e+09	7.64946083e+09	1.21e-01	6.63e-05	8.50e+03	11s
1349	##	33	3.07178844e+09	6.69287974e+09	8.62e-02	3.27e-05	6.57e+03	11s
1350	##	34	3.15455895e+09	5.38018214e+09	6.04e-02	6.75e-06	4.04e+03	11s
1351	##	35	3.19269329e+09	4.79612457e+09	4.94e-02	3.73e-09	2.91e+03	11s
1352	##	36	3.30044044e+09	4.38230821e+09	2.03e-02	3.96e-09	1.97e+03	11s
1353	##	37	3.33430417e+09	4.00788350e+09	1.37e-02	4.66e-09	1.22e+03	11s
1354	##	38	3.35950843e+09	3.89230860e+09	9.95e-03	4.66e-09	9.68e+02	11s
1355	##	39	3.36892516e+09	3.79039663e+09	8.73e-03	5.59e-09	7.66e+02	12s
1356	##	40	3.38094976e+09	3.77540393e+09	7.28e-03	3.73e-09	7.17e+02	12s
1357	##	41	3.39170383e+09	3.69430278e+09	6.10e-03	5.59e-09	5.50e+02	12s
1358	##	42	3.41210754e+09	3.64123931e+09	3.90e-03	5.59e-09	4.16e+02	12s
1359	##	43	3.41800868e+09	3.59756587e+09	3.36e-03	5.86e-09	3.26e+02	12s
1360	##	44	3.43459086e+09	3.56306785e+09	1.97e-03	5.59e-09	2.34e+02	12s
1361	##	45	3.44284408e+09	3.53224607e+09	1.36e-03	5.47e-09	1.63e+02	12s
1362	##	46	3.44906762e+09	3.51140642e+09	9.59e-04	4.66e-09	1.13e+02	13s
1363	##	47	3.45351121e+09	3.49754954e+09	6.88e-04	5.36e-09	8.00e+01	13s
1364	##	48	3.45874463e+09	3.48701899e+09	3.96e-04	5.59e-09	5.14e+01	13s
1365	##	49	3.46209218e+09	3.48086674e+09	2.23e-04	3.96e-09	3.41e+01	13s
1366	##	50	3.46328960e+09	3.47679631e+09	1.61e-04	4.25e-09	2.46e+01	13s
1367	##	51	3.46388547e+09	3.47490181e+09	1.32e-04	4.66e-09	2.00e+01	13s
1368	##	52	3.46458701e+09	3.47276652e+09	9.82e-05	4.66e-09	1.49e+01	14s
1369	##	53	3.46518064e+09	3.47094861e+09	7.08e-05	3.93e-09	1.05e+01	14s
1370	##	54	3.46606975e+09	3.46938775e+09	3.18e-05	3.83e-09	6.04e+00	14s
1371	##	55	3.46647942e+09	3.46824494e+09	1.47e-05	6.57e-09	3.21e+00	14s
1372	##	56	3.46649825e+09	3.46799145e+09	1.40e-05	5.16e-09	2.72e+00	14s
1373	##	57	3.46659097e+09	3.46774472e+09	1.04e-05	4.46e-09	2.10e+00	14s
1374	##	58	3.46667907e+09	3.46746756e+09	7.21e-06	4.63e-09	1.43e+00	14s

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1375 ## 59 3.46673707e+09 3.46725670e+09 5.16e-06 4.48e-09 9.45e-01 15s
1376 ## 60 3.46679764e+09 3.46706897e+09 2.89e-06 5.59e-09 4.94e-01 15s
1377 ## 61 3.46684957e+09 3.46697046e+09 1.19e-06 4.66e-09 2.20e-01 15s
1378 ## 62 3.46687054e+09 3.46691737e+09 4.93e-07 4.73e-09 8.52e-02 15s
1379 ## 63 3.46687618e+09 3.46690047e+09 3.86e-07 4.59e-09 4.42e-02 15s
1380 ## 64 3.46688156e+09 3.46689260e+09 2.33e-07 4.42e-09 2.01e-02 15s
1381 ## 65 3.46688321e+09 3.46688802e+09 2.36e-07 4.42e-09 8.76e-03 15s
1382 ## 66 3.46688435e+09 3.46688703e+09 6.63e-07 9.23e-09 4.87e-03 15s
1383 ## 67 3.46688468e+09 3.46688576e+09 7.49e-07 3.73e-09 1.96e-03 16s
1384 ## 68 3.46688489e+09 3.46688504e+09 4.73e-07 5.59e-09 2.75e-04 16s
1385 ## 69 3.46688499e+09 3.46688502e+09 3.72e-08 6.00e-09 5.06e-05 16s
1386 ## 70 3.46688501e+09 3.46688501e+09 1.04e-09 5.59e-09 1.61e-06 16s
1387 ## 71 3.46688501e+09 3.46688501e+09 2.91e-08 4.42e-09 5.44e-09 16s
1388 ##
1389 ## Barrier solved model in 71 iterations and 16.26 seconds (14.75 work units)
1390 ## Optimal objective 3.46688501e+09
1391 ##
1392 ##
1393 ## Root crossover log...
1394 ##
1395 ## 19809 DPushes remaining with DInf 0.0000000e+00 16s
1396 ## 0 DPushes remaining with DInf 0.0000000e+00 17s
1397 ##
1398 ## 403 PPushes remaining with PInf 1.0518248e-02 17s
1399 ## 0 PPushes remaining with PInf 0.0000000e+00 17s
1400 ##
1401 ## Push phase complete: Pinf 0.0000000e+00, Dinf 3.8078456e+04 17s
1402 ##
1403 ##
1404 ## Root simplex log...
1405 ##

```

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1406 ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
1407 ##      18841      3.4668850e+09      0.000000e+00      3.807846e+04      17s
1408 ##      19608      3.4668850e+09      0.000000e+00      0.000000e+00      18s
1409 ## Concurrent spin time: 7.74s (can be avoided by choosing Method=3)
1410 ##
1411 ## Solved with barrier
1412 ##
1413 ## Root relaxation: objective 3.466885e+09, 19608 iterations, 21.21 seconds (11.11 work units)
1414 ## Total elapsed time = 37.01s (DegenMoves)
1415 ## Total elapsed time = 41.53s (DegenMoves)
1416 ##
1417 ##      Nodes      |      Current Node      |      Objective Bounds      |      Work
1418 ## Expl Unexpl | Obj Depth IntInf | Incumbent      BestBd      Gap | It/Node Time
1419 ##
1420 ##      0      0 3.4669e+09      0 965      -0.00000 3.4669e+09      -      -      42s
1421 ## H      0      0      2.060644e+09 3.4669e+09 68.2%      -      42s
1422 ## H      0      0      2.061576e+09 3.4669e+09 68.2%      -      42s
1423 ## H      0      0      2.251800e+09 3.4669e+09 54.0%      -      43s
1424 ## H      0      0      2.347489e+09 3.4669e+09 47.7%      -      44s
1425 ## H      0      0      2.347846e+09 3.4669e+09 47.7%      -      45s
1426 ## H      0      0      2.399688e+09 3.4669e+09 44.5%      -      45s
1427 ## H      0      0      2.404597e+09 3.4669e+09 44.2%      -      45s
1428 ## H      0      0      2.442481e+09 3.4669e+09 41.9%      -      45s
1429 ## H      0      0      2.465951e+09 3.4669e+09 40.6%      -      46s
1430 ## H      0      0      2.538984e+09 3.4669e+09 36.5%      -      46s
1431 ## H      0      0      2.568667e+09 3.4669e+09 35.0%      -      48s
1432 ## H      0      0      2.609239e+09 3.4669e+09 32.9%      -      48s
1433 ## H      0      0      2.622084e+09 3.4669e+09 32.2%      -      48s
1434 ## H      0      0      2.778799e+09 3.4669e+09 24.8%      -      48s
1435 ## H      0      0      2.782675e+09 3.4669e+09 24.6%      -      48s
1436 ## H      0      0      2.789815e+09 3.4669e+09 24.3%      -      49s

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1437 ## H      0      0                2.791132e+09 3.4669e+09 24.2%      - 49s
1438 ## H      0      0                2.816012e+09 3.4669e+09 23.1%      - 49s
1439 ## H      0      0                2.819176e+09 3.4669e+09 23.0%      - 49s
1440 ## H      0      0                2.819818e+09 3.4669e+09 22.9%      - 49s
1441 ## H      0      0                2.820613e+09 3.4669e+09 22.9%      - 50s
1442 ## H      0      0                2.830808e+09 3.4669e+09 22.5%      - 50s
1443 ## H      0      0                2.835602e+09 3.4669e+09 22.3%      - 50s
1444 ##          0      0 3.4666e+09    0 1148 2.8356e+09 3.4666e+09 22.3%      - 130s
1445 ## H      0      0                3.128862e+09 3.4666e+09 10.8%      - 172s
1446 ##          0      0 3.4666e+09    0 1079 3.1289e+09 3.4666e+09 10.8%      - 193s
1447 ##          0      0 3.4665e+09    0 1107 3.1289e+09 3.4665e+09 10.8%      - 200s
1448 ##          0      0 3.4665e+09    0 1110 3.1289e+09 3.4665e+09 10.8%      - 203s
1449 ##          0      0 3.4662e+09    0 1324 3.1289e+09 3.4662e+09 10.8%      - 300s
1450 ##
1451 ## Cutting planes:
1452 ##   Gomory: 5
1453 ##   MIR: 214
1454 ##   Flow cover: 706
1455 ##
1456 ## Explored 1 nodes (43194 simplex iterations) in 300.20 seconds (279.30 work units)
1457 ## Thread count was 24 (of 24 available processors)
1458 ##
1459 ## Solution count 10: 3.12886e+09 3.12886e+09 2.8356e+09 ... 2.78981e+09
1460 ##
1461 ## Time limit reached
1462 ## Warning: max constraint violation (2.1596e-06) exceeds tolerance
1463 ## Warning: max bound violation (3.3556e-06) exceeds tolerance
1464 ## Best objective 3.128861703745e+09, best bound 3.466226980725e+09, gap 10.7824%
1465 ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 3128861703.7447 | Runtime: 300.6 s
1466 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...
1467 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)

```



```

1468 ## C1: Path establishment
1469 ## C2: Path connectivity
1470 ## C3: Biomass yield
1471 ## C4: Inventory balance
1472 ## C5: Storage capacity
1473 ## C6: Processing capacity
1474 ## C7: Demand satisfaction
1475 ## Total constraints: 27183, nnz: 977322
1476 ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
1477 ## Set parameter Username
1478 ## Set parameter LicenseID to value 2805714
1479 ## Set parameter TimeLimit to value 300
1480 ## Set parameter MIPGap to value 0.05
1481 ## Set parameter MIPFocus to value 1
1482 ## Set parameter Cuts to value 3
1483 ## Academic license - for non-commercial use only - expires 2027-04-10
1484 ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
1485 ##
1486 ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
1487 ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
1488 ##
1489 ## Non-default parameters:
1490 ## TimeLimit 300
1491 ## MIPGap 0.05
1492 ## MIPFocus 1
1493 ## Cuts 3
1494 ## LogToConsole 0
1495 ##
1496 ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
1497 ## Model fingerprint: 0x9457a8dc
1498 ## Variable types: 246642 continuous, 66265 integer (66265 binary)

```

```

1499 ## Coefficient statistics:
1500 ##   Matrix range      [1e+00, 6e+05]
1501 ##   Objective range   [3e-01, 2e+07]
1502 ##   Bounds range      [1e+00, 5e+07]
1503 ##   RHS range         [1e+00, 7e+07]
1504 ## Found heuristic solution: objective -0.0000000
1505 ## Presolve removed 3524 rows and 47233 columns
1506 ## Presolve time: 4.13s
1507 ## Presolved: 23659 rows, 265674 columns, 844553 nonzeros
1508 ## Variable types: 204919 continuous, 60755 integer (60755 binary)
1509 ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
1510 ## Showing barrier log only...
1511 ##
1512 ## Root barrier log...
1513 ##
1514 ## Ordering time: 0.69s
1515 ##
1516 ## Barrier statistics:
1517 ##   AA' NZ      : 5.958e+05
1518 ##   Factor NZ   : 7.925e+06 (roughly 200 MB of memory)
1519 ##   Factor Ops  : 8.035e+09 (less than 1 second per iteration)
1520 ##   Threads    : 10
1521 ##
1522 ##               Objective                Residual
1523 ## Iter      Primal      Dual      Primal    Dual    Compl    Time
1524 ##   0  -1.12830722e+17  1.25293753e+16  2.46e+09  1.44e+03  4.52e+12  6s
1525 ##   1  -4.82536233e+16  1.11299663e+16  1.05e+09  3.01e+05  1.97e+12  6s
1526 ##   2  -6.59994225e+15  8.60039611e+15  1.38e+08  1.76e+04  2.71e+11  6s
1527 ##   3  -1.34059642e+15  4.92218039e+15  2.59e+07  1.39e+03  5.57e+10  6s
1528 ##   4  -2.47028667e+14  2.47241194e+15  4.46e+06  4.86e+02  1.22e+10  6s
1529 ##   5  -3.93822540e+13  8.78879046e+14  5.56e+05  4.66e-09  2.69e+09  7s

```

1530	##	6	-6.90988404e+12	5.92416038e+14	3.39e+04	3.73e-09	1.25e+09	7s
1531	##	7	-2.09518253e+12	4.07755858e+14	9.36e+03	3.03e-09	7.85e+08	7s
1532	##	8	-9.44259870e+11	1.09396087e+14	3.52e+03	3.73e-09	2.09e+08	7s
1533	##	9	-5.05497792e+11	1.76093393e+13	1.33e+03	4.19e-09	3.47e+07	7s
1534	##	10	-3.76349215e+11	3.21566182e+12	7.36e+02	6.52e-09	7.15e+06	7s
1535	##	11	-3.19648232e+11	1.70615499e+12	5.37e+02	1.30e-08	4.15e+06	7s
1536	##	12	-2.47919100e+11	1.04808082e+12	3.22e+02	9.43e-09	2.65e+06	8s
1537	##	13	-1.83027892e+11	9.78011570e+11	1.89e+02	1.02e-08	2.29e+06	8s
1538	##	14	-1.05197693e+11	4.82011913e+11	1.04e+02	8.85e-09	1.17e+06	8s
1539	##	15	-7.93377100e+10	4.67690487e+11	7.43e+01	1.23e-08	1.07e+06	8s
1540	##	16	-5.96405443e+10	4.28039307e+11	6.03e+01	4.01e-03	9.46e+05	8s
1541	##	17	-3.78273989e+10	2.77874642e+11	4.04e+01	1.38e-02	6.13e+05	8s
1542	##	18	-2.24664741e+10	1.82824266e+11	2.69e+01	7.58e-03	3.98e+05	8s
1543	##	19	-1.60104553e+10	1.53416693e+11	2.07e+01	3.66e-03	3.27e+05	9s
1544	##	20	-9.68871306e+09	1.15726246e+11	1.49e+01	2.32e-03	2.40e+05	9s
1545	##	21	-3.53698777e+09	7.78409077e+10	8.06e+00	2.76e-04	1.54e+05	9s
1546	##	22	-2.12055921e+09	5.73622188e+10	6.18e+00	7.10e-09	1.12e+05	9s
1547	##	23	-1.05311464e+09	3.68791520e+10	4.96e+00	6.52e-09	7.12e+04	9s
1548	##	24	9.73902916e+08	2.34132917e+10	2.51e+00	5.36e-09	4.15e+04	9s
1549	##	25	1.61775024e+09	1.64287308e+10	1.58e+00	4.19e-09	2.71e+04	10s
1550	##	26	1.82674195e+09	1.27633798e+10	1.32e+00	3.26e-09	1.99e+04	10s
1551	##	27	2.03240106e+09	1.18693892e+10	1.08e+00	2.79e-09	1.79e+04	10s
1552	##	28	2.26195545e+09	9.93381114e+09	8.41e-01	2.79e-09	1.39e+04	10s
1553	##	29	2.34796198e+09	9.01120439e+09	7.55e-01	2.79e-09	1.21e+04	10s
1554	##	30	2.52708234e+09	8.11160960e+09	5.78e-01	3.73e-09	1.01e+04	10s
1555	##	31	2.64696717e+09	7.67980513e+09	4.73e-01	2.79e-09	9.14e+03	10s
1556	##	32	2.77664356e+09	6.20345179e+09	3.69e-01	2.79e-09	6.22e+03	11s
1557	##	33	2.94798163e+09	5.81524767e+09	2.38e-01	2.79e-09	5.21e+03	11s
1558	##	34	3.07424397e+09	5.31347446e+09	1.52e-01	2.79e-09	4.07e+03	11s
1559	##	35	3.13564723e+09	4.81827533e+09	1.17e-01	2.79e-09	3.06e+03	11s
1560	##	36	3.20265642e+09	4.49209984e+09	8.40e-02	3.73e-09	2.34e+03	11s

1561	##	37	3.25278401e+09	4.16661257e+09	6.27e-02	2.33e-09	1.66e+03	11s
1562	##	38	3.30496652e+09	3.94962698e+09	4.26e-02	2.91e-09	1.17e+03	12s
1563	##	39	3.34415259e+09	3.83638443e+09	2.94e-02	2.85e-09	8.95e+02	12s
1564	##	40	3.37182411e+09	3.70661040e+09	2.13e-02	3.26e-09	6.09e+02	12s
1565	##	41	3.39076953e+09	3.64229420e+09	1.65e-02	2.79e-09	4.57e+02	12s
1566	##	42	3.40667532e+09	3.59463837e+09	1.27e-02	2.79e-09	3.42e+02	12s
1567	##	43	3.43013602e+09	3.55691904e+09	7.45e-03	2.79e-09	2.31e+02	12s
1568	##	44	3.43973690e+09	3.52647864e+09	5.52e-03	2.79e-09	1.58e+02	13s
1569	##	45	3.44769309e+09	3.50486392e+09	4.00e-03	2.33e-09	1.04e+02	13s
1570	##	46	3.45473894e+09	3.49292270e+09	2.74e-03	2.21e-09	6.94e+01	13s
1571	##	47	3.45959351e+09	3.48550551e+09	1.87e-03	1.86e-09	4.71e+01	13s
1572	##	48	3.46267105e+09	3.48098371e+09	1.32e-03	2.33e-09	3.33e+01	13s
1573	##	49	3.46497614e+09	3.47755933e+09	9.25e-04	2.56e-09	2.29e+01	13s
1574	##	50	3.46693350e+09	3.47539716e+09	5.98e-04	2.21e-09	1.54e+01	13s
1575	##	51	3.46791800e+09	3.47394575e+09	4.32e-04	2.33e-09	1.10e+01	14s
1576	##	52	3.46889869e+09	3.47319344e+09	2.79e-04	2.33e-09	7.81e+00	14s
1577	##	53	3.46963297e+09	3.47216372e+09	1.66e-04	3.03e-09	4.60e+00	14s
1578	##	54	3.47013888e+09	3.47158489e+09	9.07e-05	2.55e-09	2.63e+00	14s
1579	##	55	3.47033927e+09	3.47121118e+09	6.27e-05	2.56e-09	1.58e+00	14s
1580	##	56	3.47054688e+09	3.47107700e+09	3.28e-05	2.33e-09	9.64e-01	14s
1581	##	57	3.47066102e+09	3.47096980e+09	1.62e-05	2.33e-09	5.62e-01	14s
1582	##	58	3.47072354e+09	3.47087887e+09	8.13e-06	2.33e-09	2.82e-01	15s
1583	##	59	3.47076439e+09	3.47083286e+09	3.14e-06	2.56e-09	1.25e-01	15s
1584	##	60	3.47078085e+09	3.47080017e+09	1.15e-06	3.26e-09	3.51e-02	15s
1585	##	61	3.47078788e+09	3.47079200e+09	4.71e-07	2.79e-09	7.48e-03	15s
1586	##	62	3.47079018e+09	3.47079068e+09	1.22e-06	3.20e-09	9.18e-04	15s
1587	##	63	3.47079036e+09	3.47079045e+09	3.78e-07	4.66e-09	1.59e-04	15s
1588	##	64	3.47079044e+09	3.47079044e+09	2.22e-10	3.26e-09	3.97e-06	15s
1589	##	65	3.47079044e+09	3.47079044e+09	5.07e-09	3.14e-09	3.40e-09	16s
1590	##							
1591	##	Barrier solved model in 65 iterations and 15.61 seconds (13.96 work units)						

```

1592 ## Optimal objective 3.47079044e+09
1593 ##
1594 ##
1595 ## Root crossover log...
1596 ##
1597 ##      19498 DPushes remaining with DInf 0.0000000e+00      16s
1598 ##          0 DPushes remaining with DInf 0.0000000e+00      16s
1599 ##
1600 ##          546 PPushes remaining with PInf 9.4402839e-04      17s
1601 ##          0 PPushes remaining with PInf 0.0000000e+00      17s
1602 ##
1603 ##      Push phase complete: Pinf 0.0000000e+00, Dinf 7.2766814e+03      17s
1604 ##
1605 ##
1606 ## Root simplex log...
1607 ##
1608 ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
1609 ##      17370      3.4707904e+09      0.0000000e+00      7.276681e+03      17s
1610 ##      17641      3.4707904e+09      0.0000000e+00      0.0000000e+00      17s
1611 ## Concurrent spin time: 6.29s (can be avoided by choosing Method=3)
1612 ##
1613 ## Solved with barrier
1614 ##
1615 ## Root relaxation: objective 3.470790e+09, 17641 iterations, 18.96 seconds (10.12 work units)
1616 ## Total elapsed time = 35.53s (DegenMoves)
1617 ## Total elapsed time = 40.05s (DegenMoves)
1618 ##
1619 ##      Nodes      |      Current Node      |      Objective Bounds      |      Work
1620 ##      Expl Unexpl |      Obj  Depth IntInf |      Incumbent      BestBd      Gap |      It/Node Time
1621 ##
1622 ##          0          0 3.4708e+09      0 1038      -0.00000 3.4708e+09      -      -      40s

```

1623	##	H	0	0		2.204955e+09	3.4708e+09	57.4%	-	41s		
1624	##	H	0	0		2.556370e+09	3.4708e+09	35.8%	-	41s		
1625	##	H	0	0		2.557841e+09	3.4708e+09	35.7%	-	41s		
1626	##	H	0	0		2.572801e+09	3.4708e+09	34.9%	-	42s		
1627	##	H	0	0		2.586045e+09	3.4708e+09	34.2%	-	43s		
1628	##	H	0	0		2.588703e+09	3.4708e+09	34.1%	-	43s		
1629	##	H	0	0		2.591809e+09	3.4708e+09	33.9%	-	44s		
1630	##	H	0	0		2.611111e+09	3.4708e+09	32.9%	-	44s		
1631	##	H	0	0		2.682935e+09	3.4708e+09	29.4%	-	46s		
1632	##	H	0	0		2.726931e+09	3.4708e+09	27.3%	-	46s		
1633	##	H	0	0		2.728997e+09	3.4708e+09	27.2%	-	46s		
1634	##	H	0	0		2.733977e+09	3.4708e+09	27.0%	-	46s		
1635	##	H	0	0		2.736554e+09	3.4708e+09	26.8%	-	46s		
1636	##	H	0	0		2.763775e+09	3.4708e+09	25.6%	-	46s		
1637	##	H	0	0		2.763930e+09	3.4708e+09	25.6%	-	46s		
1638	##	H	0	0		2.764138e+09	3.4708e+09	25.6%	-	47s		
1639	##	H	0	0		2.765702e+09	3.4708e+09	25.5%	-	47s		
1640	##	H	0	0		2.770590e+09	3.4708e+09	25.3%	-	47s		
1641	##	H	0	0		2.791031e+09	3.4708e+09	24.4%	-	47s		
1642	##	H	0	0		2.791716e+09	3.4708e+09	24.3%	-	47s		
1643	##	H	0	0		2.791887e+09	3.4708e+09	24.3%	-	47s		
1644	##	H	0	0		2.792736e+09	3.4708e+09	24.3%	-	47s		
1645	##	H	0	0		2.794726e+09	3.4708e+09	24.2%	-	48s		
1646	##	H	0	0		2.797504e+09	3.4708e+09	24.1%	-	48s		
1647	##		0	0	3.4700e+09	0	1171	2.7975e+09	3.4700e+09	24.0%	-	133s
1648	##	H	0	0		2.814974e+09	3.4700e+09	23.3%	-	147s		
1649	##	H	0	0		2.831229e+09	3.4700e+09	22.6%	-	147s		
1650	##	H	0	0		2.832151e+09	3.4700e+09	22.5%	-	147s		
1651	##	H	0	0		2.837133e+09	3.4700e+09	22.3%	-	147s		
1652	##	H	0	0		2.839771e+09	3.4700e+09	22.2%	-	147s		
1653	##	H	0	0		2.844498e+09	3.4700e+09	22.0%	-	147s		

```

1654 ## H      0      0                2.844845e+09 3.4700e+09 22.0%      - 148s
1655 ## H      0      0                2.852702e+09 3.4700e+09 21.6%      - 148s
1656 ## H      0      0                2.883902e+09 3.4700e+09 20.3%      - 148s
1657 ## H      0      0                2.885893e+09 3.4700e+09 20.2%      - 148s
1658 ## H      0      0                2.887281e+09 3.4700e+09 20.2%      - 148s
1659 ## H      0      0                2.887921e+09 3.4700e+09 20.2%      - 148s
1660 ## H      0      0                2.890277e+09 3.4700e+09 20.1%      - 148s
1661 ## H      0      0                2.892462e+09 3.4700e+09 20.0%      - 148s
1662 ## H      0      0                2.893650e+09 3.4700e+09 19.9%      - 148s
1663 ## H      0      0                3.076478e+09 3.4700e+09 12.8%      - 167s
1664 ## H      0      0                3.076597e+09 3.4700e+09 12.8%      - 193s
1665 ##          0      0 3.4698e+09    0 1156 3.0766e+09 3.4698e+09 12.8%      - 216s
1666 ##          0      0 3.4697e+09    0 1173 3.0766e+09 3.4697e+09 12.8%      - 223s
1667 ##          0      0 3.4697e+09    0 1180 3.0766e+09 3.4697e+09 12.8%      - 224s
1668 ##          0      0 3.4697e+09    0 1179 3.0766e+09 3.4697e+09 12.8%      - 225s
1669 ##          0      0 3.4688e+09    0 1333 3.0766e+09 3.4688e+09 12.7%      - 300s
1670 ##
1671 ## Cutting planes:
1672 ## Gomory: 4
1673 ## MIR: 212
1674 ## Flow cover: 724
1675 ##
1676 ## Explored 1 nodes (47028 simplex iterations) in 300.13 seconds (295.07 work units)
1677 ## Thread count was 24 (of 24 available processors)
1678 ##
1679 ## Solution count 10: 3.0766e+09 3.0766e+09 3.07648e+09 ... 2.8839e+09
1680 ##
1681 ## Time limit reached
1682 ## Warning: max constraint violation (4.3752e-06) exceeds tolerance
1683 ## Warning: max bound violation (4.8389e-06) exceeds tolerance
1684 ## Best objective 3.076597142976e+09, best bound 3.468788806673e+09, gap 12.7476%

```

```

1685 ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 3076597142.9760 | Runtime: 300.5 s
1686 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...
1687 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)
1688 ##   C1: Path establishment
1689 ##   C2: Path connectivity
1690 ##   C3: Biomass yield
1691 ##   C4: Inventory balance
1692 ##   C5: Storage capacity
1693 ##   C6: Processing capacity
1694 ##   C7: Demand satisfaction
1695 ##   Total constraints: 27183, nnz: 977322
1696 ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
1697 ## Set parameter Username
1698 ## Set parameter LicenseID to value 2805714
1699 ## Set parameter TimeLimit to value 300
1700 ## Set parameter MIPGap to value 0.05
1701 ## Set parameter MIPFocus to value 1
1702 ## Set parameter Cuts to value 3
1703 ## Academic license - for non-commercial use only - expires 2027-04-10
1704 ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
1705 ##
1706 ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
1707 ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
1708 ##
1709 ## Non-default parameters:
1710 ## TimeLimit 300
1711 ## MIPGap 0.05
1712 ## MIPFocus 1
1713 ## Cuts 3
1714 ## LogToConsole 0
1715 ##

```



```

1716 ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
1717 ## Model fingerprint: 0x74cf203b
1718 ## Variable types: 246642 continuous, 66265 integer (66265 binary)
1719 ## Coefficient statistics:
1720 ##   Matrix range      [1e+00, 6e+05]
1721 ##   Objective range   [3e-01, 4e+07]
1722 ##   Bounds range      [1e+00, 5e+07]
1723 ##   RHS range         [1e+00, 7e+07]
1724 ## Found heuristic solution: objective -0.0000000
1725 ## Presolve removed 3524 rows and 47233 columns
1726 ## Presolve time: 4.41s
1727 ## Presolved: 23659 rows, 265674 columns, 844553 nonzeros
1728 ## Variable types: 204919 continuous, 60755 integer (60755 binary)
1729 ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
1730 ## Showing barrier log only...
1731 ##
1732 ## Root barrier log...
1733 ##
1734 ## Ordering time: 0.69s
1735 ##
1736 ## Barrier statistics:
1737 ##   AA' NZ      : 5.958e+05
1738 ##   Factor NZ   : 7.925e+06 (roughly 200 MB of memory)
1739 ##   Factor Ops  : 8.035e+09 (less than 1 second per iteration)
1740 ##   Threads    : 10
1741 ##
1742 ##               Objective                Residual
1743 ## Iter      Primal      Dual      Primal    Dual    Compl    Time
1744 ##    0 -2.23836022e+17  2.14024508e+16  2.46e+09  1.44e+03  7.78e+12  6s
1745 ##    1 -9.57549264e+16  1.90182258e+16  1.05e+09  5.18e+05  3.40e+12  6s
1746 ##    2 -1.30530341e+16  1.46965996e+16  1.38e+08  3.02e+04  4.68e+11  6s

```

1747	##	3	-2.63338719e+15	8.41757600e+15	2.62e+07	2.35e+03	9.66e+10	7s
1748	##	4	-4.65337493e+14	4.26310837e+15	4.34e+06	8.24e+02	2.07e+10	7s
1749	##	5	-7.46819712e+13	1.51590403e+15	5.47e+05	1.02e-08	4.61e+09	7s
1750	##	6	-1.28724491e+13	1.02747182e+15	3.32e+04	7.45e-09	2.16e+09	7s
1751	##	7	-3.91361958e+12	6.48253409e+14	9.48e+03	1.12e-08	1.25e+09	7s
1752	##	8	-1.65010476e+12	1.83287491e+14	3.53e+03	7.45e-09	3.50e+08	7s
1753	##	9	-8.19006768e+11	6.25349294e+13	1.37e+03	8.89e-09	1.19e+08	7s
1754	##	10	-5.74348233e+11	6.93212712e+12	7.65e+02	1.12e-08	1.46e+07	8s
1755	##	11	-4.91187139e+11	5.00988215e+12	6.00e+02	1.98e-08	1.08e+07	8s
1756	##	12	-3.69527334e+11	2.42755467e+12	3.71e+02	1.50e-08	5.61e+06	8s
1757	##	13	-2.49527306e+11	1.48416624e+12	1.75e+02	1.49e-08	3.42e+06	8s
1758	##	14	-2.04152243e+11	1.26552166e+12	1.22e+02	2.05e-08	2.87e+06	8s
1759	##	15	-1.58290537e+11	6.95761880e+11	9.58e+01	1.58e-08	1.71e+06	8s
1760	##	16	-1.22482031e+11	6.75970908e+11	6.76e+01	1.86e-08	1.57e+06	9s
1761	##	17	-1.02855262e+11	6.11981052e+11	5.83e+01	2.24e-08	1.40e+06	9s
1762	##	18	-6.75302084e+10	3.05203322e+11	3.87e+01	1.86e-08	7.41e+05	9s
1763	##	19	-3.41802954e+10	2.51584613e+11	2.07e+01	2.05e-08	5.52e+05	9s
1764	##	20	-2.09777172e+10	1.33148412e+11	1.36e+01	1.82e-08	2.98e+05	9s
1765	##	21	-9.28802680e+09	1.02925579e+11	7.10e+00	1.54e-08	2.12e+05	9s
1766	##	22	-5.08044409e+09	8.05235854e+10	4.74e+00	1.49e-08	1.60e+05	10s
1767	##	23	-4.69087417e+09	7.45999836e+10	4.48e+00	1.49e-08	1.48e+05	10s
1768	##	24	-3.42256422e+09	6.14555283e+10	3.65e+00	1.16e-08	1.21e+05	10s
1769	##	25	-2.54521040e+09	4.49344647e+10	3.08e+00	1.12e-08	8.77e+04	10s
1770	##	26	-1.38654169e+09	3.26029556e+10	2.33e+00	9.31e-09	6.23e+04	10s
1771	##	27	6.80286311e+08	2.05721586e+10	9.39e-01	6.52e-09	3.62e+04	10s
1772	##	28	1.42719236e+09	1.46064445e+10	5.30e-01	5.59e-09	2.39e+04	10s
1773	##	29	1.60134212e+09	1.30717439e+10	4.45e-01	5.59e-09	2.08e+04	11s
1774	##	30	1.68461390e+09	1.12578337e+10	4.06e-01	4.66e-09	1.74e+04	11s
1775	##	31	1.76615164e+09	1.04697545e+10	3.71e-01	5.59e-09	1.58e+04	11s
1776	##	32	1.87905055e+09	9.76413677e+09	3.21e-01	5.59e-09	1.43e+04	11s
1777	##	33	2.08918321e+09	8.36788663e+09	2.28e-01	5.59e-09	1.14e+04	11s

1778	##	34	2.37972389e+09	7.14137539e+09	1.09e-01	6.52e-09	8.65e+03	11s
1779	##	35	2.53951280e+09	5.44854564e+09	5.84e-02	3.73e-09	5.28e+03	12s
1780	##	36	2.61898511e+09	4.53005305e+09	4.24e-02	3.73e-09	3.47e+03	12s
1781	##	37	2.71111978e+09	4.19958073e+09	2.62e-02	5.12e-09	2.70e+03	12s
1782	##	38	2.73004415e+09	3.98108765e+09	2.35e-02	4.19e-09	2.27e+03	12s
1783	##	39	2.74780024e+09	3.80369909e+09	2.12e-02	6.05e-09	1.92e+03	12s
1784	##	40	2.81379060e+09	3.58967213e+09	1.31e-02	5.59e-09	1.41e+03	12s
1785	##	41	2.84339904e+09	3.39515915e+09	1.04e-02	4.77e-09	1.00e+03	13s
1786	##	42	2.86747417e+09	3.27702111e+09	8.45e-03	3.84e-09	7.44e+02	13s
1787	##	43	2.91557897e+09	3.18868035e+09	4.59e-03	7.28e-09	4.96e+02	13s
1788	##	44	2.93640792e+09	3.12790829e+09	3.25e-03	4.34e-09	3.48e+02	13s
1789	##	45	2.95279414e+09	3.08359481e+09	2.30e-03	5.59e-09	2.38e+02	13s
1790	##	46	2.96686866e+09	3.04982948e+09	1.53e-03	4.42e-09	1.51e+02	13s
1791	##	47	2.98014374e+09	3.03682746e+09	8.60e-04	4.19e-09	1.03e+02	14s
1792	##	48	2.98394055e+09	3.02833451e+09	6.72e-04	4.66e-09	8.07e+01	14s
1793	##	49	2.98770539e+09	3.02065914e+09	4.89e-04	5.50e-09	5.99e+01	14s
1794	##	50	2.99013286e+09	3.01381935e+09	3.69e-04	6.05e-09	4.31e+01	14s
1795	##	51	2.99337678e+09	3.00737897e+09	2.23e-04	4.19e-09	2.54e+01	14s
1796	##	52	2.99554443e+09	3.00403143e+09	1.29e-04	4.09e-09	1.54e+01	14s
1797	##	53	2.99655093e+09	3.00139892e+09	8.53e-05	4.48e-09	8.81e+00	15s
1798	##	54	2.99734885e+09	3.00042826e+09	5.23e-05	4.25e-09	5.60e+00	15s
1799	##	55	2.99782466e+09	2.99988272e+09	3.25e-05	4.42e-09	3.74e+00	15s
1800	##	56	2.99819194e+09	2.99945264e+09	1.86e-05	5.06e-09	2.29e+00	15s
1801	##	57	2.99838971e+09	2.99914527e+09	1.13e-05	3.73e-09	1.37e+00	15s
1802	##	58	2.99852302e+09	2.99899811e+09	6.55e-06	3.99e-09	8.64e-01	15s
1803	##	59	2.99861659e+09	2.99890031e+09	3.05e-06	4.66e-09	5.16e-01	15s
1804	##	60	2.99866128e+09	2.99881347e+09	1.66e-06	3.29e-09	2.77e-01	16s
1805	##	61	2.99869100e+09	2.99875904e+09	9.57e-07	4.66e-09	1.24e-01	16s
1806	##	62	2.99870236e+09	2.99873123e+09	5.25e-07	7.27e-09	5.25e-02	16s
1807	##	63	2.99870947e+09	2.99872280e+09	5.58e-07	3.73e-09	2.42e-02	16s
1808	##	64	2.99871425e+09	2.99871960e+09	2.82e-06	7.50e-09	9.71e-03	16s

```

1809 ##      65      2.99871643e+09  2.99871821e+09  9.54e-07  3.73e-09  3.24e-03  16s
1810 ##      66      2.99871722e+09  2.99871777e+09  2.93e-07  5.12e-09  1.01e-03  16s
1811 ##      67      2.99871758e+09  2.99871760e+09  2.99e-09  4.89e-09  1.87e-05  17s
1812 ##      68      2.99871759e+09  2.99871759e+09  3.36e-09  5.59e-09  2.91e-09  17s
1813 ##
1814 ## Barrier solved model in 68 iterations and 16.77 seconds (14.09 work units)
1815 ## Optimal objective 2.99871759e+09
1816 ##
1817 ##
1818 ## Root crossover log...
1819 ##
1820 ##      19685 DPushes remaining with DInf 0.0000000e+00 17s
1821 ##           0 DPushes remaining with DInf 0.0000000e+00 18s
1822 ##
1823 ##           460 PPushes remaining with PInf 5.7748796e-03 18s
1824 ##           0 PPushes remaining with PInf 0.0000000e+00 18s
1825 ##
1826 ## Push phase complete: Pinf 0.0000000e+00, Dinf 1.3318969e+04 18s
1827 ##
1828 ##
1829 ## Root simplex log...
1830 ##
1831 ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
1832 ##      18981      2.9987176e+09  0.000000e+00  1.331897e+04  18s
1833 ##      19238      2.9987176e+09  0.000000e+00  0.000000e+00  18s
1834 ## Concurrent spin time: 6.47s (can be avoided by choosing Method=3)
1835 ##
1836 ## Solved with barrier
1837 ##
1838 ## Root relaxation: objective 2.998718e+09, 19238 iterations, 20.09 seconds (10.31 work units)
1839 ## Total elapsed time = 40.57s (DegenMoves)

```

1840 ## Total elapsed time = 46.22s (DegenMoves)

1841 ##

1842 ## Nodes | Current Node | Objective Bounds | Work
1843 ## Expl Unexpl | Obj Depth IntInf | Incumbent BestBd Gap | It/Node Time

1844 ##

1845	##	0	0	2.9987e+09	0	1064	-0.00000	2.9987e+09	-	-	47s
1846	## H	0	0				1.675607e+09	2.9987e+09	79.0%	-	47s
1847	## H	0	0				1.996396e+09	2.9987e+09	50.2%	-	47s
1848	## H	0	0				2.012783e+09	2.9987e+09	49.0%	-	47s
1849	## H	0	0				2.027420e+09	2.9987e+09	47.9%	-	49s
1850	## H	0	0				2.049300e+09	2.9987e+09	46.3%	-	49s
1851	## H	0	0				2.101456e+09	2.9987e+09	42.7%	-	50s
1852	## H	0	0				2.117499e+09	2.9987e+09	41.6%	-	51s
1853	## H	0	0				2.144168e+09	2.9987e+09	39.9%	-	51s
1854	## H	0	0				2.147679e+09	2.9987e+09	39.6%	-	52s
1855	## H	0	0				2.160926e+09	2.9987e+09	38.8%	-	53s
1856	## H	0	0				2.184323e+09	2.9987e+09	37.3%	-	53s
1857	## H	0	0				2.258573e+09	2.9987e+09	32.8%	-	53s
1858	## H	0	0				2.268774e+09	2.9987e+09	32.2%	-	53s
1859	## H	0	0				2.277223e+09	2.9987e+09	31.7%	-	54s
1860	## H	0	0				2.291178e+09	2.9987e+09	30.9%	-	54s
1861	## H	0	0				2.294975e+09	2.9987e+09	30.7%	-	54s
1862	## H	0	0				2.301497e+09	2.9987e+09	30.3%	-	54s
1863	## H	0	0				2.301576e+09	2.9987e+09	30.3%	-	54s
1864	## H	0	0				2.303263e+09	2.9987e+09	30.2%	-	55s
1865	## H	0	0				2.306486e+09	2.9987e+09	30.0%	-	55s
1866	## H	0	0				2.314636e+09	2.9987e+09	29.6%	-	55s
1867	## H	0	0				2.315561e+09	2.9987e+09	29.5%	-	55s
1868	## H	0	0				2.317962e+09	2.9987e+09	29.4%	-	55s
1869	## H	0	0				2.322088e+09	2.9987e+09	29.1%	-	55s
1870	## H	0	0				2.326215e+09	2.9987e+09	28.9%	-	55s

```

1871 ## H      0      0                2.331662e+09 2.9987e+09 28.6%      -   56s
1872 ##      0      0 2.9976e+09    0 1245 2.3317e+09 2.9976e+09 28.6%      -  159s
1873 ## H      0      0                2.608509e+09 2.9976e+09 14.9%      -  187s
1874 ##      0      0 2.9971e+09    0 1217 2.6085e+09 2.9971e+09 14.9%      -  217s
1875 ##      0      0 2.9971e+09    0 1195 2.6085e+09 2.9971e+09 14.9%      -  222s
1876 ##      0      0 2.9971e+09    0 1195 2.6085e+09 2.9971e+09 14.9%      -  224s
1877 ##      0      0 2.9971e+09    0 1205 2.6085e+09 2.9971e+09 14.9%      -  224s
1878 ##      0      0 2.9959e+09    0 1337 2.6085e+09 2.9959e+09 14.9%      -  298s
1879 ##
1880 ## Cutting planes:
1881 ##   Gomory: 2
1882 ##   MIR: 136
1883 ##   Flow cover: 586
1884 ##
1885 ## Explored 1 nodes (54379 simplex iterations) in 300.05 seconds (371.41 work units)
1886 ## Thread count was 24 (of 24 available processors)
1887 ##
1888 ## Solution count 10: 2.60851e+09 2.60851e+09 2.33166e+09 ... 2.30326e+09
1889 ##
1890 ## Time limit reached
1891 ## Warning: max constraint violation (1.7494e-06) exceeds tolerance
1892 ## Warning: max bound violation (3.0145e-06) exceeds tolerance
1893 ## Best objective 2.608509094927e+09, best bound 2.995904713522e+09, gap 14.8512%
1894 ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 2608509094.9269 | Runtime: 300.3 s

```

1895 1. Introduction

1896 Agroforestry systems that integrate trees with agricultural crops (silvoarable systems)
1897 are increasingly recognised as a promising land-use option to reconcile climate mitigation,
1898 biodiversity conservation, and rural income generation on the same hectare of land (Toens-
1899 meier, 2017). Temperate agroforestry systems (AFS) based on fast-growing species such
1900 as poplar, willow, alder, and black locust can deliver substantial biomass yields while im-

1901 proving soil quality and providing multiple ecosystem services compared to conventional
1902 monocultures (Huber et al., 2018). Biomasses produced in AFS is considered as a sus-
1903 tainable feedstock and can be used either energetically, chemically or materially (e.g. in
1904 the building sector). The latter two options are generally preferred over the energetic
1905 usage from a cleaner production perspective as biogenic carbon is stored in products
1906 during their life span which contributes to climate change mitigation.

1907 Despite substantial research on ecological advantages and pioneering initiatives to
1908 establish AFS in practice, the share of AFS in agricultural production systems is still
1909 small. E.g. in Germany about 1200 hectares of silvoarable AFS are established in 2025,
1910 which accounts for about 0.02% of total agricultural land use [DEFAP]. One of the core
1911 challenges in establishing AFS systems at large scale in practice is a lack of opportunities
1912 to generate substantial revenues from woody biomasses of AFS. On the other side, the
1913 rise of bioeconomy creates substantial demands for woody biomass and requires a reliable
1914 and cost-efficient feedstock supply. As woody biomasses are bulky feedstocks with low
1915 energy density, the design of cost-efficient supply chains is crucial to ensure the economic
1916 viability of AFS-based value chains. This requires coordinated decisions on stand estab-
1917 lishment, multi-cycle harvesting, and product cascading in spatially distributed supply
1918 chains, which can be supported by integrated optimisation models that link stand-level
1919 growth dynamics to regional supply chain design.

1920 In the last two decades, a large body of research has analysed biomass supply chains
1921 for bioenergy and biorefineries using mathematical programming models, often in the
1922 form of mixed-integer linear programming (MILP) (Zahraee et al., 2020; Sharma et al.,
1923 2013). These models typically optimise network configuration, feedstock sourcing, and
1924 logistics under cost or profit objectives. They often treat biomass availability as an exoge-
1925 nous input or rely on simplified agricultural production decisions. Moreover, downstream
1926 processing of biomasses is concentrated on one particular value chain e.g. bioenergy plants
1927 consuming all produced biomasses. Recent work has started to integrate operational de-
1928 tails such as various biomass types, biomass growth models and regeneration processes
1929 in multi-period supply chain models.

1930 The present paper contributes to this literature in four ways. First, we propose
1931 a novel MILP formulation for AFS-based biomass supply chain design that explicitly

links establishment decisions and consecutive harvest decisions, ensuring that harvesting decisions at each site are consistent with minimum and maximum rotation ages. Second, we embed allometric yield functions for different biomass types generating age-specific yields for different woody biomass types (like stems, branches, or leaves and twigs). Third, these different woody biomass types can be processed in specific downstream industries, like chemical or paper industry as well as energy sector. The model aims to provide strategic decision support for establishing regional AFS-based bioeconomy ecosystems connecting multiple industries and product markets.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on agroforestry biomass systems, allometric biomass modelling, and biomass supply chain optimisation. Section 3 summarises the MILP formulation for the agroforestry supply chain with age-lag constraints. Section ?? outlines the planned computational experiments for the case study and discusses expected insights for cleaner production strategies.

2. Literature review

Agroforestry biomass systems, stand-level growth models, and biomass supply chain optimisation have each been studied extensively, but they are rarely combined in an integrated framework that links age-dependent stand dynamics to regional product-cascading value chains (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). This section reviews (i) agroforestry systems with emphasis on system types, economics, and the cascading use of biomass as the primary revenue driver; (ii) biomass growth models at tree and stand level with explicit attention to the allometric–diameter–stand–biomass pathway; and (iii) biomass supply chain design using mathematical programming. The section concludes by identifying the modelling gap addressed in this paper.

2.1. Agroforestry systems

Agroforestry deliberately integrates woody perennials with crops and/or livestock on the same management unit, with the aim of enhancing productivity, profitability, and environmental performance (Nair, 1993; Toensmeier, 2017). In the temperate zone, three

distinct system types can be distinguished based on the spatial arrangement of trees and the rotation length at which woody biomass is harvested.

Short-rotation coppice (SRC) is defined as a monoculture plantation of fast-growing woody species covering the entire management unit, with planting densities of 8,000–20,000 cuttings $\cdot$ ha $^{-1}$. After each harvest the root system regenerates without replanting. Rotation cycles are typically 3–7~years and total stand lifetime is 20–25~years (Trnka et al., 2008; Faasch and Patenaude, 2012). SRCs are no AFSs in the narrow sense, but it is helpful to benchmark biomass production.

Short-rotation agroforestry systems (SRAFS) integrate SRC-type tree strips (typically 1–4 rows of the same fast-growing species like poplar or willow) into an otherwise arable field in an alley-cropping arrangement. Crop production continues in the inter-row strips (Huber et al., 2018; Graves et al., 2007). Thus, tree strips typically cover 10–30% of the total parcel area. Rotation lengths and total stand lifetime in SRAFS are comparable to SRCs. In SRAFS all trees are effectively edge trees, exposed to higher light and water availability than interior SRC trees. This boundary effect raises individual-tree biomass accumulation but also intensifies competition with adjacent crop rows (Huber et al., 2018).

Long-rotation agroforestry systems (LRAFS) use the same alley-cropping spatial structure as SRAFS but with rotation cycles of 12–15~years or longer (Graves et al., 2007; Nair, 1993). At this rotation length, individual trees develop substantial stem diameters and the proportion of high-value timber increases markedly, enabling premium material uses (veneer, sawlogs, engineered wood products) that are inaccessible in short-rotation systems. Therefore, often timber-quality tree species (such as walnut, cherry, oak, or high-grade poplar clones) are planted.

SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off age is hard to define and, more importantly, the intended use of AFS biomass is not always clear and can also change over time. Nonetheless, when establishing AFS, the structure of AFS and its intended use after harvest drives planting decisions and, subsequently, economic viability. Thus, in the present paper SRAFS and LRAFS are not considered as separate AFS systems, but the interplay of intended use, establishment, and harvesting decisions is analysed.

2.1.1. *Economic rationale of landowners*

The decision of a landowner to establish an AFS involves weighing substantial up-front costs and long-term opportunity costs against a diversified revenue stream. On the **cost side**, establishment requires site preparation, planting material, and planting labour, typically amounting to 1,500–3,500 €/ha for SRC or SRAFS (Faasch and Patenaude, 2012; Testa et al., 2014). Recurrent harvest and refitting costs (machine hire, chipping, transport to roadside) add approximately 200–400 €/ha per rotation depending on stand density and equipment availability (Testa et al., 2014; Spinelli et al., 2009). During the rotation, maintenance cost of AFS arise on a yearly basis. Next to expenses for establishment, maintenance and harvesting, opportunity cost arise as no revenues from crops are generated on tree strips. In regions with fertile soils and high cereal prices, yearly crop revenues may sum up to 400–600 €/ha (Faasch and Patenaude, 2012; Graves et al., 2007). Thus, opportunity cost are often the dominant cost driver in SRAFS profitability analyses.

On the **benefit side**, the primary revenue of AFS is the sale of harvested woody biomass, whose magnitude depends critically on rotation length, species, and site productivity. For poplar-based SRAFS in Central Europe, typical yields range from 10 to 25 t of dry matter biomass per ha for typical rotation lengths (Huber et al., 2018; Niemczyk and Thomas, 2021). Additionally, positive crop-yield effects in the inter-row strips of SRAFS and LRAFS may result. Tree rows act as windbreaks, reducing soil erosion and evapotranspiration; they provide partial shading that moderates heat stress during drought years; and their root systems may improve water retention in the topsoil (Graves et al., 2007; Grünwald et al., 2007). Empirical measurements in Central European alley-cropping trials indicate that these microclimate effects can increase cereal yields in adjacent strips by 5–15% relative to open-field controls, partially or fully compensating the loss of arable area occupied by the tree strips (Graves et al., 2007). Beyond direct yield effects, AFS deliver broader ecosystem services (like soil organic matter accumulation, carbon sequestration, biodiversity enhancement) that may lead to agri-environment subsidies and carbon credits (Toensmeier, 2017; Faasch and Patenaude, 2012).

Nonetheless, economic attractiveness of AFS depends primarily on revenues of AFS

biomass. Thereby, revenues are different for the types of biomass and in which value chain it is used. Often AFS biomasses are used energetically, i.e. for heat and power generation, or anaerobic digestion, which typically creates low prices per tonne of biomass. Alternatively, woody parts of the biomass (i.e., the stem and large branches) can also be used as pulp and paper grade material as well as feedstock for chemical industry or as timber achieving higher prices (Lamers et al., 2011; De Meyer et al., 2014). Thus, fostering AFS establishment by maximizing economic returns from AFS biomass requires a strategic approach to product cascading that allocates different biomass fractions to the highest-value applications, while ensuring that the rotation length is aligned with the growth dynamics of the stand and the market demand for different biomass types (Lamers et al., 2011).

Thus, for making educated decisions on AFS establishment and harvesting, the expected biomass quantities at harvest for stem, branches, and residues need to be estimated. Thereby, the shares of the biomass types change with stand age. E.g. the share of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et al., 2018; Testa et al., 2014). Therefore, in Section 2.2 growth modelling approaches are reviewed that capture the age-dependent growth dynamics of different biomass types in AFS.

2.2. Biomass growth models

Reliable estimation of stand-level biomass quantities and their age-dependent composition is a necessary prerequisite for any strategic analysis of AFS-based supply chains. Individual-tree biomass is commonly approximated by species-specific allometric power laws relating shoot dry matter to stem base diameter, and these relationships have been calibrated for Central European poplar clones under SRC and SRAFS conditions (Huber et al., 2016, 2018). For supply chain planning, however, the relevant quantity is **stand-level biomass per hectare** over a multi-decade rotation horizon, which requires a temporal growth model that captures the characteristic sigmoidal accumulation trajectory of woody biomass.

2.2.1. Stand-level growth model

Stand-level aboveground and belowground dry biomass per hectare, $M^{sl}(t)$, accumulates in a right-skewed sigmoidal pattern with a relatively early inflection point, driven

Table 1: Notation for the biomass growth and harvest yield models.

Symbol	Description
$M^{sl}(t)$	Total above- and belowground dry biomass per ha of AFS (t DM ha ⁻¹) at stand age t
$A(N)$	Stand-level Gompertz asymptote (t DM ha ⁻¹); density- and site-dependent
k	Gompertz growth-rate coefficient (yr ⁻¹)
t_0	Inflection age of stand biomass accumulation (yr)
N	Stand density (trees ha ⁻¹)
C_{site}	Site-quality constant (kg ha ^{β-1}); conservative: 4475, good site: 6471
β	Competition-density exponent; fitted $\hat{\beta} = 0.380$
ϕ_{ag}	Aboveground fraction of total stand biomass; $\phi_{\text{ag}} = 0.89$
$M^{\text{ag}}(t)$	Aboveground dry biomass per ha (t DM ha ⁻¹) at age t
$f_p(t)$	asymptotic fraction of AGB attributed to compartment $p \in \{s, b, r\}$ at age t
α_p, δ_p	Intercept and slope of the linear fraction model for compartment p
$q_p(t)$	Merchantable fraction of compartment p biomass at age t (logistic)
r_p	Logistic growth rate for merchantability of compartment p
$t_{50,p}$	Age at which 50% of compartment p biomass is merchantable (yr)
$Y_p(t)$	Merchantable dry biomass of compartment p per ha at harvest age t (t DM ha ⁻¹)
ω	Moisture content at felling (mass fraction); $\omega = 0.55$

by rapid juvenile shoot growth followed by self-thinning and competitive deceleration as canopy closure intensifies (Niemczyk and Thomas, 2021; Testa et al., 2014). This trajectory is described by a Gompertz model:

$$M^{sl}(t) = A(N) \cdot \exp(-\exp(-k(t - t_0))), \quad (1)$$

where t_0 is the age of maximum instantaneous biomass increment and k determines how quickly the stand approaches its site-specific asymptote $A(N)$. For Central European poplar under medium-rotation AFS management (10–20 yr), calibration against empirical data from Huber et al. (2018), Niemczyk and Thomas (2021), and Jha (2018) yields $k = 0.194, \text{yr}^{-1}$ and $t_0 = 9.7, \text{yr}$ for a conservative site, and $k = 0.173, \text{yr}^{-1}$, $t_0 = 10.1, \text{yr}$ for a high-productivity site, placing the growth peak close to the biologically optimal rotation age identified by Niemczyk and Thomas (2021).

The asymptote $A(N)$ is not a fixed species constant but depends explicitly on planting

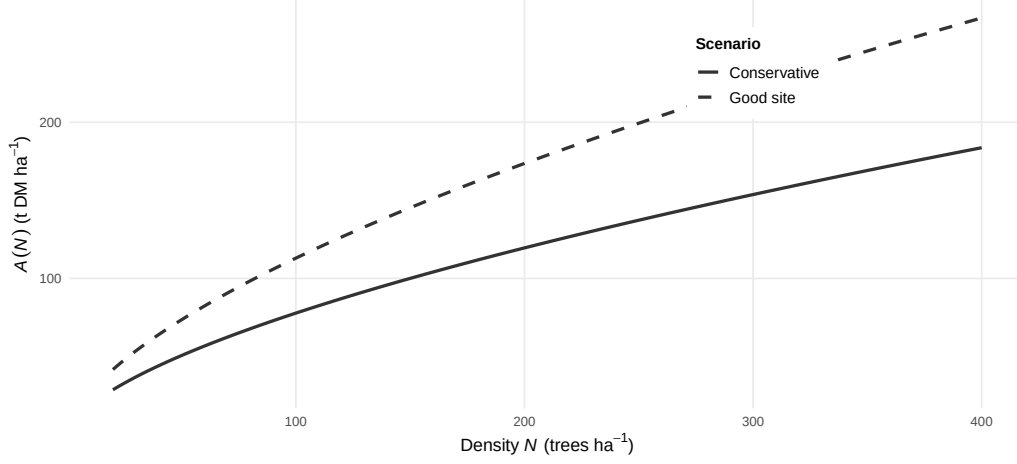


Figure 1: Stand-level Gompertz asymptote function $A(N)$ of planting density for the conservative (solid) and good-site (dashed) scenarios.

density through a competition-density power law (?):

$$A(N) = \frac{C_{\text{site}} \cdot N^{1-\beta}}{1000}, \quad (2)$$

where C_{site} encodes site quality and β is the competition exponent. Equivalently, the per-tree asymptote $A_{\text{tree}}(N) = A(N) \cdot 1000/N = C_{\text{site}} \cdot N^{-\beta}$ decreases with density as individual trees have access to less growing space. Fitting (2) to two anchor points — a conservative AFS calibration at 113 trees,ha⁻¹ (central EU; $A = 84, \text{t,DM,ha}^{-1}$) and SRC data at 1,333 trees,ha⁻¹ from Niemczyk and Thomas (2021) — yields $\hat{\beta} = 0.380$, consistent with the 0.3–0.5 range reported for managed hardwood plantations (?). Two site-quality scenarios are used throughout: a **conservative scenario** ($C_{\text{site}} = 4\,475$, 113 trees,ha⁻¹, $A = 84, \text{t,DM,ha}^{-1}$) representative of sandy loam soils in northern Central Europe, and a **good-site scenario** ($C_{\text{site}} = 6\,471$, 139 trees,ha⁻¹, $A = 138, \text{t,DM,ha}^{-1}$) representative of deep loamy soils in river valleys.

Within each compartment, only the biomass fraction whose element diameter exceeds a minimum threshold qualifies as merchantable product for industrial use. This fraction rises continuously with stand age as stems thicken and branches coarsen, and is modelled by a logistic function:

$$q_p(t) = \frac{f_p}{1 + \exp(-r_p(t - t_{50,p}))}, \quad (3)$$

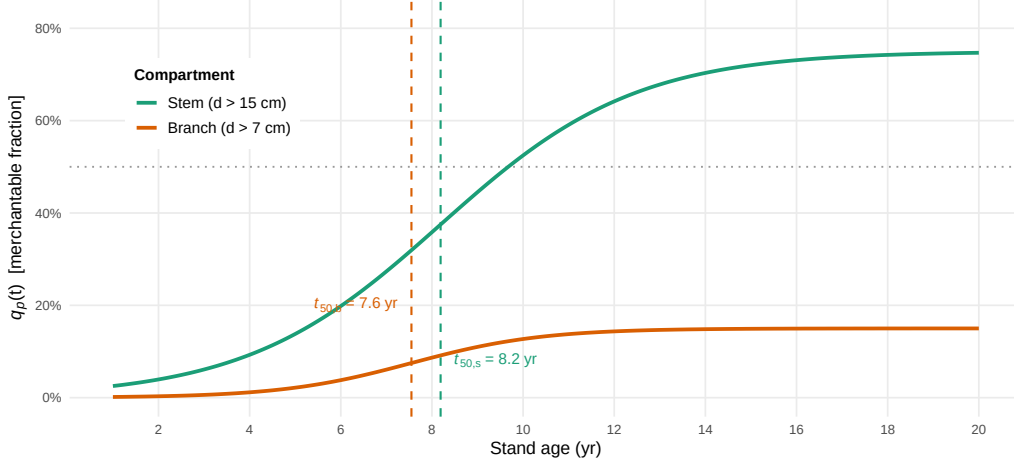


Figure 2: Logistic diameter-class merchantability functions $q_p(t)$ for stem ($d > 15$ cm, green) and branches ($d > 7$ cm, orange); the 50% threshold ages t_{50} are indicated by dashed vertical lines.

where $t_{50,p}$ is the age at which 50% of compartment p biomass surpasses the diameter threshold. f_p denotes the asymptotic share of component p for mature trees. For the merchantable stem (threshold $d > 15$ cm), calibration to data from Civitarese et al. (2019) and Niemczyk and Thomas (2021) yields $r_{stem} = 0.467, \text{yr}^{-1}$, $t_{50,stem} = 8.19, \text{yr}$, and $f_{stem} = 0.75$. For merchantable branches (threshold $d > 7$ cm at insertion), Civitarese et al. (2019) and Jha (2018) imply $r_{bran} = 0.698, \text{yr}^{-1}$, $t_{50,bran} = 7.55, \text{yr}$, and $f_{bran} = 0.15$.

Combining equations (1)–(3), the merchantable dry biomass yield of compartment p per hectare at harvest age t is:

$$Y_p(t) = q_p(t) \cdot M^{ag}(t), \quad (4)$$

with all non-merchantable fractions aggregated into the residue stream. Fresh-weight yields at felling are obtained from dry matter yields via $Y_p^{fw} = Y_p / (1 - \omega)$, where the moisture content at felling $\omega = 0.55$ is taken from Civitarese et al. (2019). Figure 2 illustrates the shapes of $A(N)$, $f_p(t)$, and $q_p(t)$; Figure 3 shows the resulting stacked AGB component trajectories compared with empirical observations.

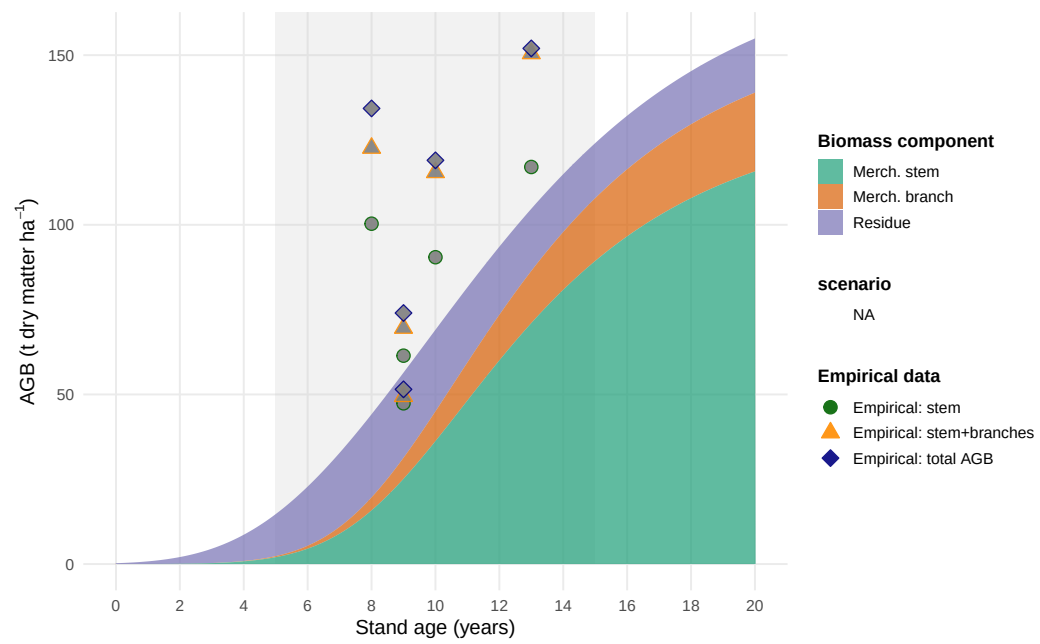


Figure 3: Stand-level aboveground biomass components for poplar under AFS management with $N = 250$ trees ha^{-1} . Empirical AGB observations compiled from Jha (2018), Civitarese et al. (2019), and Niemczyk and Thomas (2021). Grey band: typical medium-rotation harvest window (5–15 years).

2.3. Biomass supply chain design

Biomass supply chain design problems have been extensively studied in literature (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). Most models jointly optimise facility locations and capacities, biomass sourcing patterns, multimodal transport flows, and inventory policies under cost or profit objectives, sometimes augmented by greenhouse gas emission indicators (Ekşioğlu et al., 2009; Arabi et al., 2019). A key methodological feature of the more advanced formulations is the integration of long-term network design with multi-period logistics, enabling simultaneous optimisation of infrastructure and operational decisions across seasonal or annual time steps (Ekşioğlu et al., 2009; Zahraee et al., 2020).

Among generic multi-product frameworks, OPTIMASS explicitly models product quality cascades by defining a hierarchy of biomass classes and allowing higher-quality products to satisfy lower-quality demands, while the t-OPTIMASS extension incorporates biomass growth and regeneration over multiple periods (De Meyer et al., 2014, 2016). However, in virtually all existing formulations biomass availability is treated as an exogenous time-series input: stand age structure, rotation decisions, and coppice cycles are not represented endogenously, so that trade-offs between rotation length, stand management, and supply chain configuration cannot be explored in a unified model (De Meyer et al., 2016; Lamers et al., 2011). This gap is particularly pronounced for agroforestry systems with repeated coppice cycles, where establishment timing, consecutive harvest decisions, and age-dependent biomass quality jointly determine both the feasible supply profile and the achievable product cascade.

The MILP formulation developed in this paper addresses this gap by endogenising stand age dynamics through binary age-lag variables, coupling the component-specific yield coefficients η_{pa} derived from the Gompertz growth models above with supply chain flow variables, and embedding a product quality cascade that allocates stem, branch, and residue fractions to chemical, pulp, and energy markets within a single optimisation model.

2.4. Biomass supply chain design

The design and planning of biomass supply chains has attracted substantial research attention over the past two decades, driven by the need to ensure cost-efficient and

sustainable feedstock provision for bioenergy and biorefinery applications. Several systematic reviews provide comprehensive overviews of this literature. Cambero and Sowlati (2014) survey economic, social, and environmental assessment and optimisation studies for forest biomass supply chains, identifying MILP as the dominant modelling approach and noting a growing trend towards multi-objective formulations integrating life cycle assessment.

Yue et al. (2014) review biomass-to-bioenergy and biofuel supply chain models with a focus on multi-scale modelling, uncertainty treatment, and sustainability metrics, and conclude that the coupling of upstream feedstock dynamics with downstream network design remains an underexplored area. Malladi and Sowlati (2018) survey logistics-specific features of biomass supply chain models—seasonal availability, moisture content degradation, preprocessing steps—and highlight that most models treat biomass as a homogeneous commodity, neglecting quality differentiation. Zahraee et al. (2020) provide a recent meta-review of 140+ studies, confirming that cost minimisation is still the dominant objective, and that multi-product cascading formulations are rare.

The structure of biomass supply chain models typically encompasses three decision layers: (i) strategic network design (facility location, capacity sizing, technology selection), (ii) tactical planning (harvest scheduling, seasonal flows, inventory), and (iii) operational routing. The following paragraphs discuss key individual contributions and their specific methodological advances.

Ekşioğlu et al. (2009) develop one of the earliest large-scale MILP models for biomass-to-biorefinery supply chain design, integrating facility location, capacity sizing, and multimodal transport for lignocellulosic feedstocks. Their model minimises total system cost and introduces binary facility-opening variables that have since become standard in the field. A key limitation, however, is that biomass supply is treated as an exogenous input, so feedstock availability is not endogenised as a function of land-use or growth decisions. Mansoornejad et al. (2013) extend the strategic design perspective to forest biorefineries by integrating product portfolio design with supply chain configuration in a single MILP. Their model explicitly accounts for product conversion efficiencies along different processing pathways, enabling joint optimisation of process technology selection and logistics.

De Meyer et al. (2014) introduce the OPTIMASS framework, a generic MILP for biomass-based supply chain optimisation that explicitly models product quality cascades: higher-quality biomass classes can substitute for lower-quality demands downstream, whereas the reverse is infeasible. This cascade hierarchy, formalised through ordered product sets, is particularly relevant for woody biomass where stem, branch, and residue fractions command different market prices. The companion t-OPTIMASS extension (De Meyer et al., 2016) incorporates temporal dynamics by explicitly representing biomass growth and regeneration over multiple periods, enabling simultaneous optimisation of harvest timing and network flows. However, in both OPTIMASS formulations the growth dynamics are represented as exogenous time-series rather than as age-structured stand models, so rotation-length decisions and age-dependent biomass quality are not endogenised. Arabi et al. (2019) demonstrate multi-period MILP extensions for bio-based supply chains with production dynamics, but focus on microalgae rather than woody biomass and do not address product quality cascades.

Sharma et al. (2013) identify supply-side uncertainty (yield variability, seasonal availability, moisture content) as the most critical source of risk in biomass supply chains. In response, a strand of literature has developed stochastic and robust optimisation extensions. Zahraee et al. (2020) document that two-stage stochastic MILP and robust optimisation approaches now account for roughly 30% of the modelling literature, with biomass yield uncertainty and price volatility as the dominant stochastic parameters.

Biomass supply chain research focusing specifically on agroforestry or SRC contexts is comparatively sparse. Espinoza et al. (2017) note that most supply chain models assume continuous feedstock availability and do not represent the periodic harvest cycles characteristic of coppice systems. Lamers et al. (2011) analyse feedstock design principles for lignocellulosic supply chains and argue that the allocation of different biomass fractions to valorisation pathways—product cascading—is a key lever for economic viability, yet is rarely optimised jointly with network design decisions. The model developed in Section~3 of this paper directly addresses this gap by endogenising stand age dynamics through binary age-lag variables and coupling component-specific yield coefficients with a product quality cascade.

Table~2 summarises the key modelling features of the reviewed contributions.

Table 2: Key modelling features of selected biomass supply chain design studies.

Study	Objective	Multi-product	Quality cascade	Age dynamics	Uncertainty	Level
Eksioglu et al. (2009)	Cost min.	—	—	—	—	Strategic
Sharma & Taaffe (2013)	Cost min.	Partial	—	—	—	Strategic/Tactical
Mansoornejad et al. (2013)	Profit max.	✓	—	—	—	Strategic
Cambero & Sowlati (2014)	Cost/GHG/jobs	—	—	—	Stochastic	Strategic
Yue et al. (2014)	Cost/GHG	✓	—	—	—	Multi-scale
De Meyer et al. (2015)	Cost min.	✓	✓	Exogenous	—	Strategic/Tactical
De Meyer et al. (2016)	Cost min.	✓	✓	Exogenous TS	—	Multi-period
Lamers et al. (2015)	Cost/quality	✓	Partial	—	Scenario	Strategic
Espinoza et al. (2017)	Sustainability	✓	—	—	Scenario	Strategic
Malladi & Sowlati (2018)	Cost min.	Partial	—	—	—	Tactical/Oper.
Arabi et al. (2019)	Profit max.	✓	—	Exogenous	—	Multi-period
Zahraee et al. (2020)	Multi-obj.	✓	—	—	Stochastic	Strategic/Tactical
This paper	NPV max.	✓	✓	Endogenous	Scenario	Strategic/Multi-period

3. MILP Model for Agroforestry Supply Chain Design

Building on the biomass growth models introduced in Section 2.2 and the supply chain design literature reviewed in Section 2.3, this section describes the AFS supply chain design problem (AFS-SCD), a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain. The model endogenises stand age dynamics through binary age-lag variables, couples the component-specific yield coefficients η_{pa} derived from the Gompertz growth models (??)–(??) with supply chain flow variables, and assigns AFS biomass types (i.e., stem, branch, and residue fractions) to potential consumers in chemical, pulp and paper as well as energy industry.

3.1. Sets, Indices, and Time Structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/pre-treatment facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and biomass types $\mathcal{P} = \{\text{stem}, \text{branches}, \text{residues}\}$, whose growth

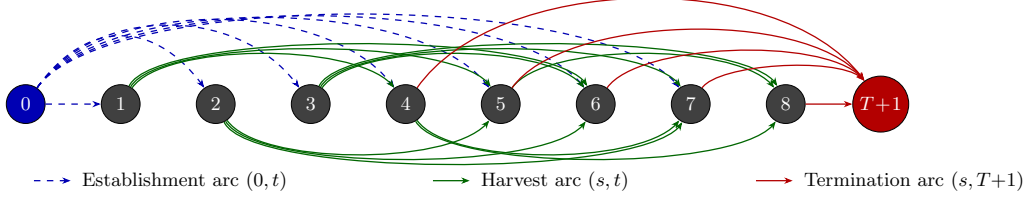


Figure 4: Network structure of set $\mathcal{S}$ for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$. Dashed blue arcs: establishment arcs $(0, t)$, $t \in \{1, \dots, T-1\}$. Solid green arcs above: harvest arcs (s, t) with odd s ; below: harvest arcs with even s . Solid red arcs: termination arcs $(s, T+1)$, $s \in \{A_{\min} + 1, \dots, T\}$.

models are outlined above in (??)–(??). Time is discretised into annual periods $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To model flexible harvest timing according to typical rotation structures of SRAFS and LRAFS, $A^{\min}$ and $A^{\max}$ define minimum and maximum stand ages. Thus, harvest are possible in periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$. To consistently model establishment and consecutive harvest decisions, the following arc sets are defined: $\mathcal{S}^{\text{est}}$ defines potential establishments of AFS. If an AFS is established in period $t \in \mathcal{T}$, arc $(0, t)$ is selected. Harvest set $\mathcal{S}^{\text{harv}}$ summarizes all potential consecutive harvest periods within the age-lag bounds $A^{\min}$ and $A^{\max}$. I.e., if an AFS is established or harvested in period s and next harvest is in period t , arc (s, t) is selected. Finally, there is no subsequent harvest after period t , the AFS is terminated by selecting arc $(t, T^{\max} + 1)$. Termination arcs are summarized in set $\mathcal{S}^{\text{term}}$. All arc sets are formally defined as follows:

$$\begin{aligned}\mathcal{S}^{\text{est}} &= \{(0, t) \mid t \in \{1, \dots, T^{\max} - A^{\min}\}\}, \\ \mathcal{S}^{\text{harv}} &= \{(s, t) \mid s, t \in \mathcal{T} \wedge t - s \in \{A^{\min}, \dots, A^{\max}\}\}, \\ \mathcal{S}^{\text{term}} &= \{(s, T^{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}, \\ \mathcal{S} &= \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}.\end{aligned}$$

Figure 4 illustrates this network structure of all arc sets for $T = 8$, $A^{\min} = 3$, $A^{\max} = 5$.

3.2. Decision Variables

Binary variables z_{ist} indicate whether an arc from set $(s, t) \in \mathcal{S}$ is selected for site i , indicating establishment, (consecutive) harvest, or termination. Taken together across all arcs, these variables trace a path of consecutive harvest cycles at each site over the

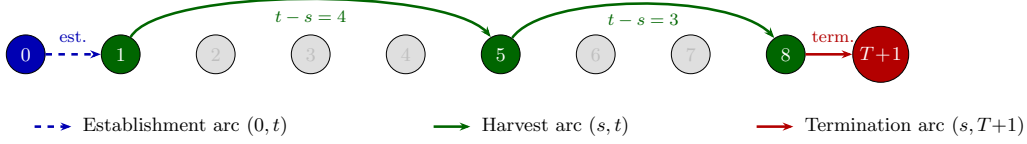


Figure 5: Example of a feasible path through the network for $T = 8, A_{\min} = 3, A_{\max} = 5$: establishment in period 1, harvests in periods 5 and 8, termination at $T + 1$.

planning horizon. Figure 5 illustrates a feasible path corresponding to the establishment of an AFS in period 1 and harvests in periods 5 and 8, whereby the harvest in $t = 8$ is final (and AFS is terminated, thus).

When a AFS is harvested in period t , continuous variables $Y_{ipt} \geq 0$ represent the harvest quantity (t) of product p at site i . Flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from AFS sites to storage/pre-treatment facilities and from there to consumers, respectively, while S_{jpt} captures end-of-period inventory at facility j .

Table 3 summarises all sets, decision variables, and parameters used in the MILP formulation. Symbols shared with the growth model of Section 2.2 — in particular the yield coefficients η_{pa} , which are derived from the component-specific Gompertz functions (??)–(??) and the stand biomass model (??) — are listed in Table ?? in Section 2.2.

3.3. Objective Function

As discussed in Section 2.1, the economic attractiveness of AFS supply chains must be assessed from a global perspective taking into account all processes, activities, and participants involved in finally serving customer markets with bio-based feedstocks. Therefore, revenues for the different biomass fractions and must be weighed against costs occurring at all value-adding stages of the supply chains. To be precise, revenues created by selling AFS-based feedstocks to different industrial consumers are counterweighted by establishment, opportunity, harvest, transport, and storage costs of landowners, agricultural firms as well as biomass logistics companies. The model therefore maximises total contribution

margin of all participants in the AFS supply chain over the planning horizon as follows:

$$\begin{aligned}
\max Z = & \sum_{k \in \mathcal{K}, p \in \mathcal{P}} R_{kp} \cdot \sum_{j \in \mathcal{J}, (p', p) \in \mathcal{Q}, t \in \mathcal{T}} X_{jkp't} \\
& - \sum_{i \in \mathcal{I}} c_i^{est} \cdot AREA_i \cdot \sum_{(0, t) \in \mathcal{S}^{est}} z_{i0t} \\
& - \sum_{i \in \mathcal{I}} c_i^{opp} \cdot AREA_i \cdot \left(\sum_{(s, T+1) \in \mathcal{S}^{term}} (s+1) \cdot z_{is(T+1)} - \sum_{(0, t) \in \mathcal{S}^{est}} t \cdot z_{i0t} \right) \\
& - \sum_{i \in \mathcal{I}} c_i^{main} \cdot (t-s) \cdot AREA_i \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
& - \sum_{i \in \mathcal{I}} c_i^{harv} \cdot AREA_i \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
& - \sum_{i \in \mathcal{I}, j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}} c_p^{tr-raw} \cdot d_{ij} \cdot X_{ijpt} \\
& - \sum_{j \in \mathcal{J}, k \in \mathcal{K}, (p, p') \in \mathcal{Q}, t \in \mathcal{T}} c_{p'}^{tr-pre} \cdot d_{jk} \cdot X_{jkpp't} \\
& - \sum_{j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}} c_j^{stor} \cdot S_{jpt}
\end{aligned} \tag{5}$$

Here, R_{kp} are product- and consumer-specific revenues (€/t). The opportunity cost term penalises dereduced crop revenues in each year an AFS is established. Transport cost terms mirror the two-stage flow structure of the AFS supply chain with differentiated rates for raw and pre-treated biomass depending on the biomass type. Cost rates differ because as e.g. the density and water content of the shipped biomasses differ or because specialized equipment (e.g. for log transports) is used [REFERENCES]. Moreover, site-specific establishment cost for planting the AFS trees is considered, next to yearly maintenance costs and harvesting cost. All these costs scale with the size of a site ($AREA_i$) measured in hectar. Maintenance cost comprise e.g. costs for trimming, replanting, and pruning unnecessary shoots. Storage costs are considered if biomasses are stored for more than a year to compensate direct costs (e.g. for storage site maintenance) but primarily opportunity cost due to potential material losses and capital commitment. The usual case is that AFS are harvested at the beginning of a year. Harvested biomass dry during spring and summer such that the biomasses can be used in autumn or winter the same year. As drying is necessary for most industrial applications and beneficial for transport economics, drying storage in the year of harvest is not associated with

2252 additional cost.

2253 3.4. Constraints

2254 Constraints (6) limit each site to at most one establishment decision over the planning
2255 horizon. Constraints (7) enforce flow conservation in each period. If an activity in period
2256 $t \in \mathcal{T}$ takes place, there needs to be a preceeding activity (harvest or establishment) and
2257 a succeeding activity (harvest or termination). This way, (7) guarantee valid (possibly
2258 empty) activity paths over the complete time horizon for each site. Constraints (8) links
2259 the binary harvest-cycle variables to physical yield quantities via the age-dependent yield
2260 coefficients $\eta_{p(t-s)}$. These coefficients are pre-computed by evaluating the component-
2261 specific Gompertz functions (??) and (??) at stand age $a = t - s$ and scaled by site
2262 area AREA_i . Thus, implicitly we assume constant yields per hectare for each site i . Note
2263 that this simplification can be dropped and site-specific yields $\eta_{ip(t-s)}$ could be intro-
2264 duced without increasing numerical complexity of the AFS-SCD . Constraints (??)–(9)
2265 link harvest quantities to transport flows and ensure inventory balance at each facility,
2266 following the multi-period network-flow structure of OPTIMASS (De Meyer et al., 2014,
2267 2016). Thereby, harvest quantities can be smaller than available AFS biomasses allowing
2268 partial harvesting. Constraints (10)–(11) enforce storage and processing capacity limits
2269 at each facility j . Constraints (12) implement product usage at multiple consumption
2270 sites k . Thereby, $(p', p) \in \mathcal{Q}$ defines a product mapping allowing high-quality product
2271 p' (e.g. stems) satisfying demand of a lower-quality products p up to maximum demand
2272 $D_{kpt}^{\max}$. This way, the cascading principle is operationalized as a central economic ratio-
2273 nale.

$$\sum_{(0,t) \in \mathcal{S}^{est}} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I} \quad (6)$$

$$\sum_{(s,t) \in \mathcal{S}} z_{ist} = \sum_{(t,u) \in \mathcal{S}} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (7)$$

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq \sum_{(s,t) \in \mathcal{S}^{harv}} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (8)$$

$$S_{jpt} = S_{jp,t-1} + \sum_{i \in \mathcal{I}} X_{ijpt} - \sum_{k \in \mathcal{K}, (p,p') \in \mathcal{Q}} X_{jkpp't} \quad \forall j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (9)$$

$$\sum_{p \in \mathcal{P}} S_{jpt} \leq \text{CAP}_j^{stor} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \quad (10)$$

$$\sum_{i \in \mathcal{I}, p \in \mathcal{P}} X_{ijpt} \leq \text{CAP}_j^{proc} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}^{harv} \quad (11)$$

$$\sum_{j \in \mathcal{J}, (p',p) \in \mathcal{Q}} X_{jkp'pt} \leq D_{kpt}^{\max} \quad \forall k \in \mathcal{K}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (12)$$

$$z_{ist} \in \{0, 1\} \quad \forall i \in \mathcal{I}, (s, t) \in \mathcal{S} \quad (13)$$

$$Y_{ipt}, X_{ijpt}, X_{jkp'pt}, S_{jpt} \geq 0 \quad \forall i, j, k, p, p', t \quad (14)$$

2274 The yield coefficients η_{pa} in (8) are the critical bridge between the stand-level growth
 2275 model of Chapter~2 and the supply chain optimisation. For stand age $a = t - s$ and
 2276 product component p , they are computed as:

$$\eta_{pa} = f_p(a) \cdot M^{sl}(a) = \frac{h_p(a)}{\sum_{p'} h_{p'}(a)} \cdot a^\infty \cdot e^{-e^{-b'^1 \cdot (a-\tau)}}, \quad (15)$$

2277 where $M^{sl}(a)$ follows from (??) and $f_p(a)$ follows from (??). For poplar clone Max~3 un-
 2278 der SRAFS conditions, the component-specific Gompertz parameterisations from Section
 2279 2.2 imply that $\eta_{stem,a}$ rises steeply between ages~4 and 10, while $\eta_{branch,a}$ and $\eta_{residue,a}$
 2280 plateau earlier, as illustrated in Figure~??. This age-dependence of product fractions is
 2281 the core mechanism through which rotation-length decisions — governed by $A^{\min}$ and
 2282 $A^{\max}$ and the arc set $\mathcal{S}$ — determine achievable product portfolios and, hence, revenues
 2283 in the objective function (5).

4. Case Study

The proposed MILP model is intended to support the design of agroforestry biomass supply chains for cleaner production in specific regional contexts. In a forthcoming case study, we focus on poplar-based alley-cropping systems in a temperate European setting, informed by empirical data from SRAFS trials in Southern Germany and SRC plantations in Central Europe (Huber et al., 2018; Trnka et al., 2008; Testa et al., 2014).

4.1. Study Region and Supply Chain Configuration

The proposed AFS-SCD model is instantiated on a real-world biomass supply chain located in the tri-state region of Saxony-Anhalt, northern Thuringia, and northern Saxony, Germany (Figure 6). This region combines the highest density of registered short-rotation coppice (SRC) Feldblöcke in eastern Germany with a cluster of major wood-processing industries that collectively represent one of the largest forest-based biomass demand centres in Central Europe. The spatial extent of the study region covers approximately 250×200 km, with road distances between potential KUP sites and consumer locations ranging from under 10 km to over 180 km.

The supply chain comprises three node types: (i) KUP production sites $\mathcal{I}$, (ii) pre-treatment and intermediate storage facilities $\mathcal{J}$, and (iii) consumer sites $\mathcal{K}$. Candidate KUP sites are drawn from the official Feldblock register of Saxony-Anhalt, filtered to parcels with a minimum area of 1 ha and classified as potentially suitable for SRC under the InVeKoS land-use scheme. Following the two-stage spatial clustering procedure described in Section-??, the $|\mathcal{I}| = 2239$ raw Feldblöcke are grouped into K super-sites using DBSCAN ($\varepsilon = 15$ km, $minPts = 3$) and Ward.D2 hierarchical clustering (maximum within-cluster radius 15 km), yielding a computationally tractable MILP instance while preserving the total eligible area. The HAC cluster assignments, visible as distinct colour zones in Figure 6, confirm a spatially coherent partitioning: dense KUP corridors along the Elbe and Saale floodplains form contiguous super-sites, while isolated parcels in the Harz foothills constitute smaller or singleton clusters.

Two types of intermediate infrastructure nodes are considered: conventional sawmills and logistics hubs. Sawmills perform primary mechanical processing (debarking, chipping) and can serve product streams P1 and P2; logistics hubs serve as consolidation

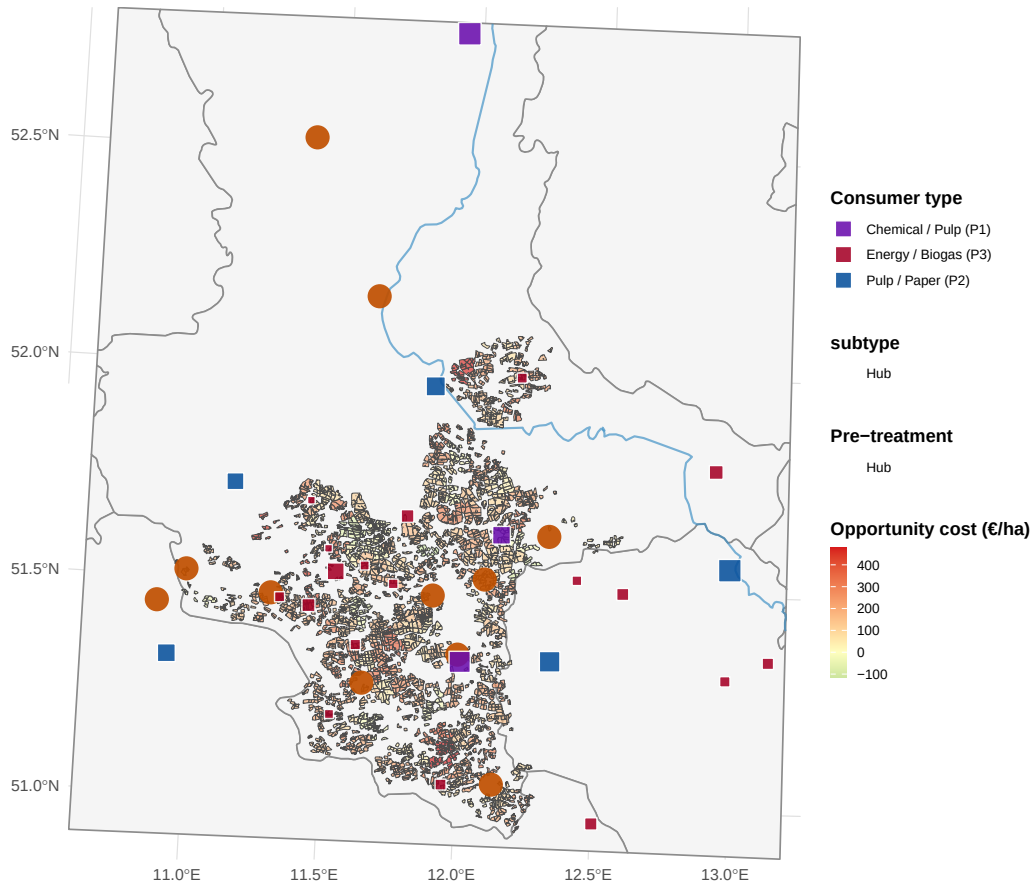


Figure 6: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Coloured polygons: potential KUP/AFS sites grouped by HAC super-site (colours encode cluster membership; no legend). Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

and temporary storage points for all three product grades and are characterised by lower capital intensity but limited processing capacity. The $|\mathcal{J}| = 11$ candidate storage/pre-treatment locations are sourced from publicly registered forestry enterprises and existing logistic nodes in the study region. Road distances d_{ij} (site i to storage j) and d_{jk} (storage j to consumer k) are computed via the OSRM open-source routing engine using the German road network, providing realistic transport cost estimates that are embedded directly in the MILP objective.

4.2. Consumer Demand and Product Classification

Nine consumer sites are included in the base case ($|\mathcal{K}| = 9$), covering the full quality cascade from high-value chemical/pulp applications down to energy use (Table 4). Consumer demands $D_{kpt}^{\max}$ are derived from publicly reported annual throughput figures, production capacities, and regulatory filings, and are held constant across periods ($D_{kpt}^{\max} = D_{kp}^{\max} \forall t \in \mathcal{T}$) to reflect stable long-term offtake contracts.

Three biomass product grades are distinguished, reflecting the cascading-use principle embedded in the AFS-SCD objective:

- **P1 – Chemical/pulp grade** ($p = 1$): long-fibre hardwood chips suitable for chemical pulping and biorefinery conversion. Dominated by Mercer Stendal GmbH (Arneburg, 1{,}500 kt/yr) and Zellstoff Rosenthal/Mercer (Blankenstein, 1{,}000 kt/yr), which together account for 403% of total P1 demand. UPM Biochemicals (Leuna) contributes 500 kt/yr of high-purity beech-fraction demand for biochemical conversion. Gate prices for P1 range from 80 to 120 €/t wood input.
- **P2 – Sawlog/pulpwood grade** ($p = 2$): roundwood and chipped material suitable for mechanical processing in sawmills. Demand is distributed across six smaller facilities with individual annual demands of 5–300 kt/yr. Gate prices range from 55 to 75 €/t.
- **P3 – Energy/pellets grade** ($p = 3$): residual wood fractions (chips, bark, off-cuts) used for heat and electricity generation or pellet production. This grade absorbs material that cannot meet P1 or P2 quality thresholds, operationalising the cascade constraint in the MILP. Gate prices range from 30 to 40 €/t.

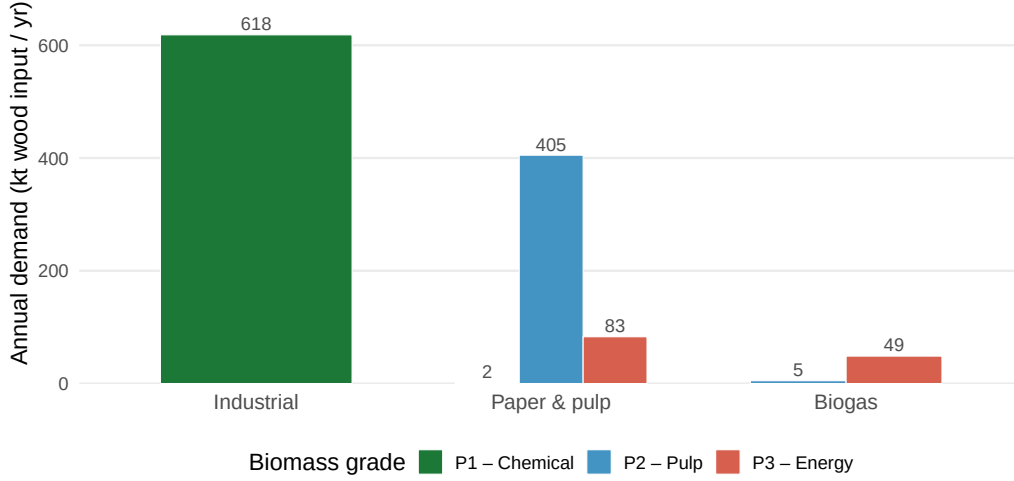


Figure 7: Total biomass demand by consumer type and grade (P1 chemical, P2 pulp & paper, P3 energy/biogas). Bars show aggregated annual demand in kt wood input per year across all individual plants within each consumer type.

2343 The demand derivation follows a three-step procedure. First, publicly available ca-
 2344 pacity data (kt wood input per year) are collected for each facility from official sources,
 2345 industry reports, and corporate disclosures (Mercer Stendal GmbH, 2024; UPM Bio-
 2346 chemicals, 2024). Second, the total annual capacity is assigned to the product grade that
 2347 corresponds to the primary feedstock of each facility (P1 for chemical/biorefinery pro-
 2348 cesses, P2 for mechanical sawmilling, P3 for thermal conversion and pellet lines). Third, a
 2349 minor P3 component is added to sawmill facilities to capture by-product streams (bark,
 2350 sawdust), consistent with observed industrial practice. The resulting demand matrix
 2351 $D_{kp}^{\max}$ (kt/yr) is shown in Figure 7 and summarised in Table 4. Aggregate demand across
 2352 the study region totals 620.5 kt/yr for P1, 410 kt/yr for P2, and 131.7 kt/yr for P3.

2353 4.3. MILP Model Parameters

2354 Table 5 summarises all scalar parameters of the AFS-SCD instance. The planning
 2355 horizon covers $|\mathcal{T}| = 15$ periods, corresponding to one full poplar rotation (years 1–15),
 2356 with a minimum harvest age $a^{\min} = 4$ and maximum productive stand age $a^{\max} =$
 2357 15 (Lehmkuhl et al., 2023). Biomass yield functions η_{pa} (odt ha⁻¹ yr⁻¹) follow the
 2358 age-structured Chapman-Richards profile calibrated in Section~??, where distinct yield

curves are assigned to each product grade to reflect the fraction of stem, branch, and foliage biomass that qualifies under each quality threshold.

Transport costs are disaggregated into two components reflecting the two-echelon logistics structure. Raw material transport from site i to pre-treatment facility j incurs a variable cost $c^{tr-raw} \cdot d_{ij}$ ($\text{€ t}^{-1} \text{ km}^{-1}$), while processed material transport from j to consumer k is costed at $c^{tr-pre} \cdot d_{jk}$. The pre-treatment rate c^{tr-pre} is set lower than c^{tr-raw} to reflect the higher load density and standardised packaging of chipped and pelletised material compared to loose roundwood. An opportunity cost parameter c^{opp} ($\text{€ ha}^{-1} \text{ yr}^{-1}$) accounts for foregone agricultural revenue on converted parcels, calibrated to the average arable land rental rate in Saxony-Anhalt (Statistisches Bundesamt (Destatis), 2023). Storage capacity at each facility j is bounded by $\bar{S}_j$, and processing capacity by $\bar{C}_j$ (kt yr^{-1}), both drawn from publicly available facility data.

4.4. Optimization results

5. Sensitivity analysis

6. Conclusion

7. Appendix

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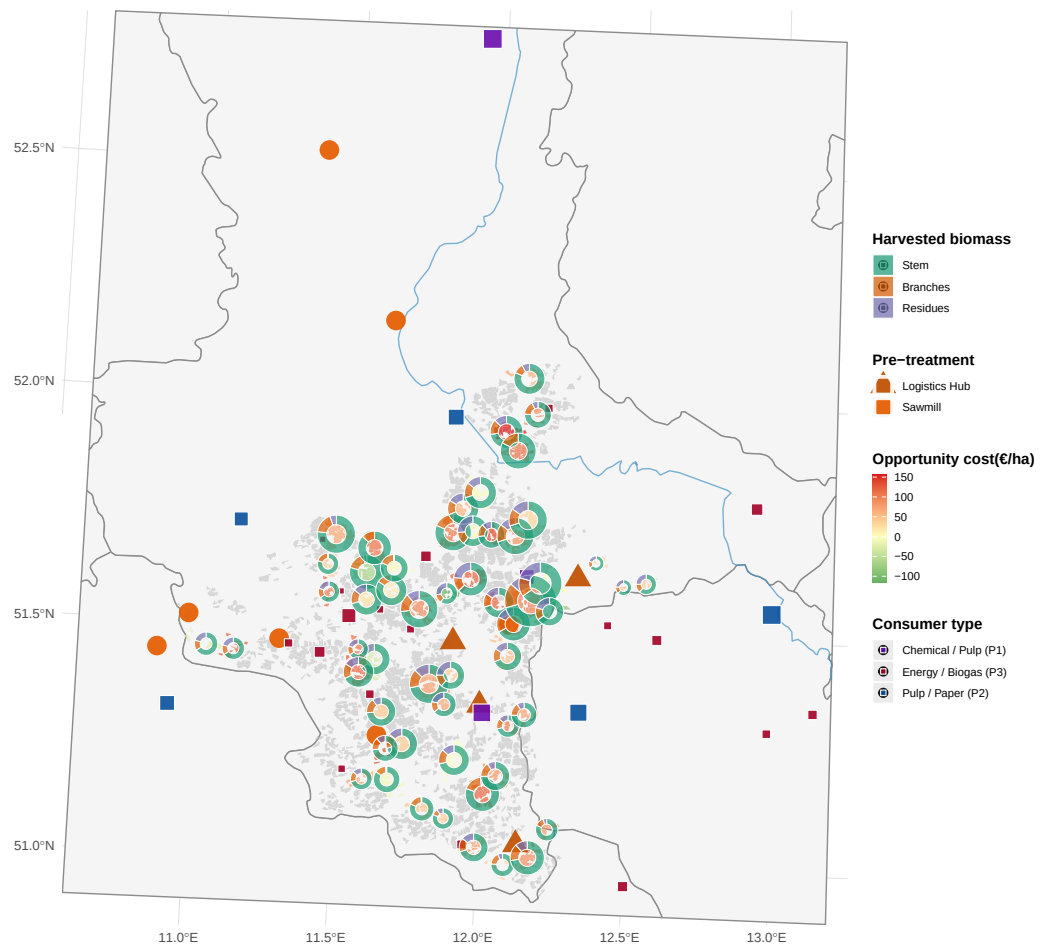


Figure 8: Optimal agroforestry supply chain network. Green polygons: active super-sites with biomass harvest; grey polygons: non-selected sites. Pie charts at active sites show the product mix of harvested biomass (stem = P1, branches = P2, residues = P3), scaled by total harvested volume. Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub). Coloured squares: consumer sites by dominant demand category.

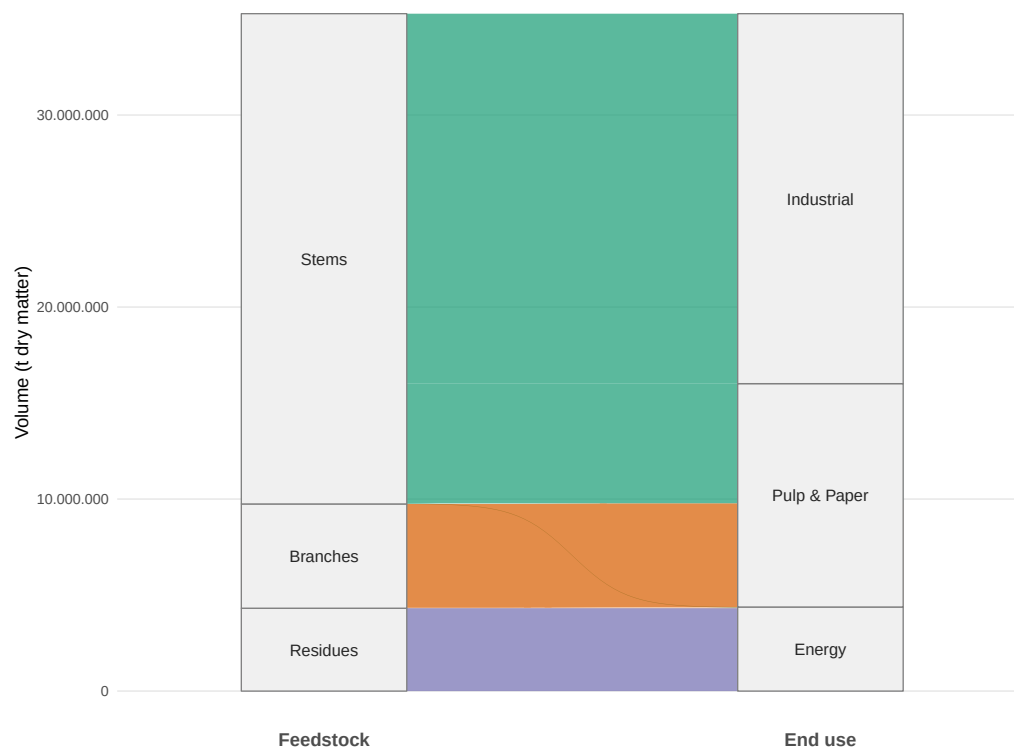


Figure 9: Total production quantity of biomass types and its industrial usage over planning horizon (in tons).

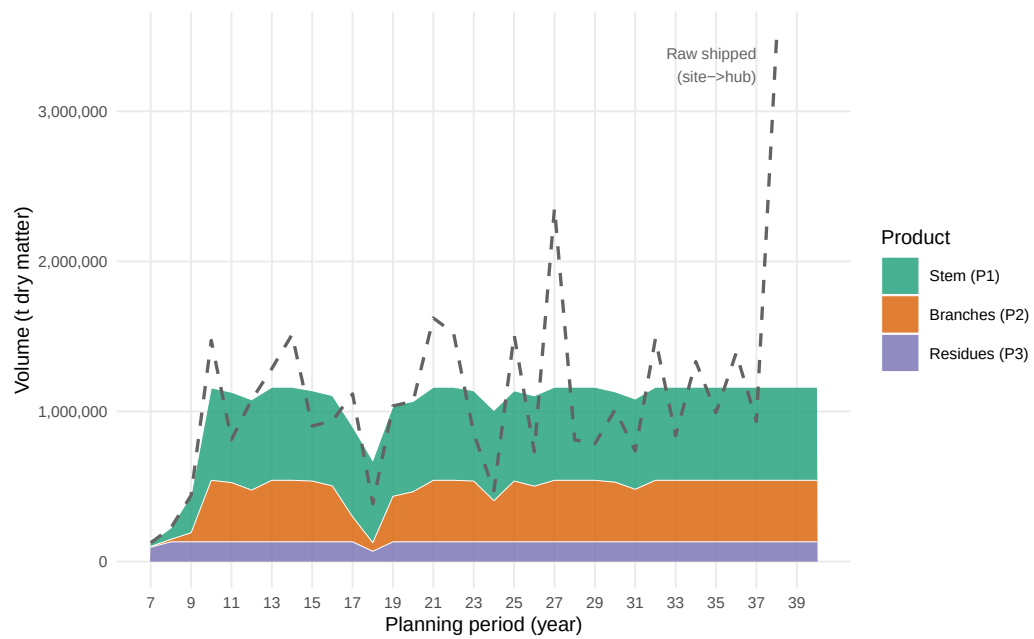


Figure 10: Total biomass delivered to consumers by product type over the planning horizon. Stacked areas show period-level volumes of stem (P1), branches (P2), and residues (P3) flowing from storage hubs to all consumers. Dashed line: total harvested raw biomass shipped from sites to hubs.

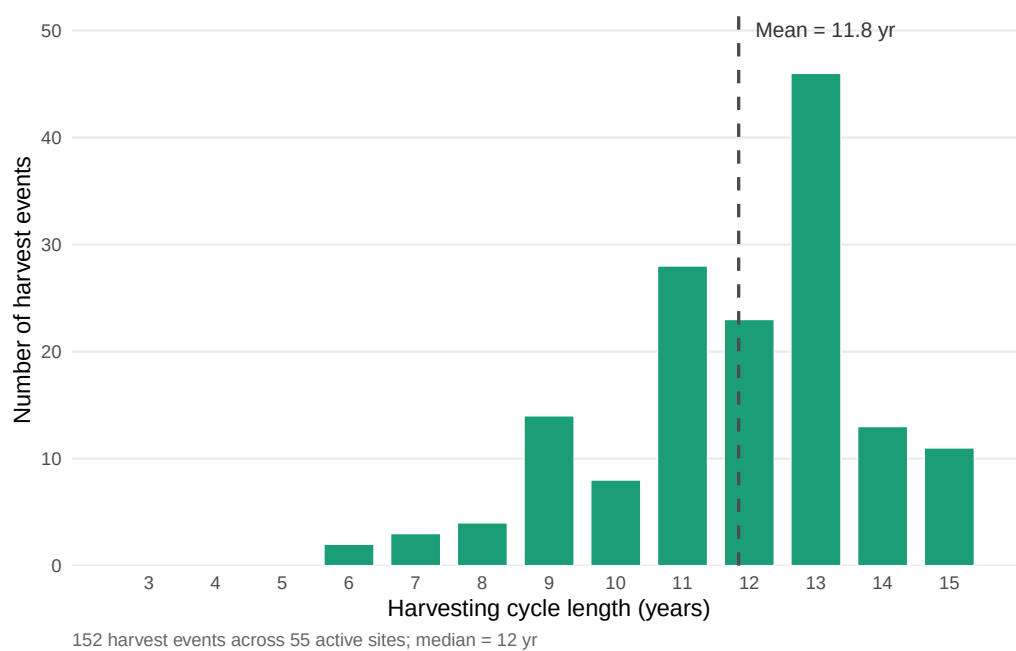


Figure 11: Distribution of harvesting cycle lengths at active agroforestry sites. Each bar represents the number of individual harvest events (over all periods and active sites) whose cycle length (stand age at harvest, i.e. $t - s$) falls in that year bin. Vertical dashed line: mean cycle length. Active sites are defined as sites with at least one selected harvest arc in the optimal solution.

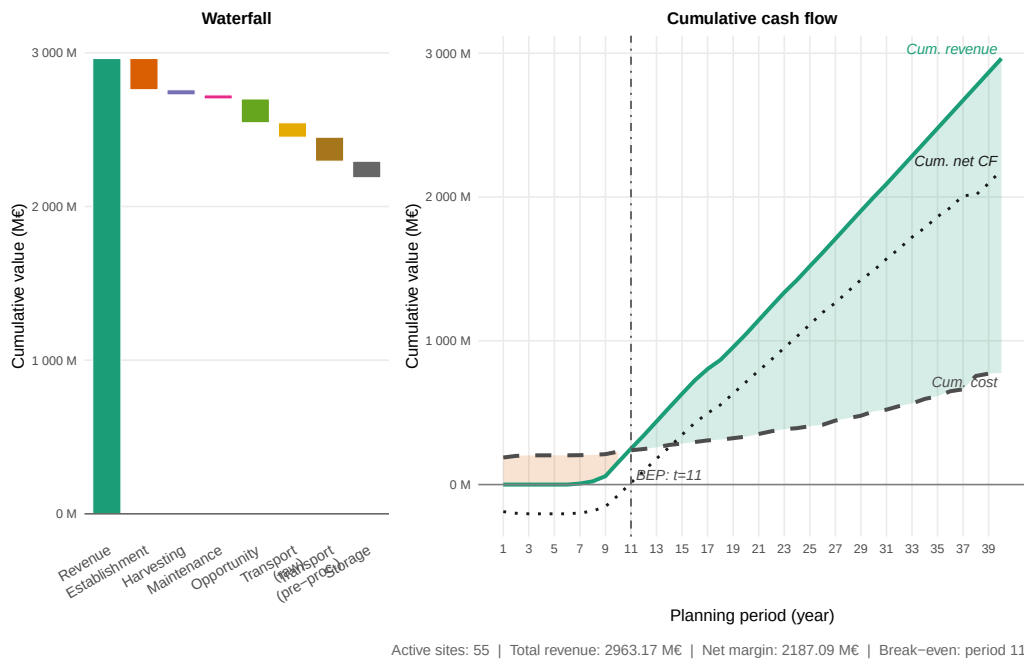


Figure 12: Total cost–profit breakdown across all active sites over the planning horizon. The left panel (waterfall) decomposes the objective value into revenue and individual cost categories; the right panel shows the period-level trajectory of revenues and stacked costs. Cultivation costs are split into establishment, harvesting, maintenance, and opportunity costs; logistics costs cover raw biomass transport (site→hub) and pre-processed transport (hub→consumer); storage costs reflect hub inventory holding charges.

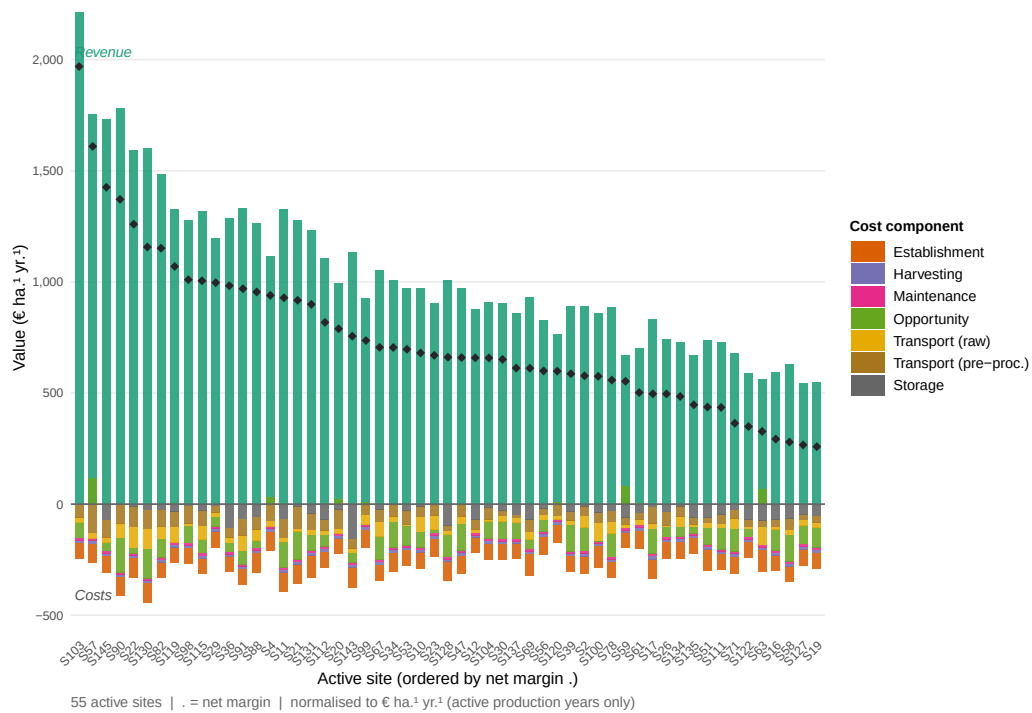


Figure 13: Total revenue and cost breakdown per active agroforestry site over the full planning horizon. Each bar represents one active site; revenue (green, above zero axis) and cost components (stacked below) are expressed in thousand euros per hectare for comparability across sites of different size. Sites are ordered by descending net margin.

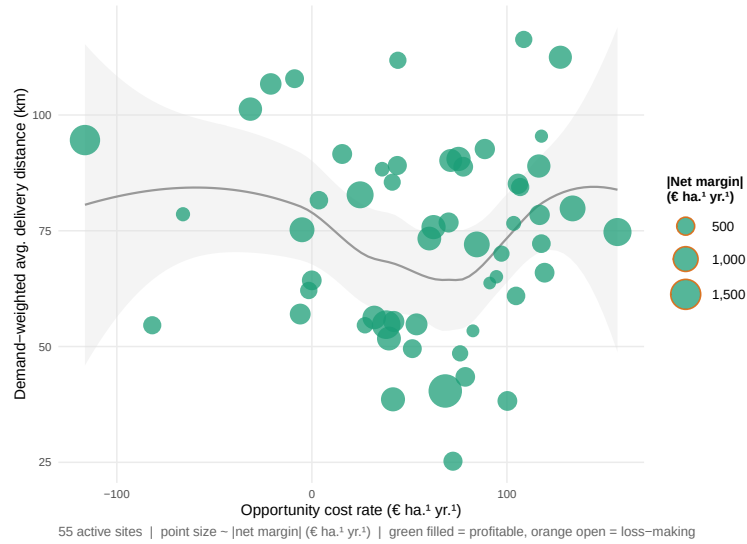


Figure 14: Scatterplot of active agroforestry sites: opportunity cost rate (€ ha⁻¹ yr⁻¹) on the x-axis versus demand-weighted average delivery distance (km) on the y-axis. Point size is proportional to the site's net margin (€ ha⁻¹ yr⁻¹); sites with negative net margin are shown as open circles. A LOESS smoothing line indicates the joint trend.

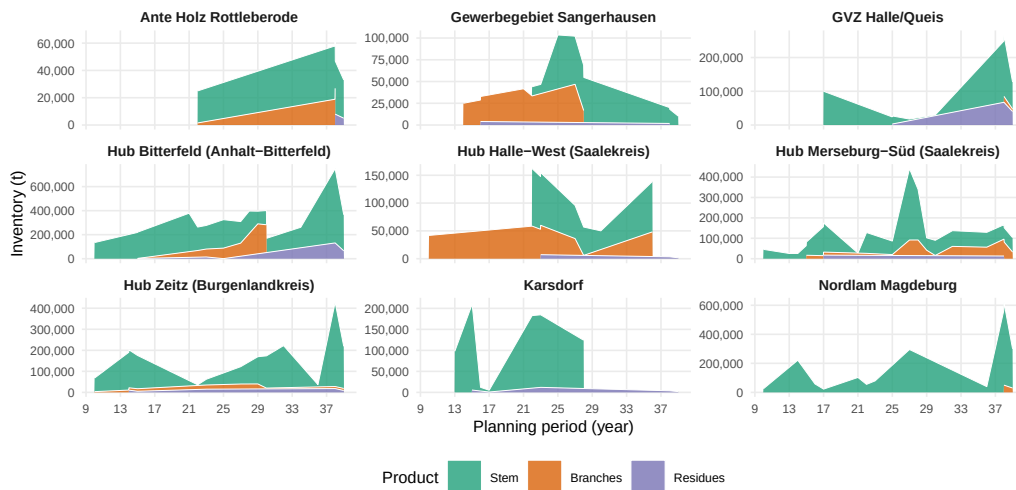


Figure 15: End-of-period inventory levels at storage/pre-treatment hubs, disaggregated by product type. Each panel corresponds to one active hub.

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Table 3: Notation for the MILP model.

Sets & Indices	
$i \in \mathcal{I}$	Agroforestry sites
$j \in \mathcal{J}$	Storage / pre-treatment facilities
$k \in \mathcal{K}$	Consumer sites
$p \in \mathcal{P}$	Product types (stem, branches, residues)
$(p, p') \in \mathcal{Q}$	Admissible product-substitution pairs (quality cascade)
$\mathcal{T} = \{1, \dots, T_{\max}\}$	Planning periods (years)
$\mathcal{T}^+ = \{0, \dots, T_{\max} + 1\}$	Extended time set incl. dummy nodes
$\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$	Feasible harvest periods
$\mathcal{S}^{\text{est}} = \{(0, t) \mid t \in \{1, \dots, T_{\max} - A^{\min}\}\}$	Establishment arcs: dummy source to period t
$\mathcal{S}^{\text{harv}} = \{(s, t) \mid s, t \in \mathcal{T}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}$	Harvest cycles within age-lag bounds
$\mathcal{S}^{\text{term}} = \{(s, T_{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}$	Termination arcs to dummy sink $T_{\max} + 1$
$\mathcal{S} = \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}$	Age-lag arc set (establishment $\cup$ harvest $\cup$ termination)
Decision Variables	
$z_{ist} \in \{0, 1\}$	1 if s and t are consecutive harvest cycles at site i
$Y_{ipt} \geq 0$	Harvest quantity (t) of product p at site i in period t
$X_{ijpt} \geq 0$	Transport flow (t) of p from site i to facility j in t
$X_{jkpp't} \geq 0$	Flow of p from facility j to consumer k satisfying demand p' in t
$S_{jpt} \geq 0$	Inventory (t) of product p at facility j end of period t
Parameters	
$T_{\max}$	Length of planning horizon (years)
$A^{\min}, A^{\max}$	Minimum / maximum stand age at harvest (years)
AREA_i	Area of site i (ha)
η_{pa}	Yield coefficient (t/ha) of product p at stand age a , derived from (??)–(??)
R_{kp}	Revenue (€/t) for product p at consumer k
C_i^{est}	Establishment cost (€/ha) at site i
C_i^{opp}	Annual opportunity cost (€/ha) at site i
C_i^{harv}	Harvest & refitting cost (€/ha) at site i
$c^{\text{tr-raw}}$	Transport cost rate (€/t km) for raw biomass
$c^{\text{tr-pre}}$	Transport cost rate (€/t km) for pre-treated biomass
c_j^{stor}	Storage holding cost (€/t) at facility j
d_{ij}, d_{jk}	Distance (km) between nodes
$\text{CAP}_j^{\text{stor}}$	Storage capacity (t) at facility j
$\text{CAP}_j^{\text{proc}}$	Processing throughput capacity (t/period) at facility j
$D_{kpt}^{\max}$	Maximum demand (t) of product p at consumer k in period t

Table 4: Consumer demand quantities and gate prices by product grade. P1: chemical/pulp; P2: sawlog/pulpwood; P3: energy/pellets. Demands in kt wood input per year; prices in €/t wood input at the factory gate.

Consumer	Annual demand			Gate price		
	P1 (kt/yr)	P2 (kt/yr)	P3 (kt/yr)	P1 (€/t)	P2 (€/t)	P3 (€/t)
consumer 1	300.0	–	–	100	–	–
consumer 2	100.0	–	–	120	–	–
consumer 3	–	300	50.0	–	75	40
consumer 4	200.0	–	–	95	–	–
consumer 5	–	30	5.0	–	70	35
consumer 6	–	15	5.0	–	65	35
consumer 7	–	10	3.0	–	60	30
consumer 8	–	5	8.0	–	55	30
consumer 9	2.0	50	20.0	80	70	38
consumer 10	–	–	1.7	–	–	35
consumer 11	–	–	2.9	–	–	35
consumer 12	–	–	2.5	–	–	35
consumer 13	–	–	2.3	–	–	35
consumer 14	–	–	1.9	–	–	35
consumer 15	–	–	2.7	–	–	35
consumer 16	–	–	1.7	–	–	35
consumer 17	–	–	1.9	–	–	35
consumer 18	–	–	1.6	–	–	35
consumer 19	–	–	1.5	–	–	35
consumer 20	–	–	1.5	–	–	35
consumer 21	–	–	1.8	–	–	35
consumer 22	–	–	3.5	–	–	35
consumer 23	–	–	1.6	–	–	35
consumer 24	–	–	2.1	–	–	35
consumer 25	–	–	4.5	–	–	35
consumer 26	–	–	1.8	–	–	35
consumer 27	–	–	3.2	–	–	35
consumer 28	18.5	–	–	55	–	–

Table 5: AFS-SCD base-case parameter values for the Saxony-Anhalt case study. Transport costs refer to road distance (OSRM). Gate prices are weighted averages across consumer sites.

Parameter	Description	Base-case value	Unit
Index sets			
$ \mathcal{T} $	Planning periods	15	yr
$ \mathcal{I} $	KUP super-sites (post-clustering)	K (clustering-dependent)	–
$ \mathcal{J} $	Pre-treatment / storage nodes	11	–
$ \mathcal{K} $	Consumer sites	9	–
$ \mathcal{P} $	Product grades	3	–
Biomass dynamics			
$a^{\min}$	Minimum harvest age (years)	4	yr
$a^{\max}$	Maximum stand age (years)	15	yr
Cost parameters			
c^{tr-raw}	Raw transport cost	0.08	€ t ⁻¹ km ⁻¹
c^{tr-pre}	Pre-processed transport cost	0.05	€ t ⁻¹ km ⁻¹
c^{opp}	Opportunity cost (foregone arable rent)	280	€ ha ⁻¹ yr ⁻¹
Revenue parameters			
$\bar{R}_{P1}$	Mean gate price P1	99	€ t ⁻¹
$\bar{R}_{P2}$	Mean gate price P2	69	€ t ⁻¹
$\bar{R}_{P3}$	Mean gate price P3	36	€ t ⁻¹